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Temporal consistency of EEG signals in affective video annotation

Pablo Muñoz Alcaide

Supervisors:
V. Javier Traver Roig
Yoelvis Moreno Alcayde

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Abstract

This thesis addresses the challenge of temporal affective consistency analysis through EEG signals, a crucial aspect of affective computing that involves the stable association between emotional states and corresponding physiological responses over time. The study leverages data from multiple observers exposed to the same stimuli (videos), in which a crowdsourcing approach was applied to collect diverse physiological responses, combined with a brainsourcing methodology that integrates brain signal acquisition for deeper insight into affective consistency across individuals.

Past work has rarely paid attention to the temporal aspect of the EEG signals, and it has been often assumed that all segments of the sequences are equally important. In this work we challenge this assumption, and we develop and evaluate novel methods for quantifying affective consistency, using three different sets of approaches, of increasing data-awareness sophistication: data-agnostic heuristic, data-aware, and learning-based. The study introduces consistency-weighted schemes for improving the classification performance of emotion recognition models, exploring whether the temporal distribution of consistency along the sequence can be meaningful or video-independent, and its relevance for emotion recognition. The proposed methodology has been experimentally evaluated on two well-known EEG datasets for affective analysis, DEAP and MAHNOB.

Our findings reveal that consistency-weighted approaches, particularly learning-based methods, outperform traditional uniform weighting in binary valence and arousal classification, leading to improved generalizability across different subjects. The tentative conclusions, while still limited in scope, reveal potential and highlight limitations, such as the need for further analysis of the influence of weighted consistency schemes, refined feature selection and the limitations of the data-aware approach. In essence, this research offers new insights into the temporal dynamics of emotions and lays the groundwork for more robust and adaptable EEG-based emotion recognition systems, opening avenues for applications in fields such as human-computer interaction, mental health, affective video annotation, and education.

Resumen

Esta tesis aborda el reto del análisis de la consistencia afectiva temporal a través de señales EEG, un aspecto crucial de la computación afectiva que implica la asociación estable entre estados emocionales y las correspondientes respuestas fisiológicas a lo largo del tiempo. El estudio aprovecha los datos de múltiples observadores expuestos a los mismos estímulos (vídeos), en los que se aplicó un enfoque de crowdsourcing para recopilar diversas respuestas fisiológicas, combinado con una metodología de brainsourcing que integra la adquisición de señales cerebrales para obtener una visión más profunda de la consistencia afectiva entre individuos.

En trabajos anteriores rara vez se ha prestado atención al aspecto temporal de las señales de EEG, y a menudo se ha asumido que todos los segmentos de las secuencias tienen la misma importancia. En este trabajo desafiamos esta suposición, y desarrollamos y evaluamos métodos novedosos para cuantificar la consistencia afectiva, utilizando tres conjuntos diferentes de enfoques, de creciente sofisticación de conocimiento de datos: heurística agnóstica de los datos, consciente de los datos, y basado en el aprendizaje. El estudio introduce esquemas de consistencia ponderada para mejorar el rendimiento de clasificación de los modelos de reconocimiento de emociones, explorando si la distribución temporal de la consistencia a lo largo de la secuencia puede ser significativa o independiente del vídeo, y su relevancia para el reconocimiento de emociones. La metodología propuesta se ha evaluado experimentalmente en dos conocidos conjuntos de datos de EEG para el análisis afectivo, DEAP y MAHNOB.

Los resultados demuestran que los enfoques basados en la ponderación de la consistencia, en particular los métodos basados en el aprendizaje, superan a la ponderación uniforme tradicional en la clasificación binaria de la valencia y la excitación, lo que lleva a mejorar la generalizabilidad a través de diferentes sujetos. Los resultados preliminares, aunque todavía limitados en su alcance, revelan el potencial y ponen de relieve las limitaciones, como la necesidad de un mayor análisis de la influencia de los esquemas de consistencia ponderada, la selección refinada de características y las limitaciones del enfoque basado en datos. En esencia, esta investigación ofrece nuevas perspectivas sobre la dinámica temporal de las emociones y sienta las bases para sistemas de reconocimiento de emociones basados en EEG más robustos y adaptables, abriendo vías para aplicaciones en campos como la interacción persona-ordenador, la salud mental, la anotación de vídeos afectivos y la educación.

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Nomenclature

Acronyms:

- **EEG**: Electroencephalography
- **HCI**: Human-Computer Interaction
- **BCI**: Brain-Computer Interfaces
- k**-Fold**: k-Fold Cross Validation
- **LOVO**: Leave-one-Video-out
- **LOSO**: Leave-one-Subject-out
- **SDM**: Subject-Dependent Model
- **SVM**: Support Vector Machine
- k**-NN**: k-Nearest Neighbours
- **EEG-CCN**: EEG Contextual Neural Network
- **GLIMPSE**: Gaze's Spatio-Temporal Consistency from Multiple Observers
- **SGD**: Stochastic-Gradient Descent
- **DEAP**: A Dataset for Emotion Analysis using EEG, Physiological and Video Signals
- **MAHNOB-HCI**: A Multimodal Database for Affect Recognition and Implicit Tagging

Symbols:

- $\mathcal{D}$: Set of videos used in an EEG dataset to gather the data.
- $\mathcal{O}$: Set of subjects who participate in the EEG emotion recognition experiment.
- $f^{o,v}$: EEG feature matrix associated with a subject $o \in \mathcal{O}$ and video $v \in \mathcal{D}$.
- ω: Typically the window size in which EEG features are calculated.
- T_ω: Number of time segments of an EEG feature sample.
- $a^v(t)$: EEG-Glimpse function for the video v at instant t.
- θ^v: EEG-Glimpse threshold for the video v.
- W^v: Consistency-score vector for the video v.

1. Introduction

This introduction sets the stage by outlining the historical context, the emergence of affective computing, the significance of Electroencephalography (EEG) in emotion research, and the specific challenges that will be addressed in the thesis. It provides a clear framework for understanding the subsequent chapters and the overall objectives of the research.

1.1. Emotion Recognition

The study of emotions has intrigued scientists and philosophers for centuries, with early theories emphasizing the evolutionary significance of emotions and their expression. In the 19th and 20th century, theories such as the proposed by James-Lange [2] or Cannon-Bard [3], which highlight the complex interplay between physiological signals and emotional experiences. These foundational theories set the stage for contemporary research in affective computing and emotion recognition.

The concept of affective computing, introduced by Rosalind Picard in 1997, marked a turning point, opening the door to integrating emotion research with computer science [4]. It established a new paradigm where computers could recognize, interpret, and respond to human emotions, transforming the landscape of human-computer interaction (HCI). This vision prompted the initiation of interdisciplinary research, which brought together experts from a range of fields, including psychology, cognitive science, artificial intelligence (AI), and neuroscience.

Machine learning (ML) and Artificial Intelligence (AI) have played crucial roles in advancing affective computing. Early efforts focused on recognizing emotions through facial expressions, voice intonations, and physiological signals. However, these modalities faced limitations due to their susceptibility to deliberate control and external influences [5]. As a result, researchers began exploring brain-computer interfaces (BCIs) and EEG-based emotion recognition as more direct and reliable means of capturing emotional states.

In this context, considering that the brain regulates and controls the nervous system to participate in emotional processes, it is specially interesting to directly utilize brain activities information (e.g., EEG) to study the emotional processes and recognize the emotional state.

Electroencephalography (EEG), which is a technique for recording electrical activity of the brain, emerged as a powerful tool in emotion recognition research, due to its relationship with higher-level cognitive processes, including emotion. EEG measurements, have proven to be effective in the diagnosis of post traumatic stress disorder (PSTD) [6] and depression [7]. However, this is in part due to the major technological advances that have enabled high-resolution and real-time monitoring of brain activity and the creation of methods for extracting meaningful features from this type of signals. These features are then used to train machine learning models capable of classifying emotional states with increasing accuracy.

In this framework, Brain-Computer Interfaces (BCIs) represent a significant advancement, breaking the gap between brain activity and external devices. BCIs are systems that facilitate direct communication between the brain and an external device, often bypassing traditional neuromuscular pathways. This technology not only allows for the decoding of brain signals into actionable commands but also opens new avenues for studying and influencing emotional states through neurofeedback and real-time brain monitoring. Moreover, among the various BCI technologies, EEG has gained significant interest among researchers due to its simplicity, portability, user-friendly, and non-invasive nature. In particular, in the field of emotional recognition and affective computing, it is particularly popular due to the proliferation of public EEG databases [8, 9, 10].

1.2. Open Problems and Problem Statement

Despite advances in EEG-based emotion recognition, there remain several major challenges that hinder the development of robust and reliable systems. In this section we will outline these open problems that hinder progress, some of which are integral to out research.

Feature Extraction and Selection. One of the main characteristics of EEG signals is their complex nature, being noisy, chaotic and non-linear, making them difficult to analyze and interpret. Therefore, is it crucial to develop methods for feature extraction that can distill meaningful information from raw EEG signals. This involves exploring different type of features, commonly associated with signals, such as time-frequency domain features and non-linear dynamical features [11]. As well as leveraging advanced signal processing techniques to enhance signal quality and reduce noise. Feature selection is another critical step in the process, as it helps reduce the high dimensionality of EEG data, minimize computational complexity, and improve classification performance by selecting the most relevant features [12, 13].

However, our work does not focus on this problem and uses features extracted from two well-known datasets DEAP [9] and MAHNOB-HCI [14] datasets by the BANANA research project [15].

Inter-Subject Variability. A primary challenge of EEG-based emotion recognition lies in the variability of EEG signals across individuals. EEG signals exhibit high specificity to each individual's brain physiology and response patterns. For instance, when evaluating emotion recognition accuracy, Zhu, Zheng, and Lu [16] have shown that a system may achieve 92% accuracy when trained and tested on data from a single individual. However, this accuracy can drop considerably when the system is tested on data from multiple individuals simultaneously, reaching only 52% accuracy across three different subjects [16]. This disparity underscores the complex nature of EEG signals and highlights the difficulty in generalizing emotion recognition models across diverse populations. Factors contributing to this variability include differences in brain structure, electrode placement, and individual emotional expression styles.

Therefore, addressing these challenges required tailored approaches that account for individual differences; for that purpose, some training protocols will be developed focused on improving and evaluating the generability among subjects of our proposed algorithms. In addition, some of these algorithms will consist of an experimental-specific part in order to improve inter-subject emotion recognition.

Affective Consistency. Affective consistency is also one of the today's growing challenges, and the main focus of analysis in this thesis. This problem refers to the stable and coherent association between emotional states and their corresponding physiological response patterns (e.g., EEG) of subjects over time. In essence, it measures how reliably emotional responses are reflected in brain signals during repeated or prolonged exposure to similar stimuli.

In the field of Brain-Computer Interfaces, affective consistency appears in a current challenge, which is affective annotation of multimedia content, usually done by means of crowd-sourcing strategies. Therefore, annotating which part of the videos is associated with some emotions, based on predicting the emotional state of multiple users who have viewed this content, is a challenge. Although some preliminary work has been done in this direction, it has been mostly conducted on static images [17]. However, emotions and stimuli may not be static and evolve over time, influenced by various internal and external factors.

Therefore, this study aims to affective consistency and if it is possible to identify which patterns in EEG signals are truly representative of specific emotional states. To this end, we will develop various solutions, such as the creation of a score that measures affective consistency over time. We will develop various affective consistency scores: heuristic-based scores, based on EEG feature data, and another whose values are obtained by backpropagation of a neural network. In addition, these scores will be used to qualitatively analyze consistency, but also to weight the temporal segments of our EEG features, to analyze whether some temporal segments are more important than others in emotion recognition and can improve algorithm predictions.

In summary, our work focuses on analyzing affective temporal consistency and exploring its potential use to improve EEG-based emotion recognition systems, while addressing challenges such as intersubject variability in emotional responses. To this end, we pose the following research questions:

RQ1: How can we develop temporal affective consistency scores that reliably quantify the stable relationship between emotional responses of multiple participants/subjects/observers and EEG signals over time?

RQ2: Can consistency-weighted approaches outperform traditional uniform weighting schemes in emotion recognition performance?

RQ3: Can consistency-based scores be leveraged to improve the generalizability of emotion recognition algorithms and reduce the impact of intersubject variability?

1.3. Research Objectives

This thesis aims to address these challenges by exploring novel approaches for improving the accuracy and affective consistency of EEG-based emotion recognition. The specific objectives of the thesis include:

- **Foundational Framework Development**: Establish a foundational understanding and framework for capturing and interpreting brain signals in response to visual stimuli.
- **Affective Consistency Analysis**: To globally and temporally analyze the consistency of brain signals in response to visual stimuli, identifying patterns that correlate reliably with emotional responses over time.
- **Consistency Solutions**: Create and evaluate novel consistency scores to quantify the stable relationship between emotional states and EEG signals, and explore how these scores can be integrated into existing models to improve emotion recognition performance.
- **Machine Learning Algorithms:** Design and implement machine learning algorithms that incorporate consistency-based weighting and other tailored solutions to optimize classification performance, reduce inter-subject variability, and increase the robustness of the models.

1.4. Structure

This thesis is organized into six chapters, each detailing a specific aspect of the research on EEG-based emotion recognition.

- **Chapter 1: Introduction.** This chapter provides a comprehensive introduction of the area of research, covering the background and motivation, historical context, the emergence of affective computing and machine learning, the significance of EEG in emotion recognition, and the challenges in the field. It also states the research objectives and provides an overview of the thesis structure.
- **Chapter 2: Literature Review.** This chapter reviews the existing literature on emotional recognition techniques, focusing on EEG-based methods. Furthermore, it discusses feature extraction, machine learning algorithms for emotion classification, and the temporal dynamics of emotion recognition. This ensures a correct exposition of the different concepts and an adequate contextual framework for the development of the thesis.
- **Chapter 3: Methodology.** This chapter describes the research methodology. This includes data collection and preprocessing, feature extraction and selection, the definition of the consistency solutions developed and the machine learning models used. It also outlines the evaluation metrics for assessing models performance.
- **Chapter 4: Experimental Work.** This chapter details the experimental work conducted in this study, covering the experimental setup, the results obtained and the qualitative analysis of affective temporal consistency. The chapter concludes with an in-depth evaluation and discussion of the findings, offering insights into the performance and implications of the proposed method.
- **Chapter 5: Conclusion.** The final chapter summarizes the thesis's key findings and contributions, discusses practical applications, and suggests directions for the future research.
- **Appendix A.** The appendix provides supplementary information to support the main findings and methodology presented in this study. It includes additional data analysis, methodological details, and a more detailed set of results.

1.5. Contributions

This thesis makes several key contributions to the field of EEG-based emotion recognition, with a particular focus on addressing the challenge of temporal affective consistency. The contributions are summarized as follows:

- **Development of Temporal Affective Consistency Scores:** Three novel methods for quantifying temporal affective consistency were introduced: Heuristic-Based, Data-Aware, and Learning-Based. These methods provide new ways to measure the common emotional responses across subjects over time, using EEG data.
- **Consistency-Weighted Approaches for Emotion Recognition:** This work proposed and implemented consistency-weighted schemes, demonstrating that such approaches outperform traditional uniform weighting in emotion recognition tasks. Learning-based consistency scores, in particular, showed superior results, enhancing the overall accuracy of emotion classification models.
- **Improved Generalization Across Subjects:** Consistency-based approaches slightly improved model performance when generalizing to new subjects, opening the door to more adaptable emotion recognition systems.
- **Comprehensive Evaluation Protocols:** A set of testing protocols, including k-Fold Cross Validation, Leave-One-Video-Out (LOVO), and Leave-One-Subject-Out (LOSO), were developed

and applied to rigorously evaluate the consistency-based solutions across different scenarios. This comprehensive evaluation framework provides a robust foundation for future studies.

- **Foundation for Future Research:** The study open new avenues for exploring more sophisticated approaches to emotion recognition based on consistency-schemes, and propose the use of advanced machine learning models like transformers with self-attention mechanisms for enhanced temporal and spatial analysis of EEG data.

2. Literature Review

This chapter provides deeper insights into the concepts mentioned in the introduction. In addition, the theoretical framework necessary for a comprehensive overview and understanding of emotional recognition techniques, EEG signal processing, feature extraction, machine learning algorithms, and the temporal dynamics of emotion recognition will be elaborated upon.

2.1. Overview of Emotion Recognition

The study of emotions has a long history, with various theories proposed to explain their origins and functions. Organizations such as the *American Psychological Association* (APA) defines the emotions as "*a complex reaction pattern, involving experiential, behavioral, and physiological elements, by which an individual attempts to deal with a personally significant matter or event*" [18]. However, scientists disagree on how emotions should be defined and on where to draw the boundaries for what counts as an emotion and what does not [19].

Furthermore, cross-cultural studies of emotions have shown the variety of ways in which emotions differ across cultures. In fact, there are observed variations across cultures in the importance, expression, and management of emotions [20]. The largest piece of evidence of these cultural differences is the language, for instance there is no corresponding translation in Polish for the emotion of 'disgust'.

For these reasons, together with the dynamic character of emotions, it becomes really complicated to find a universal way of classifying emotions. As a consequence, in order to obtain a robust framework for classifying emotions, different approaches for quantifying and discretising emotions have been proposed.

The use of quantification and discretization methods is essential, as it enables the collection of physiological data with clearly defined emotional state labels. This is vital for conducting statistical analysis in psychological research and for developing emotion recognition systems using intelligent computing. All the quantitative methods, which work by means of crowdsourcing strategies to researched users, can be classified into two main groups: discrete types or continuous dimensional.

The discrete type of emotion quantification models assume that there are an innate set of emotions that are cross-culturally recognizable. Paul Ekman *et al.* concluded that the six basic emotions are anger, disgust, fear, happines, sadness and surprise in a cross-cultural study of emotions [21]. A similar approach is proposed by Plutchik [22] in which he presents a way of quantifying emotions similar to a palette of emotions, in which more complex emotions are the result of the combination of a set of primary emotions, as described in the Figure 2.1.

On the other hand, the continuous type of emotion quantification models arise from the hypothesis of some scientists, who suggest that a common and interconnected neurophysiological system is responsible for all affective states and the boundaries between emotional states are vague and continuous. Several continuous models have been proposed, though there are just a few that remain as the most accepted models: the circumplex model, the vector model, and the Positive Activation – Negative Activation (PANA) model [23].

In this work, we will focus on the circumplex model proposed by James Rusell, who describe their modified circumplex model as representative of core affect, or the most elementary feelings that are not necessarily directed toward anything [24]. This characteristic, which makes it a very cross-culturally

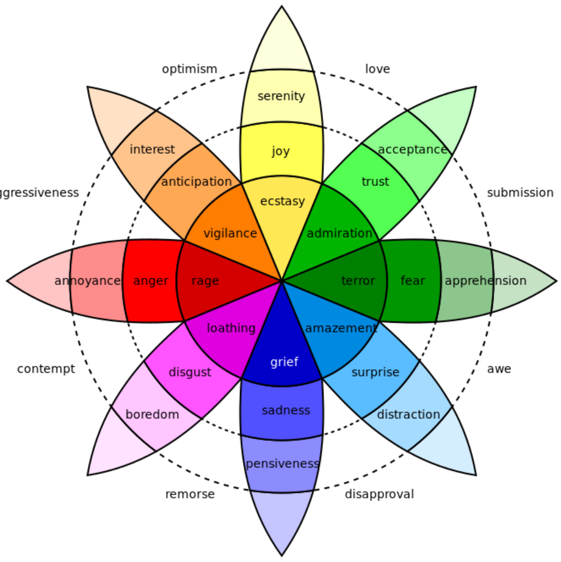

Figure 2.1: *Plutchik's original emotion's palette (source: Wikimedia Commons, Plutchik-wheel.svg).*

generalizable model, together with the fact that it is easily modelled analytically, makes it one of the most widely used models in the field of affective computing.

The circumplex model is one of the so-called dimensional models of emotions, in which the human emotions are represented on a two dimensional valence/arousal plane. Arousal, also known as intensity, refers to the degree of autonomic activation triggered by an event, spanning from calm (low arousal) to excited (high arousal). In contrast, valence represents the degree of pleasantness elicited by an event, ranging from negative (low valence) to positive (high valence).

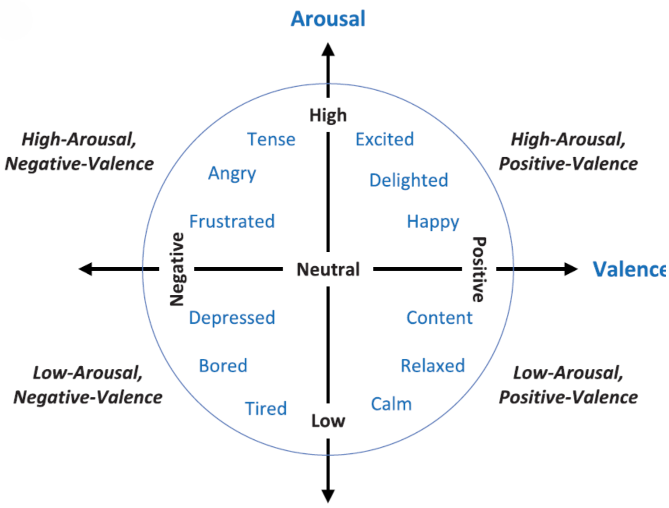

Figure 2.2: *Circumplex Model of emotion quantification. Image obtained from [1].*

Thus, human emotions can be mapped onto the circumplex model. For instance, happiness is characterized by a positive valence, whereas fear exhibits a negative valence. Conversely, anger is associated with high arousal, while sadness corresponds to low arousal. In this study, however, we will concentrate on the classification of binarized versions of valence and arousal, which is the most commonly used framework in affective computing.

Thanks to the application of emotion quantification models and the advent of intelligent computing, multiple emotional recognition techniques have been developed [25]. Some of these include physical signals, which encompass a wide variety such as facial expressions, speech, text, gestures and body postures. However, those sources can be controlled and faked by the subjects complicating the purpose of emotional recognition. For that reason, the most commonly used sources are physiological signals such as Electrocardiogram (ECG or EKG), Galvanic Skin Response (GSR), Eye Tracking (ET), Electrooculogram (EOG) and Electroencephalogram (EEG).

Among these methods, Electroencephalogram stands out due to its direct measurement of brain activity, which controls the nervous system to participate in emotional processes. A more recent alternative is the functional near-infrared spectroscopy (fNIRS), which measures blood oxygenation through light absorption [26]. EEG and fNIRS differ, among others in their different spatial resolution (higher in fNIRS) and temporal resolution (higher in EEG). There are more and bigger datasets with EEG data than there are with fNIRS, and it is a better known technology. This works focuses on EEG.

2.2. EEG-Based Emotion Recognition

Electroencephalography (EEG) is a method of recording electrical activity of the brain using electrodes placed along the scalp. These electrodes detect the electrical patterns resulting from neural activity, which can be analyzed to infer various cognitive and emotional states, such as Post Traumatic Stress Disorder [6] or depression [7].

The reasons why EEGs are so widely used in emotional recognition are manifold. [1]. These advantages includes high temporal resolution, allowing for the detection of rapid changes in brain activity associated with different emotional states. Additionally, EEG is a non-invasive method, making it more comfortable for participants and suitable for long-term monitoring without any specific risk. It captures a wide range of frequencies, providing detailed information about various brain states and activities. Furthermore, numerous publicly available EEG datasets facilitate the development and validation of new algorithms and models for emotion recognition research.

In this context, Brain Computer Interfaces (BCIs) enhance EEG by providing a framework for translating brain activity into actionable commands or new human computer interactions (HCI). BCIs enable direct communication between the brain and external devices without the need for physical movement, which is particularly beneficial for individuals with severe physical disabilities. This technology allows them to interact with their environment and control devices like wheelchairs, prosthetics, and communication systems [27]. However, a current challenge in the BCI field, which is the focus of this work, is the affective annotation of multimedia content, such as images or videos. This involves predicting the emotional state of multiple users who have viewed the content, using crowdsourcing strategies.

When we talk about emotional recognition via EEG, we are really dealing with a pattern recognition problem. This is primarily because the classification algorithm must learn to identify which patterns in the data are associated with different types of emotions (e.g high arousal and high valence). Therefore, EEG-based emotional recognition problem involves several key steps, Figure 2.3. Firstly, it is crucial to define and quantify the recognition target, allowing the problem to be framed as a computable issue. Secondly, obtaining ample and valid research data is essential for creating a comprehensive search space for model decision boundaries. Thirdly, preprocessing the data and extracting features or learning representations are necessary steps in constructing pattern recognition models. Extracting target-related representative characteristics from raw data helps eliminate redundant information that could affect model construction. Lastly, the recognition models are designed, trained, and evaluated iteratively using the processed data until a model with satisfactory recognition accuracy is achieved.

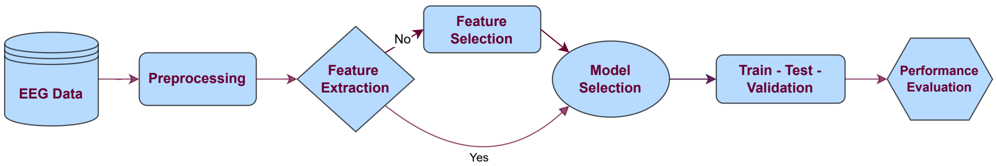

Figure 2.3: *Common Steps of a Pattern Recognition Problem.*

2.3. Feature Extraction Methods for EEG

EEG data is collected via signal acquisition using electrodes, differential amplifiers, filters and pen-type registers. This electrodes are placed following the 10-20 international system [28], in which those numbers refer to the distances between adjacent electrodes should be displayed, those are either 10% or 20% of the total front–back or right–left distance of the skull 2.4. The number of electrodes can vary

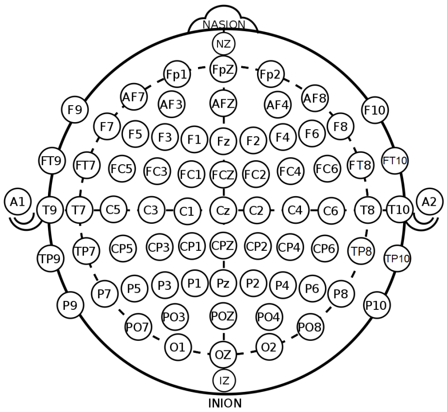

Figure 2.4: *International 10-20 system of electrode placement (source: Wikimedia Commons, International 10-20 system for EEG).*

according to the purpose, for instance the well-recognized DEAP [9] and MAHNOB-HCI [14] datasets use 32 channels of electrodes, unlike the 64 channels of the SEED dataset [29].

As a neurophysiological signal received from multiple channels, raw EEG data are very high dimensional and contain redundant and noisy information, Figure 2.5. Therefore, pre-processing the raw EEG data and extracting relevant features is essential to develop machine learning models.

The first step in the pre-processing of EEG data is to remove the noise generated by our own body. This involves eliminating electrooculogram (EOG) artifacts caused by eye blinks (with frequencies less than 4Hz), ECG artifacts from heart activity (around 1.2Hz), EMG artifacts from muscle activity (above 30Hz), and environmental power frequency artifacts (between 50 to 60Hz), among others. This is done by artifacts rejection methods, such as Independent Component Analysis (ICA) [30]. Additionally, signal filtering is performed to isolate the frequency bands of interest, such as delta Delta (<4 Hz), Theta (4–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (>31 Hz), which are critical for various cognitive and emotional analyses. Finally, Normalization techniques are also applied to ensure consistency across different recording sessions and subjects. These preprocessing steps are essential to enhance the quality

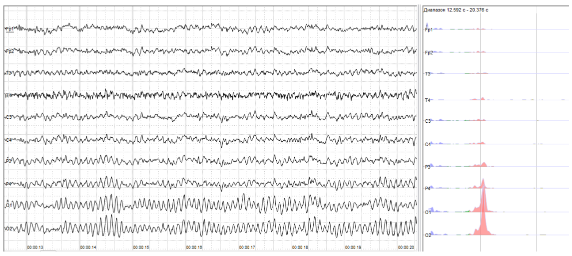

Figure 2.5: *Raw EEG data visualization of a sample of channels. Sample of human EEG with prominent resting state activity - alpha-rhythm. Left - EEG traces (horizontal - time in seconds; vertical - amplitudes, scale 100uV). Free-copyright image obtained from:* `https://en.wikipedia.org/wiki/Electroencephalography`.

and reliability of the data, making it suitable for accurate feature extraction and subsequent pattern recognition tasks.

After signal preprocessing, the feature extraction it is necessary to extract features. In this context, the feature extraction is essential due to EEG are non-stationary, non-linear, non-Gaussian, and non-short form, which complicates the signal analysis [11]. The most-common features extraction techniques used by the scientific community can be divided in two groups:

- One-dimensional feature extraction techniques:
  - Time Domain.
  - Frequency/Spectral domain.
  - Decomposition Domain.
- Multi-Dimensional feature extraction techniques:
  - Joint time-frequency domain.
  - Spatial Domain.

Time-domain feature extraction analyzes signals with respect to the time, quantifying how they change over time. This can be done by calculating simple statistics such as mean, variance or higher-order crossing (HOC), so that the data can be divided into discrete time segments or windows and the temporal properties can be analyzed.

The frequency-domain feature extraction converts time-domain signals into the frequency domain to analyze the sinusoids that make up the data. As discussed above, this can help identify specific frequency bands like alpha, delta, theta, and beta in EEG signals. Techniques include Fourier transform and power spectral density.

Decomposition methods decompose signals into its components for simultaneous filtering and feature extraction. These methods are also useful for filtering because some components describe the low-frequency parts of the signals and others the high-frequency parts and can be extracted at the desired granularity. Techniques like wavelet transform, empirical mode decomposition, and adaptive Hermite decomposition are used to retain signal characteristics while reducing noise [31].

In the group of multi-dimensional feature extraction techniques we find the joint time-frequency domain methods. These techniques analyze signals in both time and frequency domains simultaneously, capturing more detailed information. This methods include the short-time Fourier transform, the S-transform and the matching pursuit. And finally, the multi-dimensional spatial domain techniques utilize the spatial distribution of the signal and are often used in multi-channel EEG data. Common Spatial Patterns (CSP) is the most popular feature extraction technique of this type. CSP converts the brain signals into a unique space using a supervised spatial filter, which maximizes the variance for one group (e.g., one mental state or task) while minimizing it for another, enhancing the discriminability of different brain states or activities.

In summary, feature extraction is a crucial step in EEG signal analysis, involving various techniques across time, frequency, decomposition, and spatial domains to capture relevant characteristics of brain activity. These features form the foundation for further analysis and classification tasks. With a robust set of features extracted, the next step is to apply machine learning algorithms to classify emotions based on the processed EEG data. In the following section, we will explore various machine learning algorithms and their application in emotion classification, highlighting their strengths and suitability for different types of EEG data.

2.4. Machine Learning Algorithms for Emotion Classification

With the extracted EEG features providing a rich dataset representing various emotional states, the next critical step is to employ machine learning algorithms to classify these emotions accurately. Machine learning techniques are essential for uncovering patterns within the data and making predictions about emotional states based on the complex and multifaceted EEG signals. This section explores various machine learning algorithms and their application in emotion classification, highlighting their strengths and suitability for different types of EEG data.

Machine learning algorithms for EEG-based emotion recognition encompass a wide range of approaches, each leveraging different aspects of the data to achieve optimal performance. Traditional machine learning methods, such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forests, have been extensively used due to their ability to handle the high-dimensionality and complexity of EEG signals. These models often rely on handcrafted features extracted from the EEG data, which capture essential characteristics related to emotional states.

Support Vector Machines (SVM) are particularly popular due to their robustness in high-dimensional spaces and ability to find the optimal hyperplane for separating different classes of emotions [32]. K-Nearest Neighbors (K-NN) is another widely used algorithm [33] that classifies emotions based on the majority vote of the nearest data points in the feature space. Random Forests [34], which consist of an ensemble of decision trees, are also used to construct recognition models based on statistical features, nonlinear features, spectral features, etc.

In recent years, deep learning approaches have gained prominence in EEG-based emotion recognition due to their powerful feature learning capabilities. Convolutional Neural Networks (CNNs) [35] are effective in capturing spatial dependencies and patterns within EEG signals, especially when the data is represented in a two-dimensional format. Different methods have been proposed to ensure a proper representation of EEG data to fit the input format of the CNN model. From single-channel convolutional networks, thanks to recursive quantization analysis (RQA) [36], to organizing the characteristics using 2d sparse graphs [37] to maintain the topology information of the electrodes, and then using them to train a CNN. Furthermore, CNN are applied within a bi-hemisphere discrepancy CNN model (BiDCNN), which is essentially a CNN model which recognizes taking into account the asymmetry information between the two hemispheres of the brain [38].

Recurrent Neural Networks (RNNs), within a bi-hemispheric brain discrepancy model [39], and their variants, such as Long Short-Term Memory (LSTM) networks [40], are designed to capture temporal dependencies in sequential data. These models are particularly effective for emotion recognition, as they can learn from the temporal dynamics of EEG signals. By processing the data in a sequential manner, RNNs and LSTMs can identify patterns related to changes in emotional states over time.

Hybrid models that combine different types of neural networks are also explored to leverage both spatial and temporal information from EEG data. For instance, combining CNNs with LSTMs [41] can provide a comprehensive understanding of the inter-channel relationships and capturing contextual information from sequential data.

Another promising approach involves the use of Graph Convolutional Networks (GCNs) to model the functional connectivity of different brain regions. GCNs can capture the complex relationships between EEG channels, providing a more holistic representation of the brain's emotional state. Therefore, models combining GNN and LSTM have been proposed, in order to capture this interconnectivity throughout the emotional process [42].

In conclusion, the application of machine learning algorithms in EEG-based emotion recognition offers a diverse set of tools for extracting meaningful patterns from complex brain signals. By leveraging both traditional and deep learning approaches, researchers can develop robust models that accurately classify emotional states, paving the way for advanced applications in affective computing, mental health monitoring, and human-computer interaction.

2.5. Affective Consistency

In the context of signals, consistency is understood as the reliability and stability of the information transmitted by these signals over time. In particular, temporal consistency refers to the steadiness of the signal's content across different time segments. In the realm of EEG-based emotion recognition **temporal affective consistency** refers to the cross-subject stability of affective responses within specific time windows during the presentation of the stimuli. Affective consistency is not widely extended, because most research focuses on emotion recognition. However, analyzing affective consistency and developing tools to measure it can be of great help in emotional prediction and in understanding EEG signals.

In the context of affective consistency and Brain-Computer Interfaces (BCI), the affective annotation, which is in essence the process of labeling content based on the emotions it evokes, of multimedia content, such as images or videos is a current challenge. The use of crowdsourcing strategies is widely use to obtain reliable affective annotations by leveraging the collective input of multiple users [9]. Nevertheless, relying on manual annotation is not practical for scaling to a wide range of applications. For example, it is improbable that users would manually tag their emotional responses for every video clip they watch, every song they listen to, or every image they view online.

Some approaches have been explored in this direction, Ruotsalo, Mäkelä, and Spapé [43] demonstrated a novel method using fNIRS-BCI to crowdsource affective annotations. This approach involves recording brain responses from multiple participants to predict the affective dimensions that characterize the stimuli (images). By aggregating the brain signals from a group, they showed that it is possible to achieve higher accuracy in affective annotations.

However, emotions and stimuli are not static and evolve over time. It is therefore necessary to develop methods capable of determining the affective consistency of EEG signals. In the context of affective annotation of videos, this means identifying which parts of a video are most closely associated with its overall emotional label and understanding the degree of consensus for each part. Analyzing affective consistency and emotion recognition is crucial for ensuring that computational systems can accurately infer the affective states evoked by digital information. This allows for the dynamic association of content with its affective label, thereby personalizing and enhancing the user experience.

3. Methodology

This chapter describes the research methodology, including data collection, feature extraction and selections, the affective consistency analysis, the different testing methods used to deal with inter-subject variability and the Machine Learning algorithms adapted to consistency analysis.

3.1. Data Collection and Preprocessing

This section outlines the process and techniques used for gathering and preparing the EEG data for analysis.

3.1.1. EEG Datasets for Affective Temporal Consistency Analysis

To analyze affective temporal consistency, we require databases containing EEG data collected from subjects exposed to various stimuli over a period of time. Therefore, the databases considered will consist of EEG data gathered via Brain-Computer Interface (BCI) systems, along with labels representing the emotional responses elicited by specific videos in the corresponding subjects. These datasets are essential for evaluating the temporal consistency of affective responses and ensuring that the EEG data accurately reflects the subjects' emotional states throughout the stimulus presentation.

When referring to an EEG dataset of emotion recognition, we denote as $\mathcal{D}$ the set formed by the videos shown to the subjects in the data collection and $\mathcal{O}$ the set of subjects who have performed the experiment.

3.1.2. Signal Preprocessing

Data pre-processing and feature extraction were carried out according to the methodology outlined in [44]. The final set of features used in this study was provided by the BANANA project [15]. Nevertheless, this is subjected to further examination in order to gain a comprehensive understanding of the processing and extraction of EEG data.

The first **pre-processing techniques** were applied to ensure the compatibility of the databases with the different experiments carried out. Firstly, the trial lengths are standardized ensuring each sample of the different datasets are 1-minute EEG signals.

For the EEG preprocessing, several steps were implemented in [44]. Initially, all EEG signals were downsampled to 128 Hz to maintain consistency across datasets. The Artifact Subspace Reconstruction (ASR) method was then employed to remove transient and large-amplitude artifacts. A band-pass filter with a range of 4–45 Hz was applied to the signals to focus on the relevant frequency bands. Additionally, Common Average Referencing (CAR) was performed on the signals for each trial to enhance signal quality. Datasets were then standardized to have identical channel numbers and names, following the 10-20 standardized system for EEG.

3.1.3. Feature Extraction and Selection

The features have been extracted using different sampling windows of ω seconds. The different window sizes used are $\{1, 2, 4, 6, 10, 12, 15\}$ seconds. Thus, depending on the chosen window, ω seconds, we will have $\frac{60}{\omega}$ vectors of features, each corresponding to a temporal segment of the video.

Therefore, in each EEG channel and for each chosen time window $\omega \in \{1, 2, 4, 6, 10, 12, 15\}$, the Fast

Fourier transform (FFT) is applied and the Hjorth parameters (1), (2), (3), spectral entropy (4), and signal energy (6) are computed. The first step of the Fourier transform is essential, as it allows us to extract the different EEG bands, in order to calculate the characteristics for each band: Delta (<4 Hz), Theta (4–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (>31 Hz). Finally, the signal energy is computed in the frequency domain, the other features are computed after inverting the filtered Fourier transform.

Then, the following characteristics were used for the analysis. For more detailed information about feature extraction trends for EEG signals, see [11].

Definition 3.1 (Hjorth Parameters)**.** The Hjorth parameters are a set of statistical measurements, which belongs to the one-dimensional, time-domain features, used to analyze the characteristics of time series data. They consist of three parameters: Activity, Mobility and Complexity.

(D_1) The Hjorth Activity measures the signal's variance, indicating its power or energy. For a signal $x(t)$, the Activity is defined as follows:

$$\text{Activity} = Var(x(t)) = \frac{1}{N-1}\sum_{i=1}^{N}|x_i - \mu_x|^2, \tag{1}$$

where, x_i is the i-th sample of $x(t)$ expressed as a discrete time series with N values and μ_x is the mean of the signal values.

(D_2) Hjorth Mobility characterizes the frequency-dependent changes in the signal's amplitude over time. Specifically, Hjorth Mobility, is quantified using the second statistical moment.

$$Mobility = \sqrt{\frac{Var\left(\frac{d}{dt}x(t)\right)}{Var\left(x(t)\right)}}. \tag{2}$$

(D_3) The Hjorth Complexity measures the irregularity or unpredictability of the EEG signal over time. Specifically, complexity is quantified using the third statistical moment, skewness.

$$Complexity = \frac{Mobility\left(\frac{d}{dt}x(t)\right)}{Mobility\left(x(t)\right)}. \tag{3}$$

In order to define the next feature, the Spectral Entropy, it's necessary to define some concepts. The power Spectrum, $S(m)$, is defined as:

$$S(m) = |X(m)|^2, \tag{4}$$

where, $X(m)$, is the discrete Fourier transform of the signal, $X(t)$. Then:

Definition 3.2 (Spectral Entropy)**.** The Spectral entropy, which belongs to the one-dimensional, frequency domain features, quantifies the complexity or randomness of a signal's frequency spectrum. It can also be interpreted as an indicator of the distribution of spectral power. So, let $P(m) = {}^{S(m)}/_{\sum_i S(i)}$, the probability distribution of the FFT-discretized signal, $X(m)$. The spectral entropy is defined as follows:

$$H = -\sum_{m=1}^{N} P(m)\log_2 P(m), \tag{5}$$

where N is the number of values of the discretized signal.

The last feature that is used is the energy of the signal.

Definition 3.3 (Energy of Signal)**.** The Energy of a signal, $x(t)$, is defined as the sum of the square of the signal magnitude, pertaining to the one-dimensional, time-domain features. Essentially, the energy of the signal represents the total power or strength of a signal over a given period.

$$E = \int_0^N |x(t)|^2 dt. \tag{6}$$

Therefore, if we consider a subject $o \in \mathcal{O}$ and a video from the dataset $v \in \mathcal{D}$, we denote as $f^{o,v} \in \mathcal{M}_{M\times T_w}(\mathbb{R})$ the feature matrix associated with subject o and video v, where T_ω is the number of time segments extracted with the time window of size ω seconds and M is the number of features.

3.2. Affective Consistency

To analyze the temporal affective consistency, and therefore the common emotional response of subjects through EEG signals, three approaches were proposed: heuristic, data-aware, learning-based. These approaches have been developed following two objectives: to analyze temporal affective consistency, i.e. to capture common emotional response across subjects along time, and to study the potential impact of this temporal analysis on EEG emotion recognition algorithms.

In essence, the heuristic, data-aware and learning-based consistency approaches, are all about vectors, which are responsible for weighting some EEG time segments features more than others, based on temporal affective consistency. This allows us to use these weighted features to try to improve emotion recognition algorithms. So, It could be posited that weighted features with higher recognition success may indicate a greater capture of the emotional response across subjects by consistency functions. Consequently, weighting-consistency functions could represent a reliable score of temporal affective consistency over time.

Furthermore, in the case of data-aware and learning-based solutions, due to their definition, they have analytical value in their own right, independently of possible improvements in emotion recognition algorithms. Therefore, they will allow us to manually analyze the existence of a common emotional response across subjects.

We now describe each of these three approaches in detail.

3.2.1. Heuristic Consistency

The Heuristic solution is a predefined solution based on the assumption that different parts of the videos, shown to subjects, may have a significantly greater and common emotional response across subjects than other parts of the videos. Therefore, four different solutions are defined: first-consistency, last-consistency, centered-consistency and uniform-consistency.

We constructed the weighting functions—last, first, and centered—using multiple probability density functions of Gaussian distribution. This approach was chosen because segmenting a single Gaussian distribution into different time intervals would result in excessive variation in the magnitude of the weighting functions, and therefore parts of the same temporal moment of a video would be weighted with quite different values depending on the window size selected. To address this issue, we adopted a **time-based variance** approach. This method adjusts the variance in accordance with the time window to ensure that each interval retains consistent relevance relative to its position.

We use the time-based variance approach using the Gaussian function of the Python package scipy.signal.windows. This function first creates an array of length n equally spaced:

$$\begin{cases} [\frac{-(n-1)}{2}, \frac{-(n-2)}{2}, \ldots, 0, \ldots, \frac{(n-2)}{2}, \frac{-(n-1)}{2}], & \text{if } n \text{ is even,} \\ [\frac{-(n-1)}{2}, \frac{-(n-2)}{2}, \ldots, \frac{-1}{2}, \frac{1}{2}, \ldots, \frac{(n-2)}{2}, \frac{-(n-1)}{2}], & \text{if } n \text{ is odd.} \end{cases} \tag{7}$$

Then, each element is mapped by the Gaussian function defined as: $f(x) = e^{-\frac{x^2}{2\sigma^2}}$. So, if we consider a time window, $\omega \in \{1, 2, 4, 6, 10, 12, 15\}$ seconds, and we consider the time-based variance based:

$$\sigma_\omega = \frac{\left(\frac{60}{\omega} - 1\right)}{4}, \tag{8}$$

where, $^{60}/_{\omega}$ represents the length of the feature array. This adjusted variance ensures that, irrespective of the selected time window $\omega \in \{1, 2, 4, 6, 10, 12, 15\}$ seconds, the ends of the vector maintain consistent values and each time segment is weighted similarly in magnitude.

First-Consistency The first-consistency heuristic operates under the assumption that the initial parts of the stimuli (videos), trigger more pronounced emotional responses. To compute the first-consistency weights, we utilize the methodology described earlier (time-based variance), but we generate an array with twice the required segments. Subsequently, we select only the latter half of this array to form the weight vectors. This adjustment allows the weighting arrays to be finely tuned to each time window, thereby ensuring a consistent impact across varying durations, see Figure 3.1.

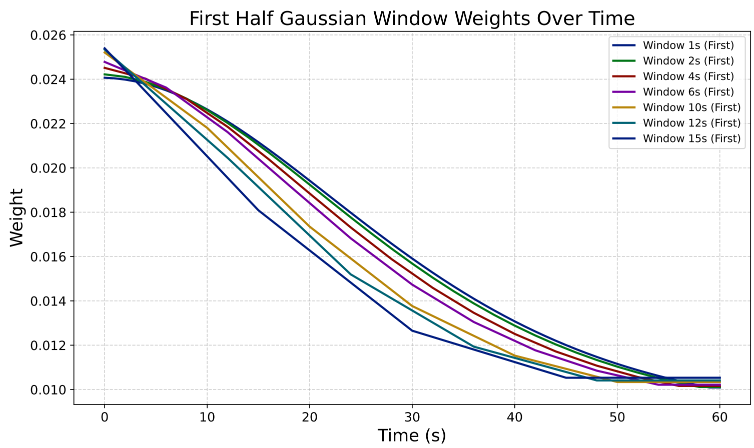

Figure 3.1: *First-Consistency vectors values for each window size, following the time-based variance approach.*

Centered-Consistency The centered-consistency heuristics operates under the assumption that the central part of the stimuli (videos), trigger more pronounced emotional responses possibly because the more affective-relevant contents roughly happen in the middle of the video. To compute the centered-consistency weights, the previous methodology is followed, see figure 3.2.

Last Consistency The last-consistency heuristic is based on the idea that the concluding segments of the stimuli (videos) hold significant impact, possibly due to the resolution or culmination of presented content. We apply the initial methodology to compute the weights but adjust the array to emphasize the final segments. This is done by reversing the array used in the first-consistency calculation, thereby aligning the higher weights with the end of the time window. Such an arrangement ensures that the end

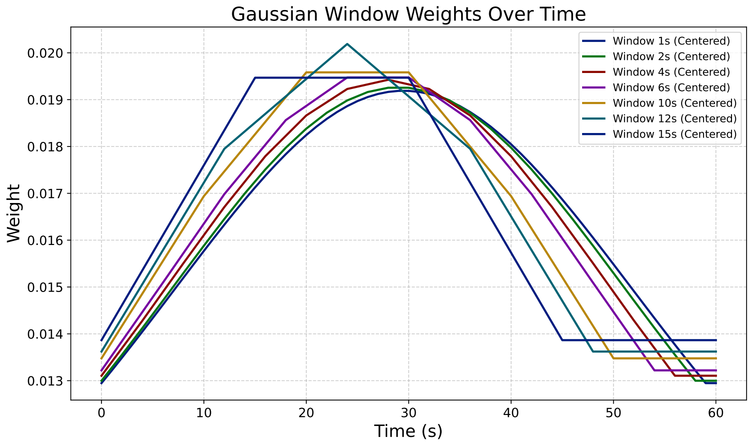

Figure 3.2: *Centered-Consistency vectors for each window size, following the time-based variance approach.*

segments are given greater importance, matching their potential impact on the viewer, Figure 3.3.

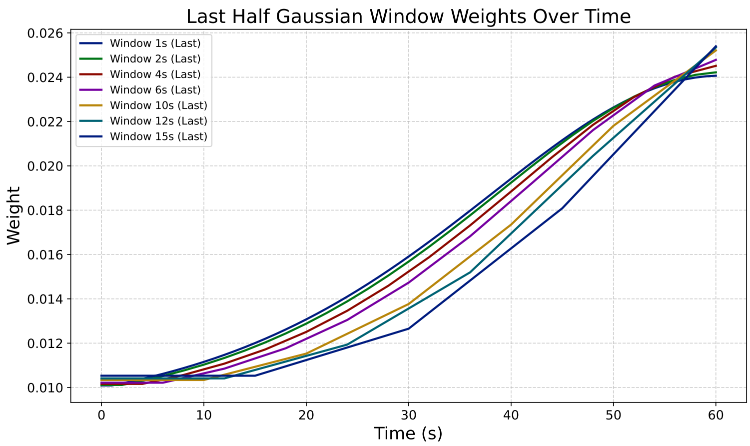

Figure 3.3: *Last-Consistency vectors for each window size, following the time-based variance approach.*

Uniform Consistency The uniform-consistency approach does not prioritize any particular segment of the stimuli. Instead, it assumes equal importance across the entire duration of the dynamic stimuli. To implement this, we use a constant value for all elements in the array, effectively distributing equal weights across the entire time window. This method is widely used in the scientific literature, reflecting its utility and acceptance for ensuring an unbiased analysis of the stimuli's overall effect, particularly when no specific part of the video is expected to dominate in terms of eliciting emotional responses [11]. As commented in the introduction, this is implicitly done by the scientific community without taking into account that some temporal segments may elicit more brain response than others, and it is for this reason that we seek to analyze the influence of other weighting schemes on emotion recognition, leading to better prediction results.

3.2.2. Data-Aware Consistency

Unlike the heuristics above, which use temporal weighting functions, we now explore a weighting function that directly depends on the data (EEG features). The data-aware solution represents an approach based on the assumption that affective time consistency can be measured directly through the data. Thus, developing a function that allows us to quantify the amount of common emotional response, across participants, in each time segment is essential. To this end, an adaptation of GLIMPSE, a measure of consistency of visual saliency, is developed. In our adaptation for affective consistency, instead of gaze points, the extracted EEG features are used, allowing us to quantify common emotional responses.

Traver, Zorío, and Leiva [45] developed GLIMPSE, gaze's spatio-temporal consistency from multiple observers, with the goal of quantifying temporal visual salience. This revolves around measuring how effectively an object or scene segment captures visual attention. The authors created a method to automatically assess temporal salience using eye-tracking data. Their core hypothesis suggested that when gaze coordinates show spatio-temporal consistency among multiple observers, it strongly indicates visual attention is concentrated at specific locations within a frame (spatial consistency) and during particular time intervals (temporal consistency). This hypothesis can be adapted to our context, where consistency in EEG features among subjects should indicate a shared emotional response during specific time segments. Before defining our adaptation of GLIMPSE, it is first necessary to formally define GLIMPSE in order to gain a comprehensive understanding of its function.

GLIMPSE works by calculating a salience score, $s(t)$, for each time segment, where higher scores indicate higher-spatio temporal and inter-observer consistency, reflecting higher temporal salience. The score is calculated using the formula:

$$s(t) = \frac{2}{n(n-1)} \sum_{\substack{i,j \in \{1,\dots,n\} \\ i \neq j}} \mathbb{1}\left[d_{ij} < \theta_s\right], \quad t \in \{1, .., T\}, \tag{GLIMPSE}$$

where d_{ij} is the pairwise Euclidean distance between the i-th and j-th points in the set of n gaze points from all observers within a temporal window centered at t. The hyperparameter θ_s denotes the spatial scale, which is in essence a distance threshold. So, a higher value of $s(t)$ represents greater spatio-temporal and inter-observer consistency, reflecting increased temporal salience 3.4.

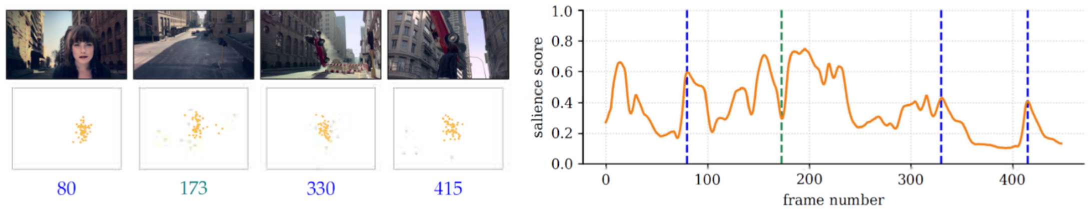

Figure 3.4: *Example of a temporal salience score $s(t)$ calculated from a video belonging to the SAVAM dataset. The image is taken from Traver, Zorío, and Leiva [45].*

To adapt the GLIMPSE function to our context, some changes have to be made. Firstly, the GLIMPSE function has to be adapted in order to consider high-dimensional vectors (feature EEG data), instead of two-dimensional vectors (gaze points). In addition, EEGs can easily vary in nature and scale depending on the context, so the fixed spatial threshold of GLIMPSE has to be revised. Finally, the original GLIMPSE uses the Euclidean distance, which is the most suitable for its context. However, when dealing with high-dimensional vectors, such as EEG features, other distance metrics might be more suitable due to the curse of dimensionality and the sensitivity of Euclidean distance to high-dimensional spaces [46].

Therefore we will define our consistency score as follows:

Definition 3.4 (EEG-GLIMPSE)**.** The EEG-GLIMPSE score, measure of affective-temporal EEG consistency among subjects, is defined for each video $v \in \mathcal{D}$ in the dataset and for each temporal segment of the video $t \in \{1, ..., T_\omega\}$ calculated over the time window, ω:

$$a^v(t) = \frac{2}{n(n-1)} \sum_{\substack{i,j \in \{1,\ldots,n\} \\ i \neq j}} \mathbb{1}\left[d_{ij}^v < \theta^v\right], \quad t \in \{1, \ldots, T_\omega\}, \tag{EEG-GLIMPSE}$$

where, again, $\mathbb{1}$ is the indicator function and θ^v is a video-dependent threshold to be determined. Moreover, d_{ij}^v represents the pairwise distance, of a metric to be selected, between the i-th and j-th points in the set, $\mathcal{F}_t$, of n-dimensional EEG feature vectors from all the observers, in a temporal window of length ω and started in $\omega \cdot t$, i.e,

$$\mathcal{F}_t = \left\{ f^{o,v}(t) \in \mathbb{R}^M \ : \ o \in \mathcal{O}, \ v \in \mathcal{D}, \right\}, \quad t \in \{1, \ldots, T_\omega\}. \tag{9}$$

Where $f^{o,v}(t)$ is the feature vector in the time segment t, corresponding to the video, v, of the dataset $\mathcal{D}$ and to the subject $o \in \mathcal{O}$.

As the original (GLIMPSE), the idea of EEG-GLIMPSE is to construct a function, $s(t) \in [0, 1]$, such that the larger its value, the greater the affective-temporal consistency. This is based on the premise that similar inter-subject feature vector values within a time segment indicate a consistent emotional response.

While alternatives methods like clustering could be used, this approach offers a computationally efficient measure of natural aggregation, reducing the need for more resource-intensive processing.

Additionally, the definition introduces two indeterminate but crucial elements: the video-dependent threshold and the distance metric. Unlike the original GLIMPSE application, where gaze points are measured within a uniform spatial scale due to their fixed focus on the screen, our adaptation involves EEG features, which vary in scale and nature. Since subjects may have different baseline emotional states depending on the video or the time of the experiment, an optimal threshold must be calculated for each video. Another important element is the distance metric. While the Euclidean metric could be used, the high dimensionality of the EEG data, resulting from different EEG bands and channels, suggests that other metrics might be more appropriate. These alternative metrics could better capture the complex relationships within the data, potentially leading to more accurate measures of emotional consistency.

To determine the optimal threshold, we propose a method based on finding a balance between two key considerations: firstly, the maximisation of the capture of common emotional responses; and secondly, the minimisation of the measurement of signals with different emotional responses that are perceived as similar. This method ensures that the threshold accurately reflects true affective consistency by optimizing the trade-off between these two factors. For this purpose, a set of n thresholds is considered, $\{\theta_1^v, \ldots, \theta_n^v\}$, for each video $v \in \mathcal{D}$. Subsequently, EEG-GLIMPSE is calculated over the thresholds and the area under the curve of EEG-GLIMPSE a^v, (AUC) is calculated over time t:

$$AUC(a^v)(\theta_j^v) = \int_1^{T_\omega} a^v(t)dt, \quad j \in \{1, \ldots, n\}. \tag{10}$$

Then considering $AUC(a^v)$ as a function being $\{\theta_1^v, \ldots, \theta_n^v\}$ the domain. The inflexion point of the curve formed by the $AUC(a^v)$ for different θs is considered the optimal threshold for the given video.

This approach might initially seem confusing or meaningless. However, the AUC of EEG-GLIMPSE allows to quantify the amount of temporal consistency measured by EEG-GLIMPSE. By considering the optimal threshold as the inflection point of the curve formed by $AUC(a^v)$ over the range of θ^v values, we effectively identify the threshold that balances two extremes: being too strict and only recognizing extremely similar emotional responses (θ^v small) or being too lenient and capturing any signal as common among subjects (θ^v large). This balance ensures that we maximize the capture of genuine common emotional responses while minimizing the inclusion of signals with differing emotional responses as consistent.

Observing the figure 3.5, we can see that the described behaviour is indeed fulfilled. In the plot it can be observed how EEG-GLIMPSE measurement for the most optimal threshold has a larger dynamic range and captures more temporal detail than other thresholds, which provide much more static measurements.

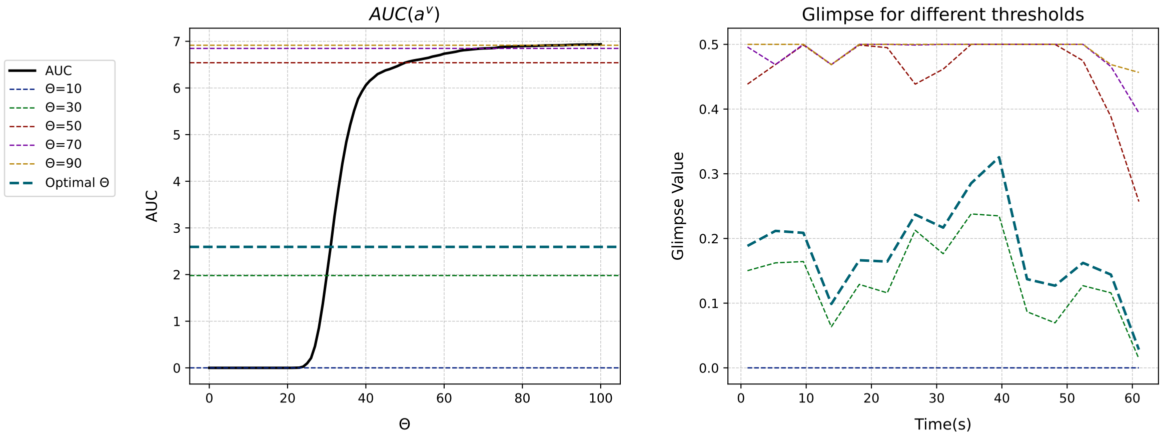

Figure 3.5: *Proposed method for selecting the optimal threshold for EEG-GLIMPSE, which balances maximizing common emotional responses while minimizing the capture of inconsistent signals. In the left subplot, the AUC of EEG-GLIMPSE for different thresholds is shown, with the optimal threshold marked as the point of maximum derivative change. The right subplot visualizes the glimpse values for the optimal threshold and several other threshold values.*

The distance metric employed is another hyperparameter really important to capture temporal consistency among subjects. The high dimensionality of the EEG data, resulting from EEG bands and channels, make Euclidean distance not very suitable for this purpose. Therefore, another metrics are considered. The appropriateness of one metric or the other has been determined by logical rather than experimental reasoning, because of the difficulties in assessing whether EEG-GLIMPSE is actually calculating a measure of time consistency. However, the values of EEG-GLIMPSE for different metrics have also been visualized to verify the hypothesis underlying our choice.

Manhattan and Euclidean distance could have the same problems due to the high dimensionality and have therefore been discarded. Cosine similarity measures directional alignment, which is difficult to interpret as a measure of the consistency where the magnitudes may also be important. Chebishev distance measures the maximum difference along any coordinate dimension, this metric is also discarded because it can be too strict, potentially overemphasizing the largest deviation in any single feature while ignoring the overall pattern of the remaining features.

Finally, Mahalanobis distance consists of calculating the distance between two vectors while taking into account the correlations of the dataset. It is defined as:

$$d(x, y) = \sqrt{(x-y)^T S^{-1}(x-y)}, \tag{11}$$

where S is the covariance matrix of the data and x and y feature vectors. Then, Mahalanobis is selected because it considers the correlations between EEG features together with the distances, providing a more accurate measure of the true structure and variability within the data. This ensures that the calculated temporal consistency accurately reflects common emotional responses across participants, capturing the complex relationships inherent in high-dimensional EEG data.

Moreover, if EEG-GLIMPSE is visualized for different metric distances and its corresponding optimal thresholds, the next results are obtained:

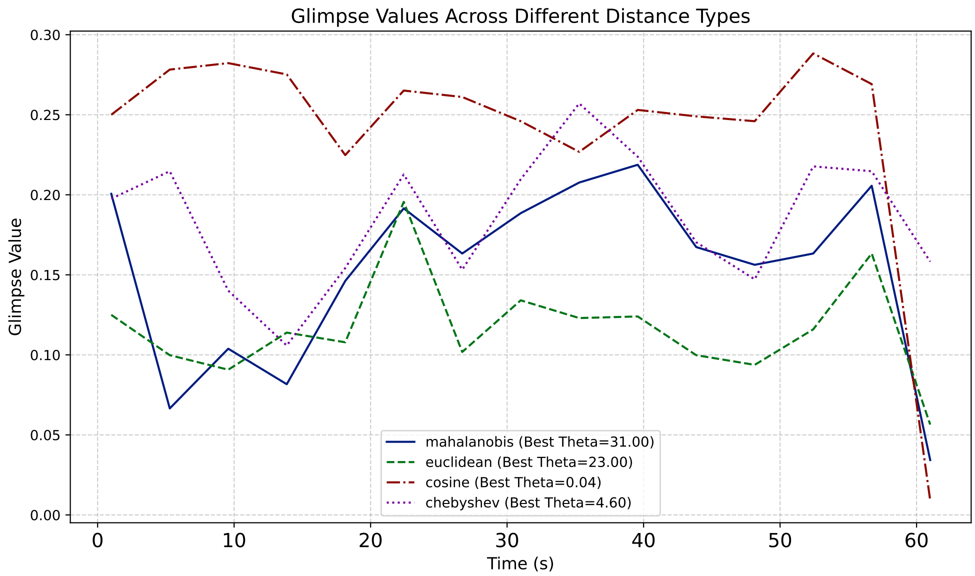

Figure 3.6: *The figure illustrates the comparison of EEG-GLIMPSE calculated using different distance metrics, including Mahalanobis, Cosine-Similarity, Chebyshev and Euclidean.*

Therefore, we can observe that EEG-GLIMPSE, when calculated using the Mahalanobis distance, Figure 3.6, exhibits more variability values over time, having a wider dynamic range and capturing more temporal details than the other distances. However, Chebyshev distance gives a similar result, but this is discarded because by its definition, it only takes into account the feature with the largest differences which can be misleading, while Mahalanobis takes into account correlations of the entire dataset.

3.2.3. Learning-Based Consistency

The data-aware method represented an improvement over the data-agnostic fixed weighting function since data is taken into account. However, the relevance of the data for a given task (such as emotion recognition) is not considered. To address this, we propose a learning-based approach which, in addition of being data-aware, it is task-aware. The learning-based solution operates on the premise that a neural network trained for emotion recognition can inherently capture affective consistency across subjects. In this approach, one of the layers in the neural network is designed to weight each time segment of data, with weights that are dependent on the context, in this case the video to which the data corresponds. During training, the weights associated with this layer will adapt through backpropagation to ensure correct classification, while simultaneously capturing the measure of temporal affective consistency at each time point. This process effectively allows the network to learn the weights allowing to identify the temporal patterns of emotional responses across different subjects.

This method uses a dense neural network that has two main components: the contextual weighting part and the dense layered part. The contextual weighting part involves multiplying the feature matrix, which is associated with a video $v \in \mathcal{D}$ and a subject $o \in \mathcal{O}$, by a video-dependent weighting vector.

Through backpropagation, this multiplication allows the model to assign weights to different temporal segments based on their influence on emotion recognition. Consequently, the learned weights can serve as an indicator of how consistently emotions are expressed over time in the video.

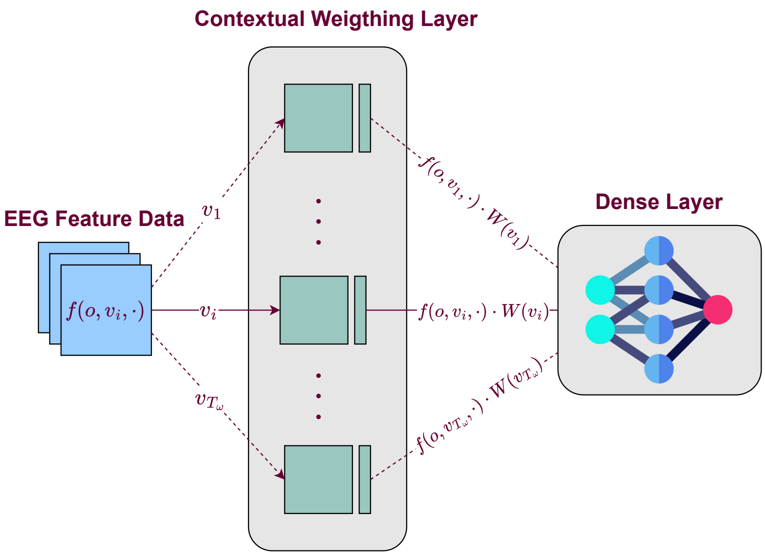

Figure 3.7: *Forward-pass of the contextual neural network used to capture the temporal affective consistency across subjects, highlighting the contextual weighting.*

Formally, given a video $v \in \mathcal{D}$, and a ω-second time window, over which the features have been extracted, the contextual weighting part is defined as follows:

$$f^{o,v} \cdot W^v = \sum_{t=1}^{T_\omega} f^{o,v}(t) \cdot W^v(t), \quad o \in \mathcal{O} \quad \text{and} \quad v \in \mathcal{D}, \tag{12}$$

where $f^{o,v} \in \mathcal{M}_{M \times T_w}(\mathbb{R})$ is the feature matrix correspoding to the subject, $o \in \mathcal{O}$, and the video v of the dataset $\mathcal{D}$ and the $W^v \in \mathbb{R}^{T_\omega}$ is the weighting vector, which depends on the video and is in charge of weighting the different time segments of the feature matrix.

Using the PyTorch framework, the contextual part can be integrated as a layer within the neural network. This integration allows the weights, W^v, to be selectively adjusted through backpropagation. Consequently, this adjustment enables the network to capture temporal affective consistency effectively.

Although this definition might seem confusing when visualizing the adjustment of the weights due to its non-traditional neural network structure, a closer examination of the contextual layer reveals that it functions essentially as a dense layer with V neurons and no activation function, where only one of the neurons is activated through the forward-pass, Figure 3.8. The unique aspect of this layer is that the weights are dependent on the specific video to which the data belongs and the input layer receives feature vectors, which are contextually weighted by the weights associated with the neuron corresponding to the video.

Therefore, the Learning-Based Consistency approach aims to learn a measure of the affective consistency by creating a context-adaptive model, which determines the importance it assigns to each time segment via backpropagation and thus maximizing emotional recognition. The detailed structure of the EEG-Contextual Neural Network is explained in the following section.

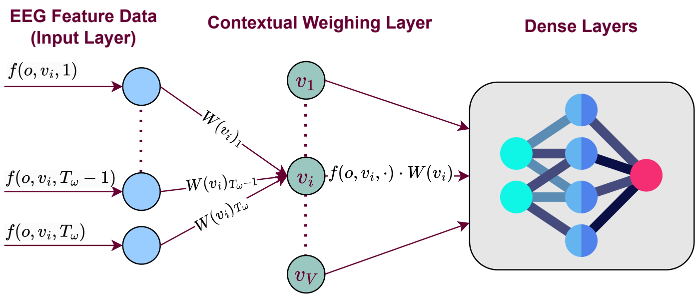

Figure 3.8: *Alternative representation of the forward-pass of the contextual neural network, Figure 3.7. This shows that the contextual layer consists of V neurons without activation function, where only the neuron corresponding to the video associated with the data is activated and the input layer receives feature vectors.*

3.3. Experimental Protocols

In this section, we present the protocols used to evaluate our emotion recognition systems, which are based on our affective consistency solutions. These protocols include traditional approaches such as K-fold cross-validation, adaptations of the leave-one-out method to address the cross-subject recognition challenge, and a subject-dependent method to evaluate recognition for each subject independently. These diverse evaluation strategies ensure a comprehensive assessment of our models' performance across different scenarios and subject variations.

3.3.1. Leave-one-Subject-out (LOSO)

The Leave-One-Subject-Out (LOSO) protocol is an adaptation of the classical leave-one-out method designed to evaluate the generalization of the algorithms across individuals.

In this protocol, data from $N-1$ subjects is used to train the algorithm, while all data from the remaining single subject is used to train the algorithm, while all data from the remaining single subject is used for testing. This process is repeated until each subject has been used as the test subject exactly once. This approach allows for a thorough evaluation of the model's performance in a cross-subject emotion recognition problem, ensuring that the model generalizes well across different individuals.

3.3.2. Leave-one-Video-out (LOVO)

The Leave-One-Video-Out (LOVO) protocol is another adaptation of the classical leave-one-out method designed to avoid potential shortcomings of EEG emotion recognition algorithms, such as generalization to new subjects.

In this protocol, the data is divided into V blocks (V is the number of videos), where each block corresponds to the data of all the subjects from a single video. The algorithm is trained on $V - 1$ blocks and tested on the remaining block. This process is repeated until each video data has been used as the test block exactly once. By evaluating the model's performance on unseen videos and using data from all the subjects to train, the LOVO protocol guarantees a reliable analysis of how well the algorithm can generalize to new and previously unseen video data, providing a robust measure of its effectiveness.

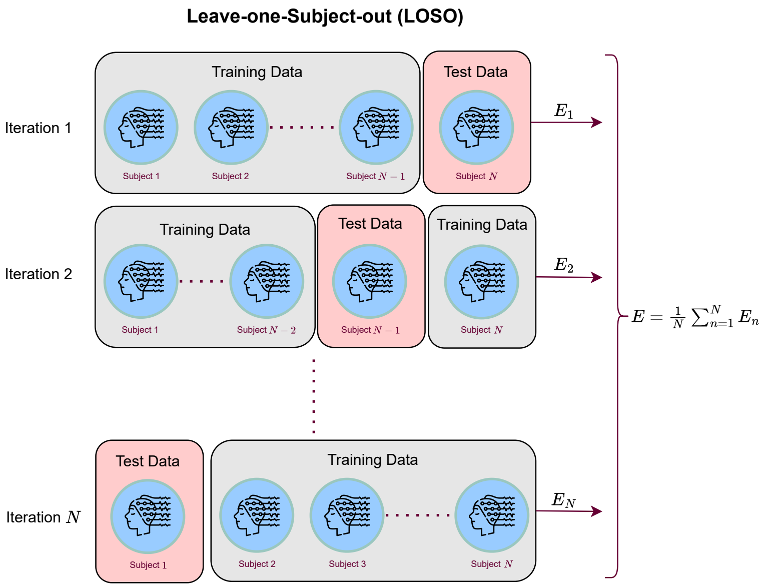

Figure 3.9: *This diagram shows the LOSO protocol, where data from $N-1$ subjects is used for training and data from the remaining subject is used for testing. This process is repeated until each subject has been tested once, ensuring the model's generalization across different individuals.*

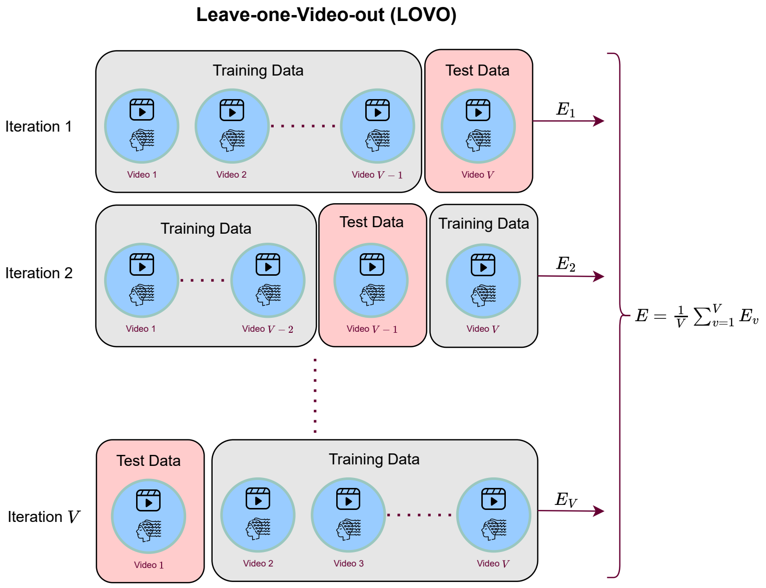

Figure 3.10: *This diagram shows the LOVO protocol, where data from $V-1$ videos is used for training and data from the remaining video is used for testing. This process is repeated until each video's data has been tested once, ensuring the model's generalization to new and unseen video data.*

3.3.3. Subject-Dependent Model (SDM)

The Subject-Dependent Model (SDM) protocol involves training and testing the models using data from only a single subject. According to Zhu, Zheng, and Lu [16], an emotion recognition system can achieve up to 92% accuracy when trained and tested on data from a single individual, while only achieving 52% accuracy when tested across three different subjects. Therefore, the SDM protocol ensures a robust evaluation of the consistency solutions by focusing on the individual variability in emotional responses.

This protocol allows for the fine-tuning of models to capture the unique affective patterns of each subject, leading to more accurate and personalized emotion recognition.

Following the previous approaches, the Subject-Dependent Model (SDM) protocol considers all the data belonging to a single subject. It then performs Leave-one-Video-out (LOVO) within each subject's data.

3.3.4. K-Fold Cross Validation

The classical K-fold Cross Validation method is also used, due to its widespread use for evaluating the performance of machine learning models. In our implementation the data is divided into 90% training and 10% test using stratified sampling to maintain a balance in target classes (high/low arousal and valence).

The training set is then further divided into 10 folds for 10-fold Cross Validation, allowing us to fine-tune the hyperparameters of each model. This iterative process ensures that each fold serves as a validation set once while the remaining folds are used for training, thereby providing a comprehensive evaluation and optimizing the model's performance.

3.4. Machine Learning Algorithms

In this section we present and explain the algorithms used for our emotion recognition and affective consistency analysis.

3.4.1. Support Vector Machine (SVM)

One of the algorithms selected is the Support Vector Machine (SVM), which is a widely used classifier in EEG emotion recognition systems. SVMs are favored in numerous studies [1], [11] due to their simplicity and properties well-suited for emotion recognition via EEG. Notably, SVMs are particularly effective in high-dimensional spaces, a crucial advantage given the inherently high-dimensional nature of EEG data, especially in our case, where the characteristics for all EEG bands and channels are considered. Additionally, SVMs perform well on small to medium-sized datasets, such as the available public datasets MAHNOB [14] and DEAP [9] datasets. Therefore, by finding an optimal hyperplane that separates different emotion classes, as in the figure 3.11, with maximum margin, SVMs effectively handle the complexity of EEG signals, ensuring robust classification performance.

In addition, the chosen hyperparameters that we will use in the work are: kernel, the regularization parameter C, gamma (γ) hyperparameter (with kernels: radial-basis-function and polynomial), and the degree for polynomial kernel.

It is crucial to make the right choice of kernel, because the kernel defines how the original feature space is transformed into a higher-dimensional space where a linear separator might exist, even if the data is not linearly separable in the original space. Due to the complexity of EEG features and their high dimensionality nature, EEG data tipically exhibits complex patterns that are not linearly separable in the original space. For this reason, different kernel functions will be explored to address non-linearities in the data: Linear Kernel, Radial Basis Function (RBF) or Gaussian Kernel and polynomial kernel.

The linear kernel, defined as $k(x, y) = x \cdot y$, is the simplest kernel, as it directly computes the dor product of the input vectors. This option is explored because of its computational efficiency and simplicity.

The polynomial Kernel, $k(x, y) = (\gamma x \cdot y + c)^d$, where c is a constant and d is the degree of the polynomial,

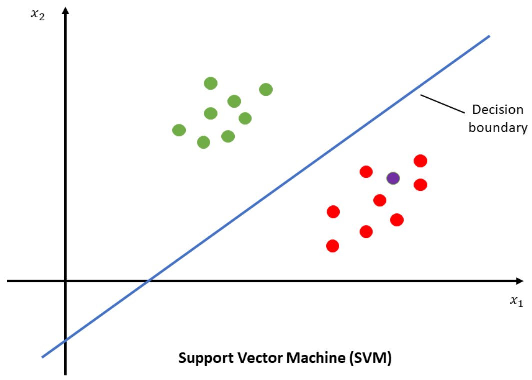

Figure 3.11: *Illustration of how a Support Vector Machine (SVM) classifier separates two classes of data points (green and red) using a decision boundary (blue line). The SVM aims to find the optimal hyperplane that maximizes the margin between the two classes, ensuring robust classification performance. The axes x_1 and x_2 represent the feature dimensions, while the decision boundary effectively separates the data into their respective classes.*

is also explored because polynomial kernel is well-suited for capturing complex, non-linear structures by expanding the input feature space through polynomial transformations, allowing the model to handle non-linear EEG patterns up to a specified degree.

Finally, the Radial Basis Function or Gaussian Kernel,

$$k(x, y) = e^{-\frac{\gamma\|x-y\|^2}{2\sigma^2}}, \tag{13}$$

is also explored because of its high effectiveness for capturing intricate, non-linear relationships by mapping the data into an infinite-dimensional space.

The regularization parameter C is also studied, because it is a central component of the SVM, playing a crucial role in determining the trade-off between maximizing the margin and minimizing the classification error. Mathematically, C appears in the optimization problem that defines the SVM, which objective is to minimize the following expression:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i, \tag{14}$$

where $\|w\|^2$ represents the margin (how far apart the classes are in the feature space), and ξ_i are slack variables that allow for misclassification of some training examples. Therefore, the parameter C determines how much penalty is assigned to these misclassified points, which is critical for balancing the bias-variance trade-off. A smaller C will result in a softer margin, allowing more misclassifications to improve generalization. Conversely, a larger C enforces a stricter margin, aiming to correctly classify more training points but increasing the risk of overfitting. For these reasons, to achieve optimal performance in EEG emotion recognition, C must be carefully tuned, to strike the right balance between model flexibility and accuracy.

Finally, the Gamma (γ) hyperparameter is also considered, because it is of great importance in RBF and polynomial kernels, since it controls how much influence a single training sample has, controlling the spread of the kernel. Gamma γ, scales the dot product in the case of the polynomial kernel and scale inversely the radius of influence in the RBF kernel. Therefore, a small γ value reduce the influence of the input feature and increase the radius of influence of the input features, respectively, resulting in a smoother decision boundary. Conversely, a larger γ value increases feature sensitivity in the polynomial kernel and narrows the radius of influence in the RBF kernel.

3.4.2. K-Nearest Neighbours Algorithm (k-NN)

K-Nearest Neighbours Algorithm (k-NN) is also selected for the emotion recognition analysis, which is also a widely used classifier in EEG-based emotion recognition [47], [48], [44]. The main reason for its widespread use is its simplicity, as well as its ability to model complex distributions without making assumptions about the underlying data. Yin, Zhao, Wang, Yang, and Zhang [47] achieve up to 85% of accuracy in binary classification for valence and arousal, following a subject-dependent strategy. Therefore, this ability to achieve high accuracies and its particular efectiveness for small to medium-sized datasets, such as the DEAPand MAHNOB, makes it a suitable choice for our purpose.

The operation of the k-NN is very simple, yet powerful. This works by finding the k closest data points (neighbours) to a given query point and assigning it the most common class among these neighbours, see Figure 3.12. Therefore, k is the main hyperparameter of the model, significantly influencing the performance of the classifier: a smaller k could make the model very sensitive to noise and local variations in the data, while a larger k value smoothens the decision boundary, potentially leading to underfitting. Consequently, k is one of the hyperparameters selected to study in order to ensure an optimal performance of the classifier.

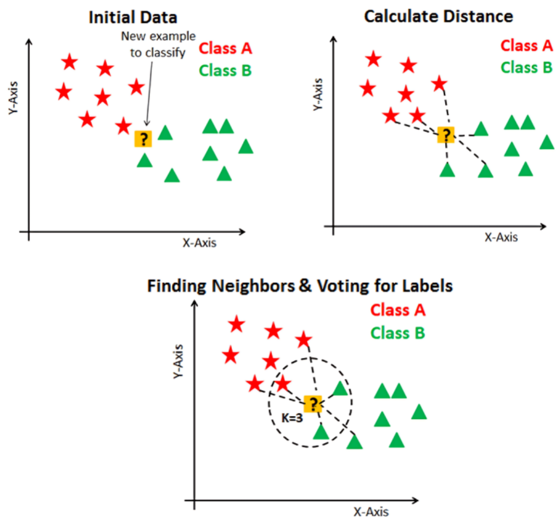

Figure 3.12: *Visualization of how a k-Nearest Neighbors classifier works: (1) Initial data with two classes, (2) Distance calculation, and (3) Finding the k nearest neighbors and voting for labels. Source: [49].*

Determining the neighbors of a data point involves identifying the closest data samples to that point. This process depends on selecting an appropriate distance metric, which defines how "closeness" is measured. Therefore, the choice of distance metric is crucial, as it directly influences which samples are considered neighbors and, consequently, the performance of the k-NN algorithm. For that reason, the distance metric is another of the hyperparametrs selected in order to achieve the best possible performance.

Finally, another important hyperparameter is the weight function, which determines how much influence each of the k-nearest neighbours has on the final classification. We evaluate different weighting schemes, such as uniform weights, where all neighbors contribute equally, and distance-based weights, where closer neighbors have a stronger influence. This selection allows the algorithm to leverage local information effectively and improves classification accuracy, especially when dealing with non-uniform data distributions.

3.4.3. EEG Contextual Neural Network (EEG-CNN)

We designed the EEG Contextual Neural Network with two clear objectives: to analyze the temporal affective consistency through the temporal weights learned by the network, as previously explained (Learning-Based Consistency), and to capture the complexity of the EEG data by means of a neural network that adapts its first layer according to the context (the video to which the data belong). For this purpose, the EEG-CNN is made up of two distinct parts, the contextual layer and the dense layers.

As explained previously, the contextual weighting part involves multiplying the feature matrix, which is associated with a video $v \in \mathcal{D}$ and a subject $o \in \{1, \ldots, N\}$, by a video-dependent weighting vector, in order to obtain a single vector of features weighted according to the importance of each time segment, Figure 3.7. This weighting part can be interpreted as a dense layer with V (number of videos) neurons and no activation function, where only one of the neurons is activated and the input layer receives feature vectors, see Figure 3.8.

The dense part is, in essence, a traditional dense neural network, which receives the sum of the pre-weighted feature vectors and passes through a few dense layers and performs the classification of positive/negative arousal or valence.

Specifically, the contextual part receives the matrix formed by the vectors, $\{f^{o,v}(t_1), \ldots, f^{o,v}(t_\omega)\}$, which represents the extracted EEG features of the observer, o, and the video, $v \in \mathcal{D}$ in the different time segments, extracted over the time window of size ω seconds. Then, the feature matrix is multiplied by a contextual vector, that is video-dependent, W^v of size t_ω, thus obtaining a weighted sum of the feature vectors according to their importance in emotion recognition:

$$\overline{f^{o,v}} = \sum_{j=1}^{t_\omega} f^{o,v}(t_j) W^v(t_j), \quad o \in \mathcal{O} \quad and \quad v \in \mathcal{D}. \tag{15}$$

Therefore, $\overline{f^{o,v}} \in \mathbb{R}^M$ is a vector of size M (number of features), which will be the input data of the dense layer part of the Contextual Neural Network.

The dense part receives the weighted sum of feature vectors, vector of size M (representing the number of features). This input is processed through three fully connected layers, each one followed by a ReLU activation function, $f(x) = max(0, x)$, which introduces non-linearity. The number of hidden units in each layer is determined using the K-fold Cross Validation protocol, and the specific values are discussed in the next section. Finally, the processed feature vector is passed through an output layer that classifies the emotional state into either positive or negative arousal, or valence.

To train the neural network, each feature matrix, denoted as $f^{o,v}$, corresponding to subject $o \in \mathcal{O}$ and video $v \in \mathcal{D}$, is treated as a batch. This approach addresses the limited sample size of EEG datasets and the necessity to encapsulate the complexities inherent in each subject and video.

The training process employs Binary Cross Entropy Loss as the loss function and Stochastic Gradient Descent (SGD). The loss function selected for backpropagation is the Binary Cross Entropy Loss (BCELoss), due to its suitability for binary classification problems, measuring the binary cross-entropy between the target and the output probabilities. Mathematically, BCELoss for a single sample (our case,

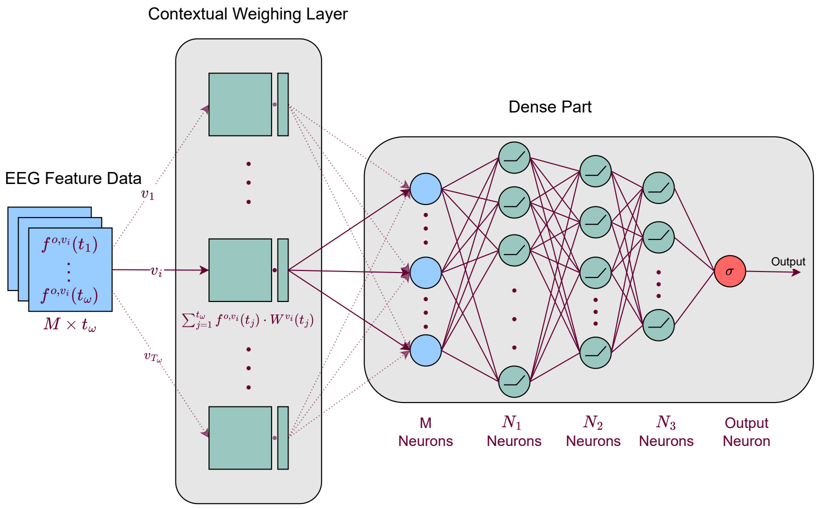

Figure 3.13: *This diagram illustrates the forward-pass of the Contextual EEG Neural Network, comprising the contextual layer and the dense layers. The contextual layer multiplies the EEG feature matrix, associated with each video and subject, by a video-dependent weighting vector, producing a weighted sum of feature vectors. These weighted features are then processed by the dense layers to classify emotional states into arousal or valence categories.*

as each batch is a sample) is defined as:

$$BCELoss = -\left[y\log(p) + (1-y)\log(1-p)\right], \tag{16}$$

where y represents the true label (positive or negative valence/arousal) and p denotes the predicted probability of the label being 1 (positive). This loss function penalizes incorrect predictions more heavily as they diverge from the true labels, thus guiding the optimization process towards minimizing the divergence.

With regard to the optimizer, before considering SGD, other optimizers were tested, like Adam, because of their efficiency and effectiveness. However, to ensure the update, in each batch, of the weights associated with the video to which the data corresponds, Adam required the implementation of masking techniques and Torch hooks, which significantly increased the computation time. Therefore, the simplicity of SGD allows to ensure a correct backpropagation of the corresponding weights, due to the update rule for the weights w_t at each time step t in SGD is defined as follows:

$$w_{t+1} = w_t - \gamma \nabla L(w_t), \tag{17}$$

where γ is the learning rate, and $\nabla L(w_t)$ is the gradient of the loss with respect to the weights. Thus, by applying a split technique during backpropagation, Code A.1, the weights of the contextual layer are updated following the standard SGD, while those of the dense layers are updated incorporating hyperparameters such as momentum or weight decay for faster convergence. This approach ensures the correct update of the corresponding weights in a very efficient way and the parameter update in the same direction for both layers.

Momentum and weight decay are considered for backpropagation of the dense layers to ensure more stable convergence, avoiding local minima and oscillations and to prevent overfitting.

Momentum is a technique used in optimization algorithms like Stochastic Gradient Descent (SGD) to accelerate convergence by incorporating information from past gradients. The idea is to smooth out the updates by adding a fraction of the previous update to the current one, which helps avoid oscillations and speeds up learning, especially in the direction of consistent gradients. Mathematically, the momentum update is defined as:

$$v_{t+1} = \beta v_t + (1 - \beta)\nabla L(w_t), \tag{18}$$

where v_t is the velocity or momentum term, which accumulates past gradients and β is the momentum coefficient, that controls how much of the past gradient to include. This momentum term is then applied into the weights update as follows:

$$w_{t+1} = w_t - \gamma v_{t+1}. \tag{19}$$

Furthermore, weight decay is considered due to small amount of data, which make the model tend to overfitting. For that, weight decay is suitable because is a regularization technique that penalizes large weights by adding an additional term to the loss function.Specifically, weight decay helps prevent the model from relying too heavily on specific features, which can lead to overfitting, by shrinking the weights during training. During optimization, weight decay modifies the gradient of the loss function as follows:

$$\nabla L(w_{t-1}) = \nabla L(w_{t-1}) + \lambda w_{t-1}, \tag{20}$$

where $\nabla L(w_{t-1})$ is the gradient of the original loss function with respect to the weights, and λw_{t-1} is the additional weight decay term. Pseudo-algorithm of this version of SGD is provided in the Appendix, see Algorithm 1.

In summary, while more complex algorithms like Adam offer certain advantages, the simplicity and flexibility of Stochastic Gradient Descent (SGD) allow for a more direct implementation of contextual weight updates without the need for additional masking techniques. By combining standard SGD with momentum and weight decay, the model achieves faster, more stable convergence while also mitigating the risk of overfitting, making this approach well-suited for the EEG emotion recognition task at hand.

4. Experimental Work

This chapter presents the experimental work, including experimental setup and results obtained from our study. It begins with an overview of the datasets used to analyze our objectives, then is followed by an explanation of the experimental setup and finally is provided an extensive and comprehensive evaluation and discussion of the findings. Through detailed analysis, we aim to provide insights into the performance and implications of our proposed methods.

4.1. Description of Datasets

The proposed methodology is evaluated on two well-known datasets which include EEG signals in the context of affective analysis: DEAP [9] and MAHNOB-HCI [14]. These datasets are widely used in affective computing research and provide multimodal recordings, including EEG signals, which serve as the foundation for analyzing emotional responses to visual stimuli. A detailed description of each dataset, including the data description, preparation and analysis, is provided below.

4.1.1. DEAP

The DEAP dataset comprises multimodal data collected from 32 participants. This data has been collected through showing participants 40 selected music videos of 1 minute each, and their physiological signals were recorded. Specifically, 512Hz EEG peripheral physiological signals through 32 channel. In addition, facial expressions were recorded for 22 participants, however this was not used in this thesis.

The stimuli, comprising 40 music videos, were selected using a novel method involving retrieval by affective tags from the last.fm website, video highlight detection, and an online assessment tool. The selection process aimed to ensure a broad range of emotional content that could elicit diverse emotional responses from the participants.

Participants, after each video, rated their emotional state using 5 different dimensions. The arousal-valence rating, based on the Russell and Barrett [24] theory, Dominance, Liking, which is how much did they like the video, and Familiarity, which is how well did they know the video. These emotion states are quantified on a continuous scale from 1 to 9, except for familiarity which is quantified on a discrete scale from 1 to 5. However, only the arousal and valence measures are used in this study, due to their widespread use in the field of emotion recognition.

In terms of data collection and preprocessing, EEG data was recorded using 32 channels following the international 10-20 system. EEG and peripheral physiological signals were measured at a 512 Hz sampling rate. then downsampled to 128 Hz. We omit the other details of data acquisition and preprocessing, because is not the aim of this thesis master, for more information of the data acquisition, see the original article [9] and for more information of data preprocessing see [44].

4.1.2. MAHNOB-HCI

The MAHNOB-HCI dataset [14] is a multimodal database recorded to study emotion recognition and implicit tagging. It includes data from 27 participants, both male and female, from diverse cultural backgrounds, in order to avoid the cultural bias in emotion [20].

The dataset features two experiments. In the first experiment, 27 participants watched 20 emotional

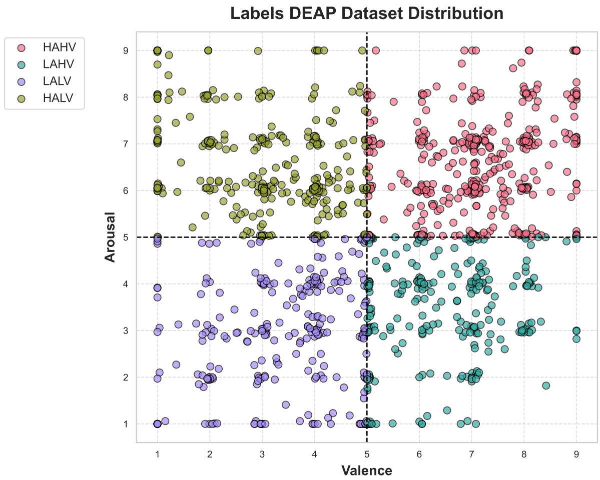

Figure 4.1: *Distribution of emotional labels (from 0 to 9) in the DEAP dataset, categorized by arousal (y-axis) and valence (x-axis) scores. The four quadrants represents the different emotional states: High Arousal and High Valence (HAHV), Low Arousal and High Valence (LAHV), Low Arousal and Low Valence (LALV) and High Arousal and Low Valence (HALV).*

video clips and self-reported their felt emotions using dimensions of arousal, valence, dominance, and predictability, as well as emotional keywords. The second experiment involved showing short videos and images first without any tags and then with correct or incorrect tags, assessing participants agreement with the displayed tags.

The videos used in the MAHNOB-HCI dataset were carefully selected to elicit a broad range of emotional responses from the participants. A total of 20 video clips were chosen, each varying in length from 34.9 to 117 seconds. However, following [44], where the data is provided, in this thesis clips with a duration of less than 1 minute were excluded, and clips longer than 1 minute were truncated to ensure consistency with the DEAP dataset. Therefore, the total number of samples in the dataset is 449 trials.

These clips included scenes from commercially produced movies, as well as popular online resources, designed to induce specific emotions such as joy, sadness, fear, and disgust. The selection process involved a preliminary study where 155 video clips were shown to over 50 participants, who then annotated these clips with emotional tags. Based on the results of this study, the final set of videos was chosen to maximize emotional diversity and intensity. Additionally, neutral clips, such as past weather forecasts, were included to serve as a baseline for emotional responses.

The data modalities recorded in the MAHNOB-HCI dataset include face videos captured from multiple angles, audio signals recorded using room and head-worn microphones, eye gaze data including gaze position, pupil diameter, and blink data, and physiological signals such as EEG, ECG, GSR, respiration amplitude, and skin temperature.

In terms of data collection and preprocessing, EEG data was recorded using 32 channels following the international 10-20 system. Peripheral physiological recordings included ECG, GSR, respiration, and skin temperature with a 1024 Hz sampling rate, downsampled to 256 Hz. We omit the other details of data acquisition and preprocessing, because is not the aim of this thesis master, for more information of the data acquisition, see the original article [14] and for more information of data preprocessing see [44].

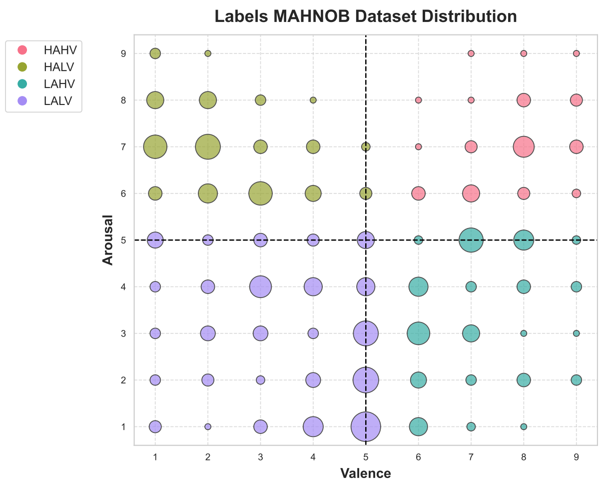

Figure 4.2: *Distribution of emotional labels in the MAHNOB dataset, categorized by arousal (y-axis) and valence (x-axis) scores, which are measured on a discrete scale from 1 to 9. The size of each point indicates the number of labels associated with that specific value—the larger the point, the higher the number of labels. The four quadrants represents different emotional states: High Arousal and High Valence (HAHV), Low Arousal and High Valence (LAHV), Low Arousal and Low Valence (LALV), and High Arousal and Low Valence (HALV).*

4.1.3. Analytical Data Overview

Both datasets use the extracted features explained in the previous section: Hjorth Parameters, Spectral Entropy and the energy of the signal. These features have been calculated for each band: Delta (<4 Hz), Theta (4–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (>31 Hz) and for each EEG channel of the 32 chosen. Since there are 5 different frequency bands, 5 features for each band, and 32 EEG channels, this results in a total of $32 \times 5 \times 5 = 800$ unique features.

In addition, as the data is calculated for a time window of $\omega \in \{1, 2, 4, 6, 10, 12, 15\}$ seconds, in the case of DEAP, $\omega \in \{2, 4, 6, 10, 10\}$ seconds, in the case of MAHNOB, for each video $v \in \mathcal{D}$, and for each subject $o \in \mathcal{O}$, each sample, $f^{o,v}$ has size $800 \times T_\omega$, where $T_\omega = \frac{60}{\omega}$. Thus, in the Figure 4.3 we can see how the data is distributed in the dataset, which is, in fact, transposed but in practice each sample is transposed to be consistent with the previous mathematical reasoning:

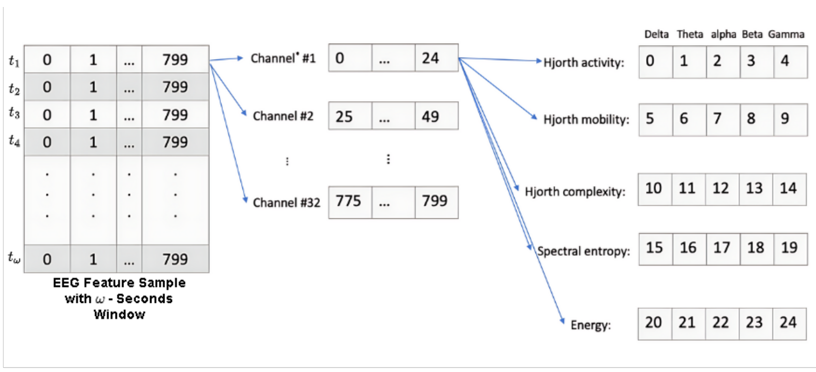

Figure 4.3: *Distribution of EEG features of a single video sample across our processed DEAP and MAHNOB datasets, calculated using a time window of ω seconds.*

It is important to note that the number of samples is quite limited: 1,280 samples for DEAP and 449 for MAHNOB, which complicates the generalization of the machine learning models used. As a result, these models tend to overfit and show large variations in prediction across subjects.

Regarding the labels, this work focuses independently on the prediction of both arousal and valence. In particular we focus on binary classification, in contrast to other works [50], then the labels are binarized, assigning a value of 1 when the rating is greater than 5 (High Valence or High Arousal), and 0 when it is less (Low Valence or Low Arousal). Samples with a neutral emotion are removed from the model's training for that emotion, due to their ambiguity and potential to introduce noise into the learning process, thereby affecting the model's performance. In the case of DEAP dataset the Neutral Valence and Arousal represents a 1.2% of the data, each other. While, neutral emotion represents 18% in the case of Valence and 13% for Arousal in MAHNOB dataset.

Furthermore, the data is slightly imbalanced, as shown in the Figure 4.4, which illustrates the distribution of the classes. Then, DEAP is slightly biased towards high arousal and valence, while MAHNOB leans towards low arousal and valence. This imbalance can further complicate model training, as it may lead to biased predictions favoring the majority class, as we will see in the Emotion recognition result.

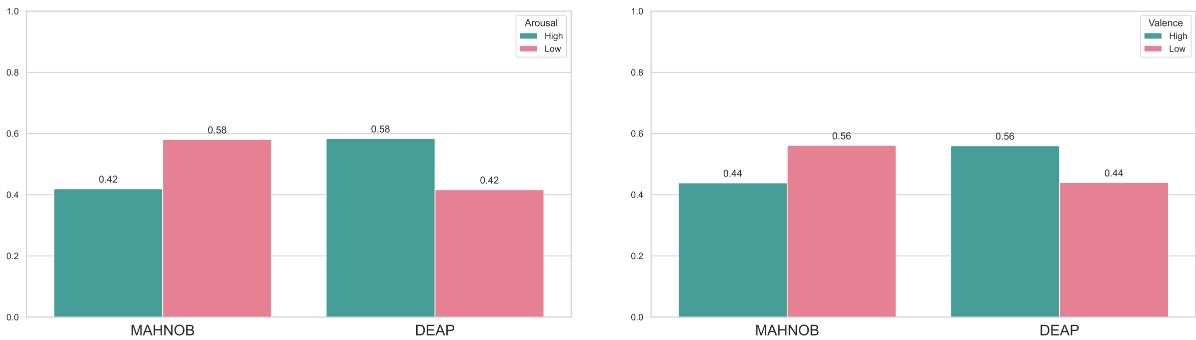

Figure 4.4: *Proportion of high and low Arousal (Left) and Valence (Right) states in the MAHNOB and DEAP datasets.*

4.2. Experimental Setup

In this section we explain how the different protocols, algorithms and consistency solutions are set up and combined to evaluate both emotional prediction and to analyze affective temporal consistency.

4.2.1. Training Protocols and Setup

To evaluate emotional prediction we use the protocols explained in the previous chapter, Leave-one-Subject-Out (LOSO), Leave-one-Video-out (LOVO), Subject-Dependent Model (SDM) and K-fold Cross Validation, together with the classification algorithms described: SVM (traditional classifier used in EEG emotion recognition), and EEG Contextual Neural Network (EEG-CNN). Therefore, for each protocol and for each classifier, two models are trained, one for each emotion: Arousal and Valence.

However, in order to evaluate the consistency solutions we applied these to the different algorithms, in each protocol. In the case of the SVM, this classifier works by finding a hyperplane separating vectors (samples). However, in our case, each sample, $f^{o,v}$ of a given video and subject, is a matrix of size $M \times T_\omega$, where M is the number of extracted features and T_ω is the number of time segments over which the features are extracted with window size ω seconds. Therefore, it is necessary to convert this matrix data into a vector format suitable for SVM processing. To this end, many studies [11], [17], average the feature data vectors associated with the time segment $t \in \{1, \dots, T_\omega\}$, obtaining the following vector of size M (number of features).

$$\overline{f^{o,v}} = \frac{1}{T_\omega} \sum_{t=1}^{T_\omega} f^{o,v}(t), \quad \text{where} \quad \overline{f^{o,v}} \in \mathbb{R}^M. \tag{1}$$

However, we study this notion of averaging, since it implicitly assumes that the data from all temporal segments are equally important or discriminative, which might not be true, due to dynamical nature of the visualized contents by the participants. Therefore, to analyze temporal affective consistency, we will not only uniformly average the data, but also use the different consistency solutions defined in the previous chapter. Thus, the time segments of a sample will be weighted by the different weight vectors calculated from the Heuristic, Data-Aware and Learning approaches, as illustrated in the accompanying diagram.

Formally, if W^v is the weight vector of one of the solutions proposed, for the video $v \in D$, the vectors used in order to train the SVM will be:

$$\overline{f^{o,v}} = \sum_{t=1}^{T_\omega} f^{o,v}(t) \cdot \sigma\left(W^v(t)\right), \quad o \in \mathcal{O}. \tag{2}$$

Here, σ denotes the sigmoid function, which is used to normalize the weights, ensuring that the influence within each time segment is appropriately scaled. This allow the model to emphasize features of segments of the data that might be more discriminative of the emotional state.

In the EEG Context Neural Network, the layer responsible for weighting each temporal segment, represented by $W^v \in \mathbb{R}^{T_\omega}$, is learned through backpropagation. However, to apply the different consistency solutions, we replace this layer with the predefined solutions: Heuristic-Based (First, Centered, Uniform, Last) and Heuristic-Based (Glimpse). Then, the weights in this layer are then frozen to evaluate how each solution affects the network's emotional prediction performance.

However, the EEG Context Neural Network cannot be directly applied using the LOVO protocol because, in each iteration, all data corresponding to a single video is used for testing. This means the network does not have access to the data from the test video during training, making it impossible to learn the specific weights associated with that video. To address this limitation, we modify the structure of the EEG Context Neural Network by making the weighting mechanism subject-dependent rather than video-dependent. This adjustment allows the network to generalize across different individuals while still capturing experiment variability in emotional responses.

4.2.2. Hyperparameter Optimization and Model Configuration

The hyperparameters used in the LOVO, LOSO and SDM protocols are those provided by the k-fold cross-validation protocol. In this case, to perform the hyperparameter optimization, we utilized the HyperOpt [51] library and Bayesian optimization, via Tree-Based Parsen Estimators [52], to ensure a correct and efficient selection of the hyperparameters.

For the Support Vector Machine (SVM), the hyperparameter search involved exploring various kernel types, including polynomial, linear, radial basis function (RBF) and sigmoid kernels. The search also considered different values for the regularization parameter C (0.01, 0.1, 1, 10 and 100), which controls the trade-off between achieving a low training error and maintaining a larger margin for classification, to prevent overfitting. Also, the γ parameter, which defines the influence of each training example, was explored with values in $\{0.01, 0.1, 1, 10, 100\}$, Finally, for the polynomial kernel, the degree was varied between 2 and 6, allowing the model to capture more complex decision boundaries when appropriate.

In the case of the k-Nearest Neighbours (k-NN), the hyperparameter search is performed within a small search space. Firstly, the number of neighbours, which is the main hyperparameter of the model, was explored with values in $\{1, ..., 20\}$. The other hyperparameters considered were the distance metric, where both Euclidean and Chebyshev distances were evaluated, and the weighting scheme, where uniform weighting (all neighbours are weighted equally) was compared with distance-based weighting (where closer neighbours have more influence).

On the other hand, the parameter space defined for the neural network has been designed to explore various architectures and training configurations. The number of hidden units in the first two layers was tested with 64, 128 and 256 neurons, while the third layer was set to vary between 32, 64 and 128 units.This space has been chosen following a descending structure in the number of neurons to progressively reduce the dimensionality of the network, promoting feature extraction in the early layers and refining the learned representations in the later layers. For the weighting layer and the dense layers, the learning rate was selected from values of 0.0001, 0.001 and 0.01, ensuring an optimal balance between fast convergence and avoiding overshooting during training. In the dense layers, momentum was tested with values 0.9, 0.95 and 0.99 to determine the value that best avoid getting stuck in local minima. Lastly, weight decay was applied to regularize the model and prevent overfitting by penalizing large weight values. The weight decay parameter was explored within a range from $1e-4$ to $1e-2$, promoting simpler and more generalized models.

After performing the hyperparameter optimization using the Bayesian approach with HyperOpt, the optimal parameters for each model were selected, Table 4.1. In order to simplify the process, and do not choose a unique hyperparameter selection for each configuration (model, consistency, window size and dataset), the same configuration was chosen for each one, based on the most repeated hyperparameter configuration throughout the k-Fold validation process.

Table 4.1: *Summary of models and their corresponding hyperparameters.*

Model	Category	Hyperparameters
SVM	Valence	**Kenel**: Radial Basis Function (RBF) **C**: 100 **Gamma** (γ): 1.0
	Arousal	**Kernel**: Radial Basis Function (RBF) **C**: 10 **Gamma** (γ): 0
k-NN	Valence	**Num. Neighbours**: 4 **Metric**: Chebyshev **Weights**: Distance
	Arousal	**Num. Neighbours**: 7 **Metric**: Euclidean **Weights**: Distance
EEG-CNN	Valence & Arousal	**Momentum**: 0.9 **Learning Rate**: 0.0001 **Gamma** (γ): Three layers: (256,256,23) hidden units **Weight Decay**: 1e-5

4.2.3. Performance Metrics

To evaluate the performance of the machine learning models in classifying emotions from EEG data, we used three key metrics: accuracy, balanced accuracy, and F1-Score. Additionally, we tracked the training loss for the neural network to monitor the learning progress.

Accuracy, as it measures the percentage of correct classifications, is the metric of choice in this type of problems. It is calculated using the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) predictions, as shown in the formula below:

$$\text{Accuracy} = \frac{TP + TN}{Total}, \tag{3}$$

where $Total$ represents the sum of all classifications (i.e., $TP + TN + FP + FN$). While it is the most popular, accuracy may be misleading in imbalanced datasets, as is the case here, where it tends to favor the majority class.

In class-imbalanced contexts, one possible alternative is the balanced accuracy is also chosen, which is designed to avoid this problem by averaging the true positive rates across clasess. This ensures that performance on minority classess is equally weighted:

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right). \tag{4}$$

Furthermore, F1-Score is one of the chosen metrics. F1-Score measures a balance between precision (how many of the predicted positives are actually correct) and recall (how many of the actual positives the model correctly identified). The formula is expressed as follows:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}. \tag{5}$$

This score ranges from 0 to 1, where 1 means perfect precision and recall. In the context of emotion recognition, the F1-Score is useful because it ensures the model is not just good at predicting the more common emotion class/label, but also accurately identifies the less frequent one.

4.2.4. Consistency Evaluation and Analysis

Finally, a comparison between the emotional response obtained by EEG-GLIMPSE and the Learning-Based approach will be made to check if a similar behaviour is obtained, thus concluding that a certain consistency is captured. In addition, these will be manually analyzed with a manual check of the associated videos, to verify that indeed, the video contains a meaningful stimulus in the places comprising the consistency solution.

4.3. Emotion Recognition

This section details the experimental results evaluating the performance of the proposed machine learning models, incorporating the developed protocols and consistency strategies. The results are organized into subsections that cover model evaluations, performance comparisons, and the impact of different configurations. Detailed results, including accuracy, F1-Score and balanced accuracy for all cases, are given in the appendix. In this section we will discuss the most important results for each protocol. The section concludes with a summary highlighting the best-performing models and key insights drawn from the experimental outcomes.

To present the analysis more effectively, we will use graphs to display the results. This section focuses exclusively on the performance of our algorithms and aims to draw meaningful conclusions. The graphs will summarize the outcomes by highlighting the best results for each time window and for each category (valence and arousal). Specifically, we will showcase the top scores in terms of balanced accuracy and F1-Score, as these metrics provide an unbiased evaluation of the algorithm's performance. Additionally, we will present the best average performance achieved by each time window across both categories for a more comprehensive overview.

4.3.1. K-Fold Results

Firstly, the protocol evaluated is the K-Fold Cross Validation presented in the previous chapter. As explained, the protocol is evaluated for the SVM and k-NN with all the consistency solutions, while for the EEG-CNN is only evaluated within its default configuration (learned consistency). The detailed results including accuracy, F1-Scores and balanced accuracy for the k-fold protocol, for each case can be viewed in the following tables: Table A.8, Table A.6, Table A.7, Table A.4 and Table A.5.

Firstly, we will analyze the results for the DEAP dataset, which can be seen in Figure 4.5.

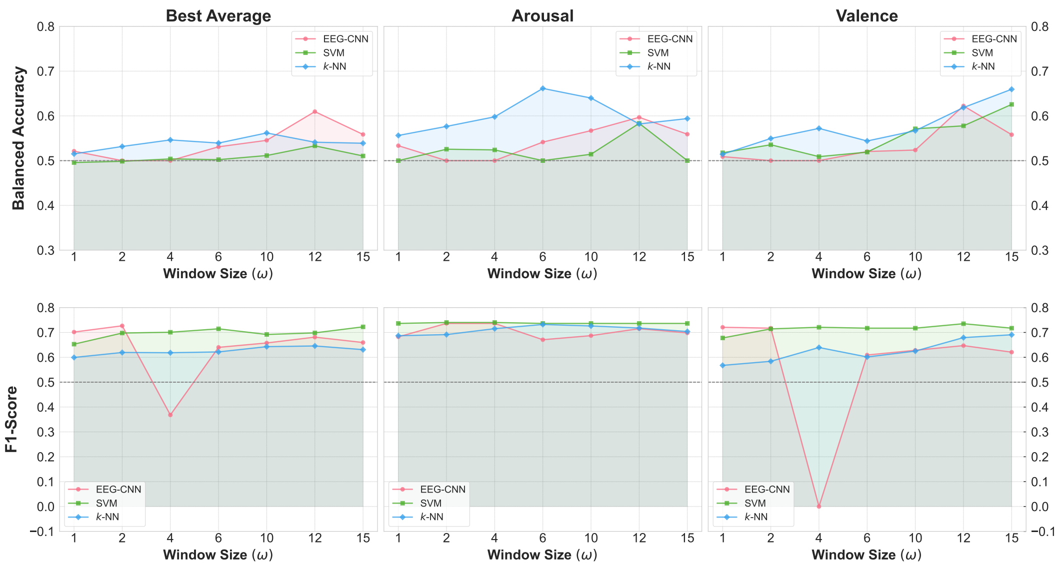

Figure 4.5: *Best results per window size for the DEAP dataset using the k-Fold protocol, including top balanced accuracy, F1 scores for each category, and average performance across both.*

Thus, by looking at the Figure 4.5 we can see that the best balanced accuracy results are achieved by the k-NN, which achieve a maximum peak of balanced accuracy of 0.661 for arousal with a window size of 6 seconds and 0.659 for valence with a window size of 15 seconds. However, k-NN is not the model that, for a time window, achieves the best balance between arousal and valence balanced accuracies, which is in fact the EEG Context Neural Network, achieving up to 0.610 of best mean balanced accuracy.

It's important to highlight that these results were calculated across all consistency scores (except for the EEG-CNN). Additionally, better performance in one category does not necessarily affect the overall mean of the balanced accuracy or F1-Score, since the model that optimizes the mean score may not achieve the highest performance in each category individually.

In addition, what we can see is that smaller time windows do not achieve the best results, as they conclude in the paper [44]. Achieving the best results for the SVM in the 12s and 15s time windows for arousal and valence, respectively, for the EEG-CNN at 12s and for the k-NN at 6s and 15s, respectively.

However, if wee look at the F1-Score, we notice that it remains relatively stable across different time windows compared to the balanced accuracy, which varies more noticeably. For example, despite k-NN achieving a peak balanced accuracy of 0.661 for arousal with a 6-second window and 0.659 for valence with a 15-second window, the F1-Score for these configurations does not show a significant increase or decrease. This stability indicates that changes in balanced accuracy may be driven more by variations in sensitivity across classes rather than by significant shifts in precision or recall individually.

In the case of MAHNOB dataset, Figure 4.6 shows the summary of the results.

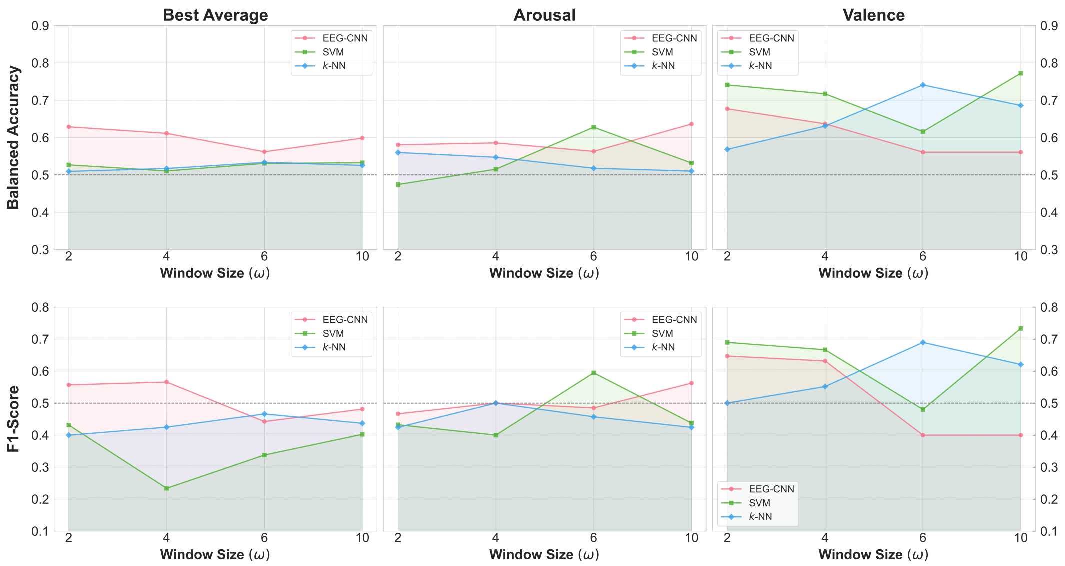

Figure 4.6: *Best results per window size for the MAHNOB dataset using the k.Fold protocol, including top balanced accuracy, F1 scores for each class, and average performance across both classes.*

In the case of MAHNOB, the results are much better, especially in the case of balanced accuracy. Where the best performance is achieved by the EEG-CNN model, achieving up to 0.636 of balanced accuracy in arousal prediction and by the SVM for valence classification achieving up to 0.772 of balanced accuracy. Additionally, as in the previous case, the model that best balances arousal and valence is the EEG-CNN.

Moreover, in this case there is no clear evidence that some time windows provide better prediction across all the models. However, the best performances are achieved with a window size of 10 seconds, which is the maximum.

Additionally, the F1-Score in this case is considerably lower than in the DEAP dataset, even though it achieves higher performance in terms of balanced accuracy, especially in the Arousal. This could imply a general lack of precision. In other words, while the model may be correctly identifying more samples of the minority classes (improving recall), it is also misclassifying a greater number of samples from other classes (leading to an increase in false positives). This trade-off results in a lower F1-Score despite the increase in balanced accuracy.

For the Valence, the results for k-NN and SVM show relatively high F1-Scores, indicating that the model is able to predict both positive and negative valence classes with better balance between precision and recall. This suggests that, in Valence, the model is not only capturing the true positives but is also maintaining lower false positive rates, contributing to a more stable F1-Score.

In summary, the results indicate that medium and larger window sizes achieve the best results and achieve better performance in most cases. Moreover, the k-NN model shows the highest balanced accuracy for the DEAP dataset, while the EEG-CNN model achieves the best overall balance between arousal and valence. For the MAHNOB dataset, the SVM model performs best in terms of balanced accuracy for valence classification and EEG-CNN for Arousal classification. However, despite higher balanced accuracy, the F1-Scores are lower, suggesting a trade-off between recall and precision.

4.3.2. LOSO Results

As explained previously, one of the objectives of the LOSO protocol is to evaluate the generalization of the models across subjects. This enables to assess the possible impact of consistency solutions in addressing the problem of inter-subject prediction, which drops significantly the performance, unlike subject-dependent protocols.

The LOSO protocol is evaluated for the SVM and k-NN for the EEG-CNN across all the consistencies. The detailed results including accuracy, F1-Scores and balanced accuracy for the k-fold protocol, for each case can be viewed in the appendix in the Tables A.9 to A.14.

In the case of DEAP dataset Figure 4.7, gives a summary of the top performance results.

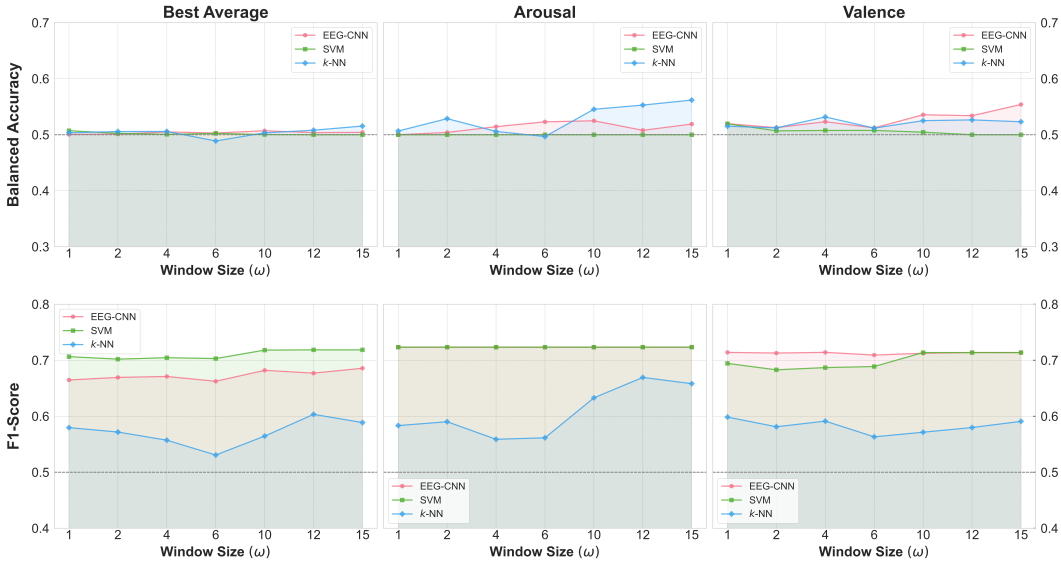

Figure 4.7: *Best results per window size for the DEAP dataset using the LOSO protocol, including top balanced accuracy, F1 scores for each category, and average performance across both states.*

In general, we can see that, as in the previous case, the best results of the best models are achieved in the 3 scenarios for the largest time windows, in this case 15s. Also, there is a trend towards improved balanced accuracy as the window size increases, with some points of stabilization. However, the quality of the prediction is much lower than in the previous case, reaching only 0.554 of balanced accuracy for valence and EEG-CNN, 0.562 for arousal and k-NN and 0.543 of the best balanced mean of the k-NN between both categories.

However, despite the poor results in terms of balanced accuracy, the F1-Scores are much more higher, particularly in the EEG-CNN and SVM. This is a consequence of the fact that the model is actually, in almost all cases, only predicting the positive class, i.e. the majority class for both emotional dimensions. And therefore suffering from overfitting. Furthermore, if we look at the standard deviation, which is extremely low, Figure 4.8, we can also intuit this behaviour.

Regarding the MAHNOB dataset, Figure 4.9, we will analyze it in less detail since similar conclusions can

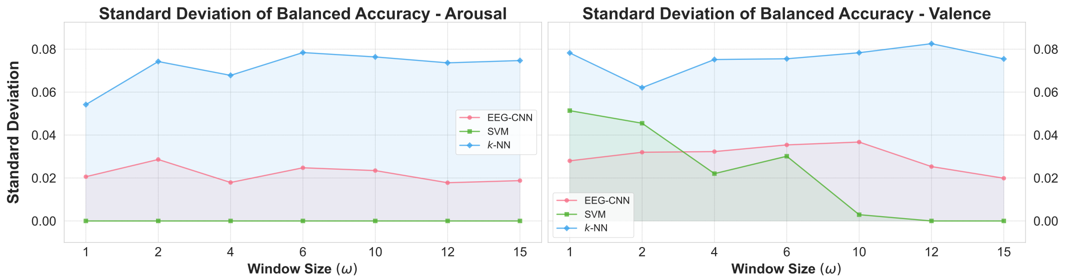

Figure 4.8: *Average standard deviation of balanced accuracy per window size for the DEAP dataset using the LOSO protocol.*

be drawn as with the DEAP dataset. For the SVM there is also complete overfitting towards the majority

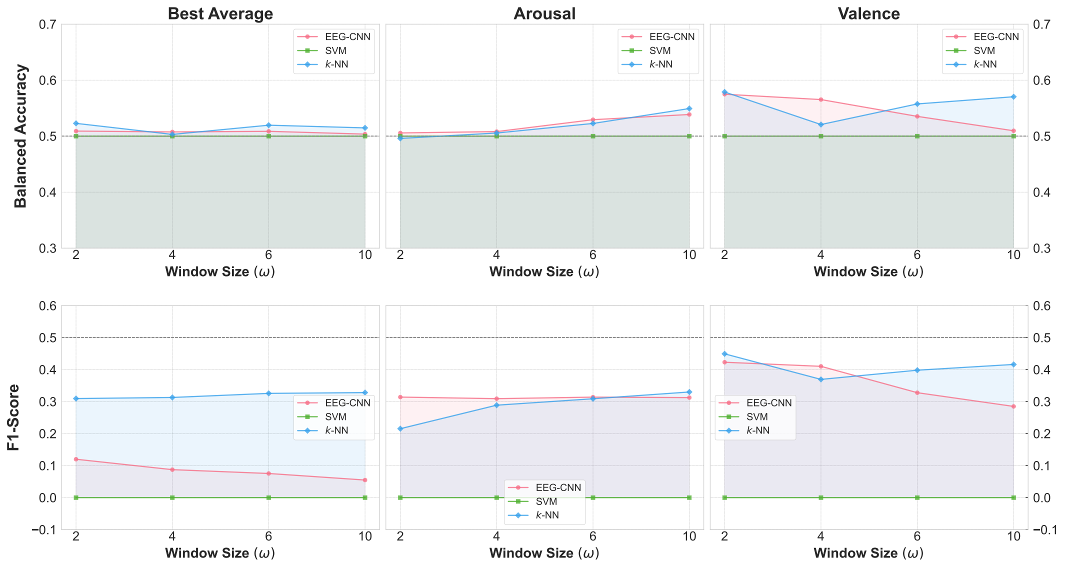

Figure 4.9: *Best results per window size for the MAHNOB dataset using the LOSO protocol, including top balanced accuracy, F1 scores for each category, and average performance across both.*

class, while for the other models the F1-Score is slightly higher, indicating that the model does not only predict one class. Although the balanced accuracy is still relatively low, reaching a maximum value of 0.549 for Arousal and k-NN and 0.575 for valence and EEG-CNNN, so the model is not performing well across classes.

It is worth mentioning that certain strategies, such as SMOTE (Synthetic Minority Over-sampling Technique), oversampling (repeating samples with more extreme labels before binarization), and under-sampling (removing samples with labels closer to neutral emotion), were tested to address the issues caused by the slight class imbalance in the SVM. However, these methods did not improve the obtained results.

In summary, we can conclude that even with the consistency solutions applied, the models are still unable to effectively solve the inter-subject prediction problem. They cannot generalize well enough to accurately predict emotions for new subjects not included in the training set. In the next section, we will analyze the results in more detail based on the consistency solutions used, where we will discuss whether they have had any impact in the performance, and we will suggest potential strategies to address this issue in the conclusions.

4.3.3. LOVO Results

Having analyzed the results for the LOSO protocol, in this subsection we analyze the results for LOVO. This protocol allows us to solve the inter-subject recognition problems, due to the fact that the training set contains data from all the subjects, and to evaluate how the models behave with the data corresponding to new videos seen by the subjects. The LOVO results can be found in detail in the appendix, including accuracy, F1-Scores and balanced accuracy for each case, in Tables A.15 to A.19.

In the case of LOVO protocol, if we observe Figure 4.10 and Figure 4.11, unlike the k-Fold, in this case it cannot be concluded that there is a trend indicating that the larger the time window sizes the better the performance of the models. However, part of the best results are obtained for large time windows, both for arousal and valence.

For the DEAP dataset, the best result for Arousal is achieved by the k-NN result with a 15 second time window, achieving up to 0.618 of balanced accuracy. Valence classification achieve its best performance with the EEG-CNN model with 12 second time window and a 0.589 of balanced accuracy.

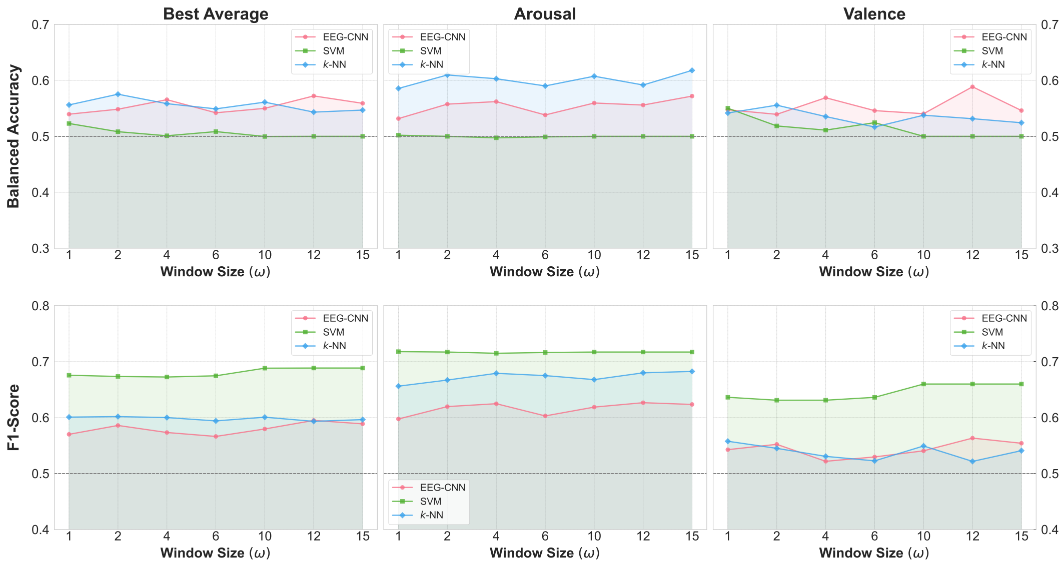

Figure 4.10: *Best results per window size for the DEAP dataset using the LOVO protocol, including top balanced accuracy, F1 scores for each category, and average performance across both.*

MAHNOB dataset, see Figure 4.11, achieves slightly higher results in terms of Balanced Accuracy, with 0.615 of balanced accuracy for arousal classification and the EEG-CNN with a 10 seconds time window and 0.608 of balanced accuracy for valence classification and the k-NN model with 2 seconds time window. However, we can see how the SVM has a stagnant prediction for practically all cases and a null F1-Score. This indicates that predicting only one of the classes in each video has a minimum value of 0.529 for Arousal and 0.594 for Valence. Therefore, the prediction achieved by all models for valence category is very low.

In summary, the results indicate that the models do not generalize well when using all the data from a video for testing. Including information from all subjects in the training set is not sufficient, highlighting that the real challenge lies not in intra-subject prediction but in intra-subject video prediction. Additionally, the best average results across categories are very low, suggesting that there is no specific configuration (model-consistency-window size) that performs consistently well in both categories.

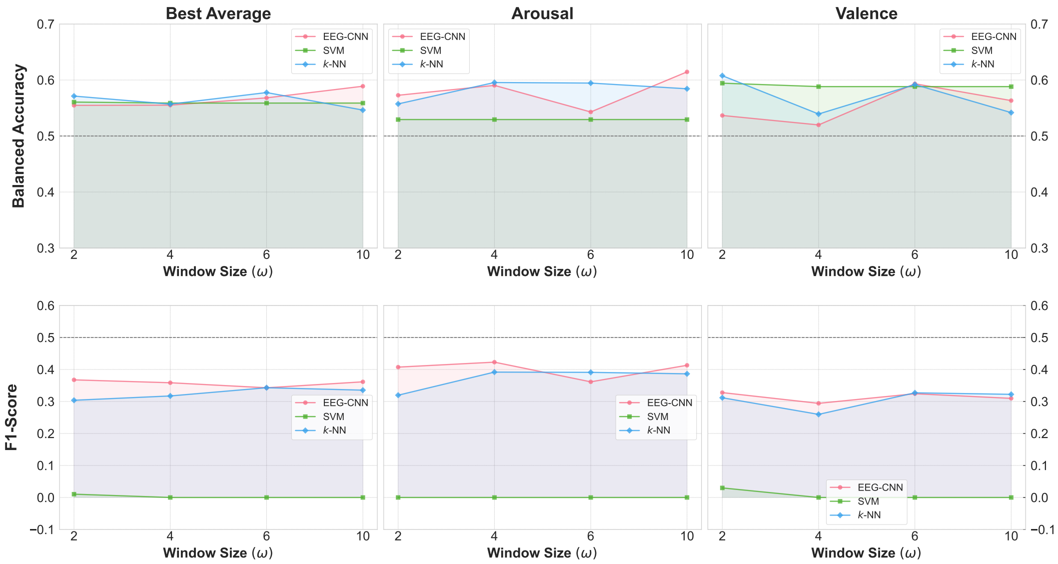

Figure 4.11: *Best results per window size for the MAHNOB dataset using the LOVO protocol, including top balanced accuracy, F1 scores for each category, and average performance across both.*

4.3.4. SDM Results

One of the objectives of the Subject-Dependent Model (SDM) protocol is to look at intra-subject prediction, on which algorithms have started to achieve high performance in EEG based emotion recognition [1]. In this subsection we will see if our algorithms, with their different configurations (model-consistency-window size) achieve higher performance, being consistent with the existing scientific literature. Detailed results,including accuracy, F1-Scores and balanced accuracy for each case, can be found in the appendix, in Tables A.20 to A.25.

Firstly, it is necessary to clarify that in this case the balanced accuracy represents the same as the accuracy, because in each test set there is only one sample associated to a video and therefore only one possible label. Furthermore, the $F1$-score is not analyzed because it would only indicate the pressence of true positives.

In general, the results are slightly better than k-Fold protocol, where more than one subject are in the test and train set. For DEAP dataset, see Figure 4.12, k-NN achieves the best result for Valence classification with 0.703 of balanced accuracy with a window size of 1 second. On the other hand, best arousal classification result is achieved by the SVM which achieves up to 0.753 with a window size of 6 seconds.

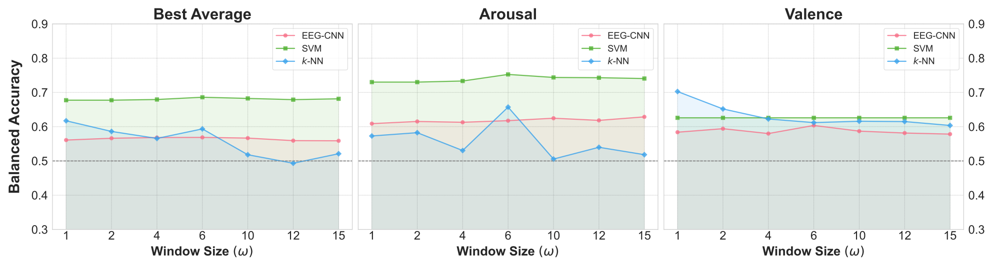

Figure 4.12: *Best results per window size for the DEAP dataset using the SDM protocol, including top balanced accuracy for each category, and average performance across both.*

On the other hand, best MAHNOB results, see Figure 4.13, are achieved by k-NN model, which achieves up to 0.747 of accuracy in arousal classification and 0.661, with a window size of 2 seconds and the same consistency configuration.

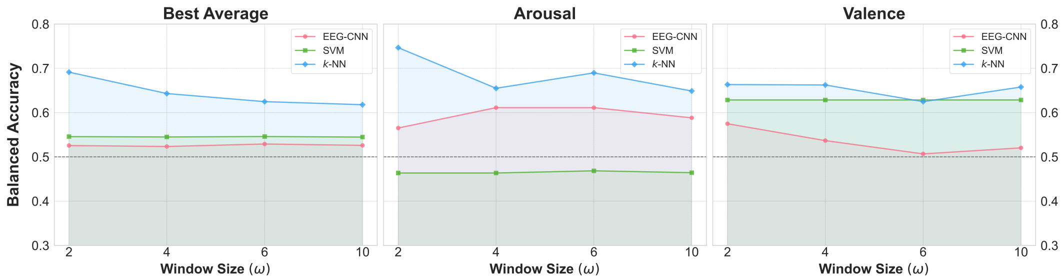

Figure 4.13: *Best results per window size for the MAHNOB dataset using the SDM protocol, including top balanced accuracy for each category, and average performance across both.*

In general, we can conclude that the intra-subject results are slightly better than in the other protocols where more subjects are involved in each training session. However, the results do not reach the very high accuracies achieved by the scientific community [1, 16]. Therefore, some improvements such as incorporating feature selection methods [12, 13] may increase considerably the performance of the models. Furthermore, SVM and k-NN reach the best results in this case, while EEG-CNN got outperformed in all cases, therefore adding hyper-parameter optimization for this specific problem may increase considerably the performance. Since, the parameters of the model are those obtained by the k-Fold in which all subjects are involved, so the parameters are not optimized for a specific subject.

4.3.5. Per-Band Analysis

The aim of this subsection is to assess the effect on EEG recognition of selecting a single EEG band. This will help us to determine whether feature selection or more accurate feature extraction is necessary to improve the performance of the proposed models. For this purpose, in this subsection we present the results obtained per band for the k-Fold method. In this case, the protocol has been applied as explained in the previous chapter, for SVM and k-NN. However, for the EEG-CNN, we have considered the selected hyperparameters for all bands and performed the stratified sampling: 90% training set and 10% test without performing the k-Fold. This is due to the lack of computational power and therefore the very high time required for this.

Firstly, we will analyze the overall results by window size and datasets, which can be viewed in Figure 4.14.

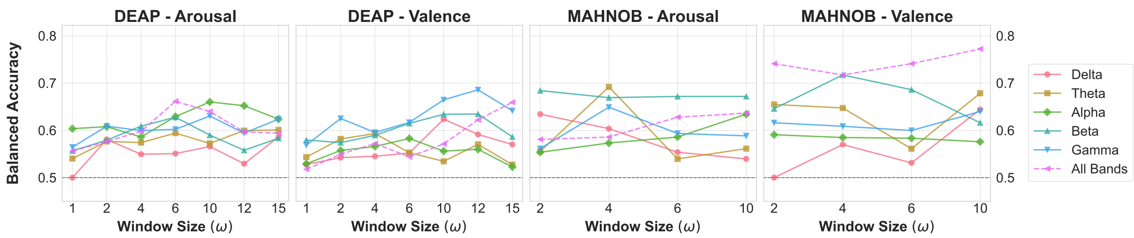

Figure 4.14: *Best results per window size and EEG band for DEAP and MAHNOB dataset.*

Overall, the "all-band" configuration achieves the highest balanced accuracy for the DEAP dataset in the Arousal classification task using a 6-second window, and for the MAHNOB dataset in the Valence classification task using a 10-second window. However, for the remaining two scenarios, the best performance is reached with different configurations: the Gamma band for the DEAP dataset in the Valence task with a 12-second window, and the Theta band for the MAHNOB dataset using a 4-second window.

Looking at the results by model, see Figure 4.15, some additional conclusions can be drawn. For the SVM model, the selection of a specific band, only provides a small improvement in prediction for the MAHNOB dataset with a time window of 2 seconds, while in the rest of the classification tasks, the all bands configuration achieve a better result. Unlike the SVM, EEG-CNN increase its performance in arousal classification and DEAP dataset, and Arousal and Valence for the MAHNOB datasets, achieving significantly better results. Additionally, the k-NN slightly improve for valence classification for DEAP dataset, considering the Gamma band and improve significantly in arousal classification of the MAHNOB dataset with the beta band.

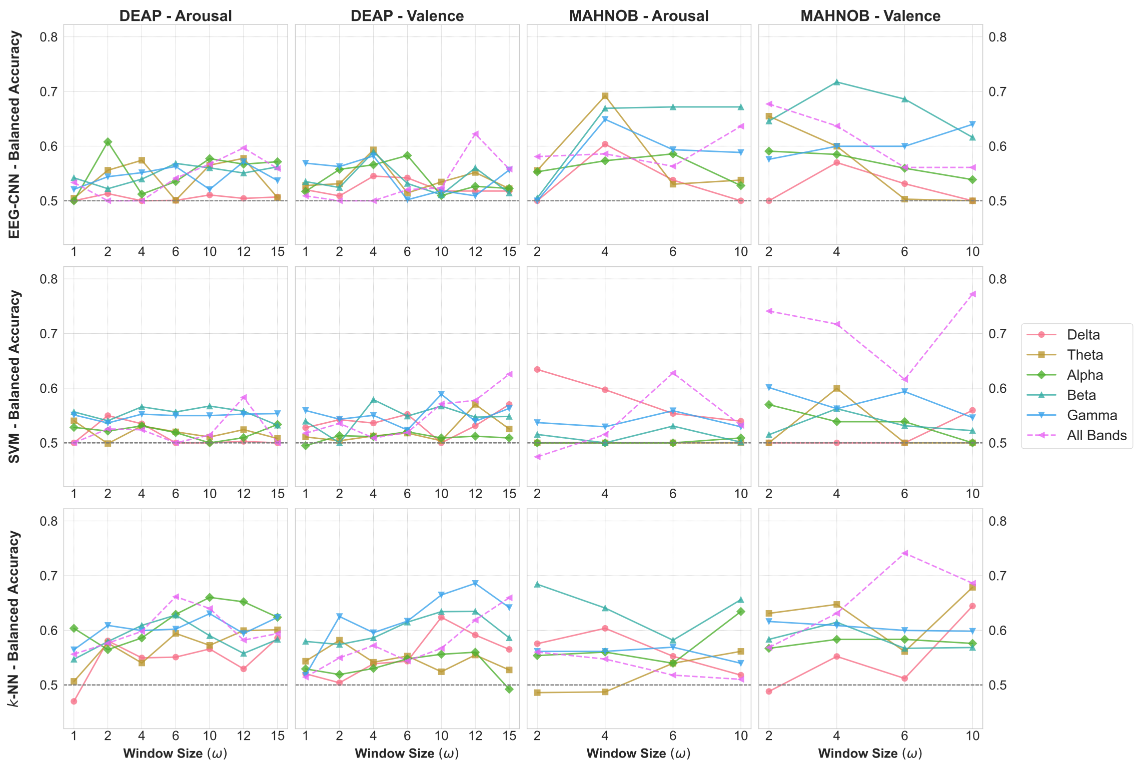

Figure 4.15: *Best results per window size, EEG band and model for DEAP and MAHNOB dataset.*

In summary, we can observe that the models often benefit from selecting specific frequency bands, although each model performs differently depending on the band chosen. This suggests that accurately reducing dimensionality could significantly improve the models' ability to predict new data. Therefore, combining our models with feature selection methods that operate not only at the feature level but also across EEG bands and channels, along with the consistency solutions calculated based on these, could substantially enhance the performance of these algorithms.

4.4. Temporal Affective Consistency

This section will analyze the results (classification and analysis) related to the proposed consistency solutions. First, we will examine the impact of these solutions on EEG-based emotion recognition. Next, we will compare the different consistency solutions to identify whether there is agreement on which time segments produce the strongest emotional response across subjects. Finally, we will conduct a subjective analysis using video samples from the various datasets.

4.4.1. Impact of Consistency Solutions on Emotion Recognition

To analyze the impact of the consistency solutions, the best results of the models, obtained for the different consistency solutions, will be analyzed. Only the balanced accuracy will be considered because it provides a more comprehensive measure of raw performance. For this purpose, K-Fold, LOSO and SMD protocols are considered because of its capability to incorporate that solutions. Moreover, it should be noted that, as in the results per band, the EEG-CNN in k-Fold we have considered the selected hyperparameters for default configuration (learned consistency) and then is performed the stratified sampling: 90% training set and 10% test without performing the k-Fold. This is due to the lack of computational power and therefore the very high time required for this.

The first thing we can observe for the K-Fold protocol and the different datasets, see Figure 4.16 and Figure 4.17, is that feature vector weighting can significantly impact prediction performance. In fact, we see variations of 10 to 15 percentage points depending on the type of consistency used. For the DEAP dataset, each model shows distinct results. For the SVM, the data-aware consistency, EEG-Glimpse, provides a clear improvement for Arousal prediction, while the learning-based solution performs best for Valence prediction. Both solutions also achieve the best balance between the two classifications. On the other hand, the learning-based approach provides the best classification results for valence and arousal classification for the k-NN model. Finally, for the EEG-CNN model, the learning-based solution performs best for Valence, while surprisingly, the heuristic-based "Last" solution performs best for Arousal. However, the best overall balance between Arousal and Valence classification is achieved with the learning-bases configuration.

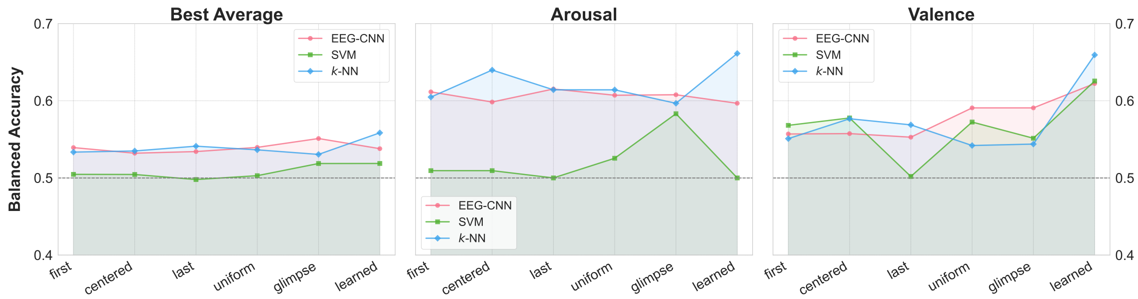

Figure 4.16: *Best balanced accuracies grouped by consistency solution across models for DEAP dataset and k-Fold protocol.*

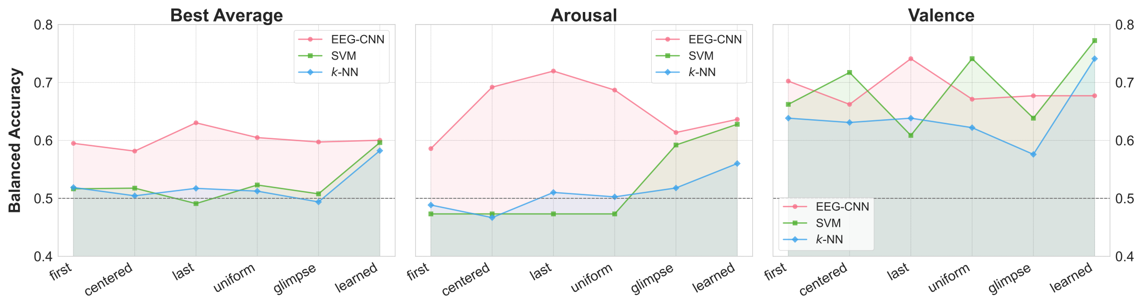

Figure 4.17: *Best balanced accuracies grouped by consistency solution across models for MAHNOB dataset and k-Fold protocol.*

Regarding MAHNOB dataset, results are similar, see Figure 4.17. EEG-CNN shows better results in all cases using the heuristic "Last" solution, similar to the previous scenario. k-NN also performs best with the learning-based solution, while SVM achieves its highest performance in all cases using the learning-based solution as well. It also reaches its second-highest performance with the data-aware EEG-Glimpse solution, as observed for the DEAP dataset.

The results for the LOSO protocol are much less significant, because the results are much poorer due to the very limited hability of the models to generalize to new subjects. However, a common behavior can be observed across models and datasets, see Figure 4.18 and Figure 4.19. Overall, the learning-based solution achieves better results in the vast majority of cases, except for SVM in the Valence classification, where the best result is obtained using the heuristic-based "Last" solution, although the difference is only 0.006, which is negligible. Similarly, for k-NN in the Valence classification, the best result is reached with the heuristic-based “First” solution, but with only a 0.009 difference compared to the learning-based solution. Therefore, despite the small improvements in classification, the consistency-weighted approach using learning-based consistency slightly improves the uniform weighing and thus the generalizability of the models to new subjects.

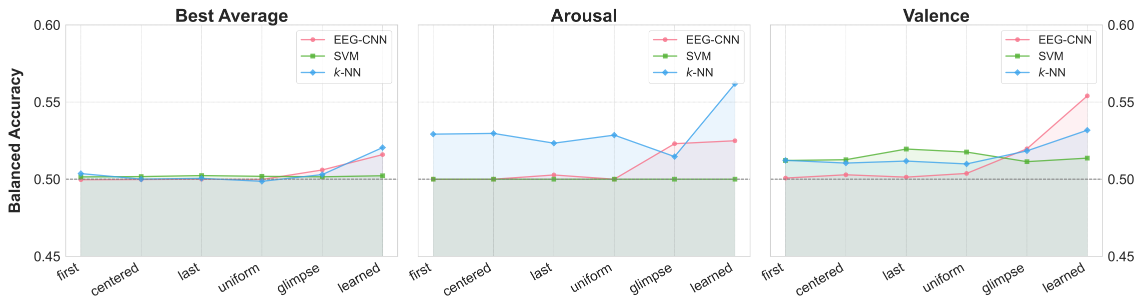

Figure 4.18: *Best balanced accuracies grouped by consistency solution across models for DEAP dataset and LOSO protocol.*

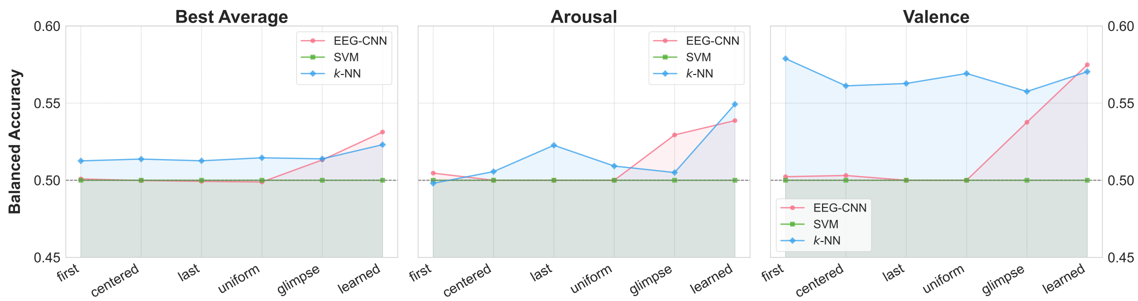

Figure 4.19: *Best balanced accuracies grouped by consistency solution across models for MAHNOB dataset and LOSO protocol.*

Finally, the SDM results grouped by consistency, despite having better results, are much more stable across consistencies, see Figure 4.20 and Figure 4.21.

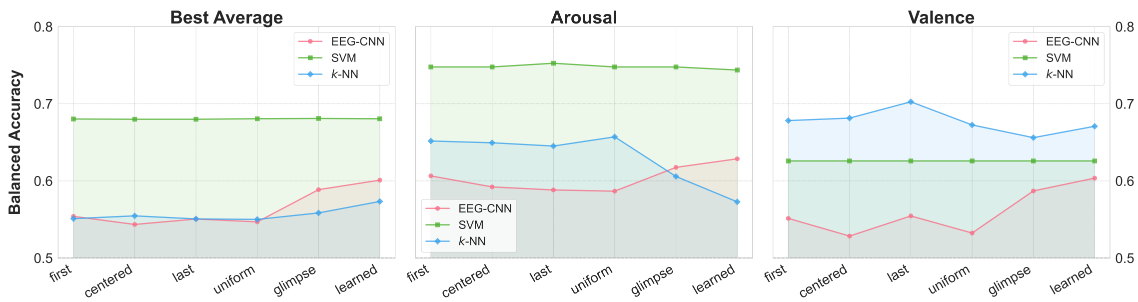

Figure 4.20: *Best balanced accuracies grouped by consistency solution across models for DEAP dataset and SDM protocol.*

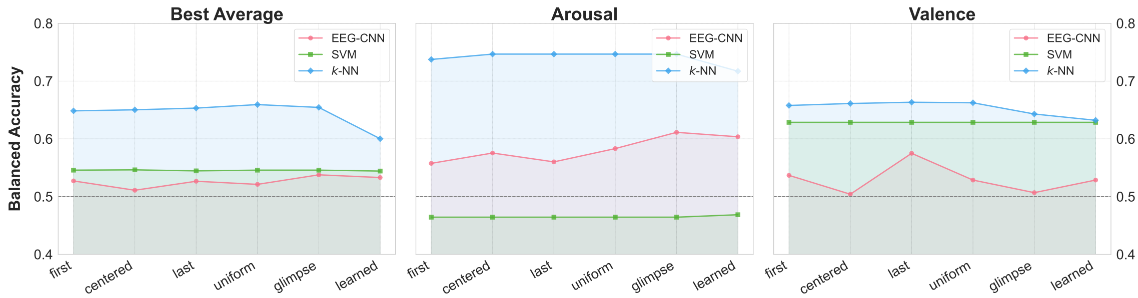

Figure 4.21: *Best balanced accuracies grouped by consistency solution across models for MAHNOB dataset and SDM protocol.*

Therefore, obtaining for the SVM model almost identical results across all consistency solutions. On the other hand, k-NN exhibits slightly more variability, with heuristic approaches outperforming the data-aware and learning-based methods. For EEG-CNN, the learning-based approach delivers the best results for the DEAP dataset, with a noticeable margin. In contrast, for the MAHNOB dataset, the best results are achieved with the data-aware EEG-Glimpse solution for Arousal classification and the heuristic-based "Last" solution for Valence.

In short, considering all the results collectively, we can conclude that giving more importance to some time segments than others can significantly affect the performance of our classification algorithms. In general, it was observed that in all cases, one or another of the proposed solutions outperformed the uniform weighing method that is typically employed by the scientific community. This may suggests that some time segments may be more important for prediction than others. Moreover, EEG-Glimpse and Learning-Based scores have obtained the best results in most of the cases, especially the learning-based one, even slightly improving the generalization to new subjects. This could indicate an effective capture of the common temporal emotional response by both solutions, and thus suggesting that they could be an effective score of the temporal affective consistency.

4.4.2. Comparison of Consistency Solutions

In this subsection we will qualitatively compare the consistency solutions obtained by the data-aware or learning-based approach. In order to identify whether there is agreement on which time segments produce the strongest emotional response across subjects. It is important to note that both scores are on different scales, even after applying softmax to both. Therefore, they are normalized to facilitate comparison. It is also worth mentioning that this does not alter the outcome since our goal is to capture the emotional response qualitatively rather than quantitatively. In other words, the focus is on identifying patterns and peaks, rather than on absolute values or dimensions.

We should also mention that this comparison will be performed across several time windows to see if a possible agreement is observed throughout different time windows. Finally, it is important to note that the learning-based score depends on the specific classification task, and therefore, for each video, we have two separate scores: one for Valence and one for Arousal.

The first comparison between data-aware EEG-Glimpse and learning-based consistency solution is made with DEAP video with index 7 in the dataset, see Figure 4.22. First, we can observe that the learning-based scores behave somewhat more erratically than EEG-Glimpse, generating more peaks. Nevertheless, we can also see that the trend for the learning-based scores in Valence is notably consistent across both time windows, showing a pattern of rises and falls that align in most time segments. Even when the specific segments do not match perfectly, the overall trend remains consistent. In the case of learning-

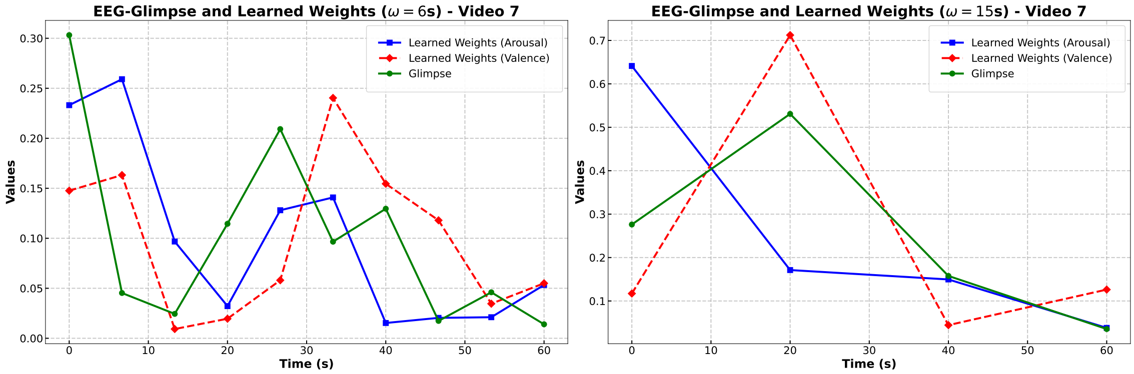

Figure 4.22: *Comparison between EEG-Glimpse and Learning-Based scores obtained for the video with id = 7 in DEAP dataset.*

based for Arousal, this agreement is lost for the 15-second time window, however for the 6-second time window, this agreement can also be observed.

Secondly the video with index 9 is employed to compare the data-aware EEG glimpse and learning-based consistency scores, see Figure 4.23 In this case, we can drawn the same conclusions as in the previous

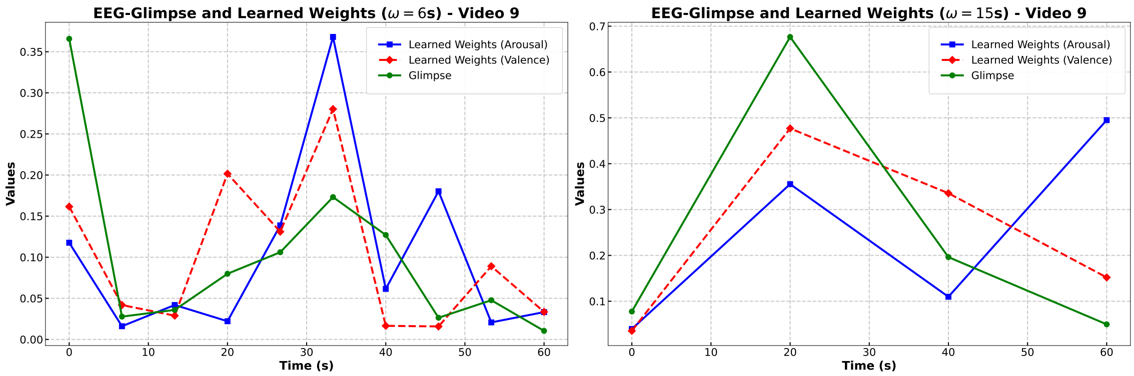

Figure 4.23: *Comparison between EEG-Glimpse and Learning-Based scores obtained for the video with id = 9 in DEAP dataset.*

case. Once again, the behaviour is more erratic in the case of the 6-second and learning-based time window, but coincides with EEG-Glimpse in general trends. While in the case of the 15-second time window, the learning-based Arousal score differs significantly in the last segment. However, the Valence score follows the same pattern as EEG-Glimpse across all segments, although with a slightly greater dynamic range.

Finally, the video with index 36 is analyzed, Figure 4.24. This video was selected because in the next section is chosen to verify whether the consistency solutions accurately capture the emotional response triggered by a specific event in the video. In this case, we again observe a high level of agreement between the learning-based Valence and EEG-Glimpse for the 6-second time window. We can see that the trend matches in almost all time segments except for the 24-second mark, where the learning-based score shows a steep downward peak. The Arousal score, on the other hand, differs more significantly from both but still captures a similar emotional response in some segments. However, for the 6-second time window, it is the Arousal score that shows much greater agreement with EEG-Glimpse, unlike the previous cases where it was the learning-based Valence score. This discrepancy could be due to the neural network's poorer performance for that particular video, resulting in a less accurate capture of the emotional response across subjects.

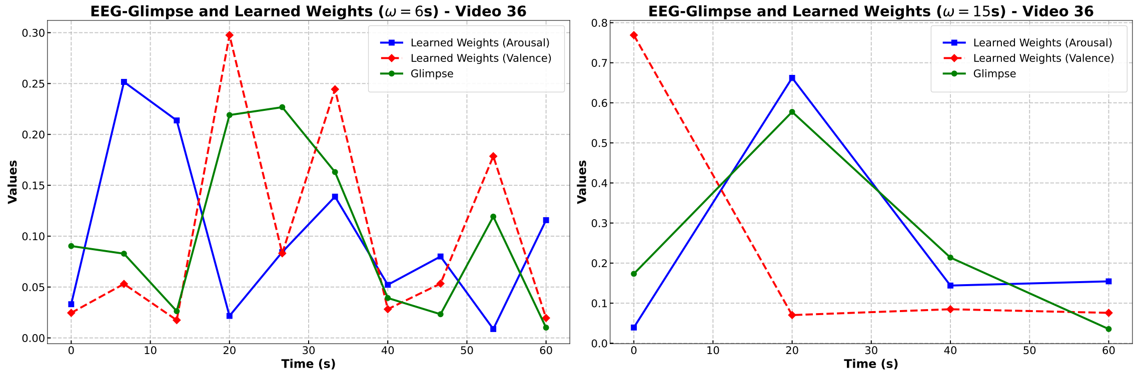

Figure 4.24: *Comparison between EEG-Glimpse and Learning-Based scores obtained for the video with id = 36 in DEAP dataset.*

Overall, there is a noticeable level of agreement, especially between the EEG-Glimpse and the learning-based Valence scores, which also demonstrates better prediction results for the EEG-CNN compared to Arousal scores for DEAP dataset. Suggesting that better predictions lead to better capture of affective temporal consistency. This agreement, together with the small improvements produced by the feature weighting of EEG-Glimpse and Learning-Based features produced in many cases, over the heuristic approaches, may indicate that these scores are indeed succeeding in capturing emotional response across subjects. Therefore, if these scores were calculated based on more relevant features, it would likely result in a much better capture of temporal affective consistency and a greater advantage over traditional uniform weighting. For this reason, refining the algorithms used through the application of feature selection will improve the performance of the model and more accurately identify the segments of the videos that elicit the strongest emotional responses.

4.4.3. Video-Based Subjective Analysis

In this subsection, we watched some videos from the dataset to test whether the consistency solutions that capture an emotional response across subjects match some relevant events in the video that might explain those brain responses. We must stress the difficulty of this analysis, due to its subjectivity, and therefore we will simply try to analyze whether, when emotional responses are elicited, there is one event in the video that stands out above the rest.

Firstly, the first video analyzed is which corresponds to the id = 9 of DEAP dataset, where consistency scores can be observed at Figure 4.23. This video corresponds to the song I Want to Break Free of Queen. Firstly, as observed in many of the videos, the initial emotional response captured by the scores occurs at the beginning, which could be due to the first reaction triggered upon hearing the video—in this case, a song familiar to much of the audience. The second peak, noted in both the Learning-Based Valence score and the Glimpse score, aligns with the start of the music, while the next peak captured by Glimpse, which has almost the same value as the Learning-Based score, coincides with the start of the vocal melody. What it might suggest is that both scores are indeed capturing the common emotional response.

The other video analyzed corresponds to the Refuse/Resist song of Sepultura. In this video it is much more difficult to identify the importance of the segments obtained by the consistency solutions, see Figure 4.24, with the content of the video. However, the peak obtained by the Learning-based Valence and EEG-Glimpse corresponding to the end of the video coincides again with the beginning of the vocal melody. Moreover, the peaks obtained in the central moments of the video also coincide with the appearance of images related to protests or wars that are relatively shocking.

In conclusion, no additional videos were included due to the difficulty of rationalizing when emotionally evocative events occur. However, in the analyzed videos, we were able to observe that the emotional responses captured by the consistency solutions often align with key events in the video, such as the start of a familiar song, the appearance of impactful images or the start of vocal melodies, supporting the validity of the observed patterns.

5. Conclusions and Future Work

This thesis focused on addressing the challenge of temporal affective consistency analysis in EEG-based emotion recognition, which refers to the stable and coherent association between emotional states and corresponding physiological response patterns over time. Using the DEAP and MAHNOB datasets, this research introduced novel methods for analyzing and quantifying affective consistency, trying to provide a new perspective on understanding when emotional responses are reflected in brain signals during repeated or prolonged exposure to stimuli. In addition, these quantification methods were used to try to improve EEG-based emotion recognition and address the challenge of intersubject variability in emotion recognition algorithms.

5.1. Key Aspects of the Work

To achieve these objectives, this thesis introduced and evaluated different methods of quantifying temporal affective consistency aimed at addressing the three research questions posed in chapter 1, and repeated here for convenience:

RQ1: How can we develop temporal affective consistency scores that reliably quantify the stable relationship between emotional responses of multiple participants/subjects/observers and EEG signals over time?

RQ2: Can consistency-weighted approaches outperform traditional uniform weighting schemes in emotion recognition performance?

RQ3: Can consistency-based scores be leveraged to improve the generalizability of emotion recognition algorithms and reduce the impact of inter-subject variability?

In order to answer **RQ1**, we developed and compared three main consistency scores mechanisms: Heuristic-Based (based on the assumption that certain video segments elicit stronger emotional responses), Data-Aware (based on the assumption that affective time consistency can be measured directly from the data), and Learning-Based (a model learns how to best temporally weight the brain features). To verify if these solutions truly capture emotional responses, a combination of quantitative (impact on emotion recognition) and qualitative analysis was conducted.

In order to answer **RQ2** and **RQ3**, the study aimed to evaluate if consistency-weighted EEG features could outperform traditional uniform weighting schemes. Thus, the objective was to assess whether certain time segments in the EEG signals were better predictors of emotional states by capturing more substantial emotional responses, and therefore, improve the algorithm's classification performance. For this purpose, several testing protocols were developed: K-Fold Cross Validation with stratified sampling (pure performance of Emotion Recognition), Leave-One-Video-Out (LOVO) for emotion recognition on a new video viewed by subjects, Leave-One-Subject-Out (LOSO) for generalizing to a new subject, and Subject-Dependent Model (SDM) for intra-subject predictions. These protocols were paired with machine learning algorithms, including SVM, k-NN, and a custom EEG Context Neural Network.

5.2. Main Findings

The main findings of this MSc thesis can be summarised as follows:

Data-aware and Learning-based scores could reliably quantify temporal affective consistency. The qualitative analysis shows a remarkable agreement in the emotional capture of both scores:

EEG-Glimpse and Learning-Based, especially in valence scores, where the EEG-Neural Network achieved the best results. This finding may indicate that better predictions could lead to a more accurate capture of affective temporal consistency, and that EEG-Glimpse can effectively capture common emotional responses without relying on available ground-truth labels. Moreover, when analyzing certain videos from the DEAP dataset, the peaks in these consistency scores (when both are in agreement) coincided with key emotional moments, such as starting of familiar songs, the appearance of a striking image or the beginning of vocal melody, indicating that both methods may be effectively capturing common emotional responses across subjects.

Consistency-weighted approaches outperform traditional weighting. Although the emotion recognition performance was not higher than most state-of-the-art works, the research demonstrated that weighting based on consistency scores significantly enhances binary valence/arousal classification performance. Overall, it was observed that in all cases, some of the consistency-based solutions outperformed traditional uniform weighting. In particular, the learning-based consistency score achieved better results in most scenarios, especially with the k-Fold protocol. This suggests that this approach more effectively captures the temporal emotional response. While EEG-Glimpse also outperformed uniform weighting in many cases, the learning-based approach generally delivered superior results. This might seem contradictory to the qualitative analysis, where both scores were normalized and placed on the same scale. However, EEG-Glimpse operates within a smaller dynamic range, meaning it assigns less variation in importance to different segments, which could affect its performance. Additionally, heuristic-based approaches, such as “Last” or “First”, performed well in specific situations but did not consistently achieve the same performance across all models and datasets. This suggests that while heuristic solutions can provide a baseline understanding of temporal affective consistency, more advanced learning-based methods are needed for effective generalization across different subjects and stimuli.

Consistency-weighted slightly improves generalization to new subjects. The experimental results showed that emotion recognition within the same subject remained consistent across different consistency approaches. However, when generalizing to new subjects (using the LOSO protocol), the impact of consistency became slightly more noticeable. Although the overall performance was limited and the differences were small, learning-based solutions and some heuristic approaches performed better when applied to new subjects.

5.3. Limitations and Future Directions

One of the main limitations of this study is the ***suboptimal performance*** of the models, which could be attributed to several factors. Firstly, the absence of ***feature selection***; our models were trained using all available features, whereas analyses focusing on specific frequency bands have shown improved results in some cases. By incorporating more refined feature selection techniques at the level of features, frequency bands, and EEG channels, we could achieve better performance and more reliably assess the influence of weighting contributions through consistency scores. For example, methods like Recursive Feature Elimination (RFE), Mutual Information (MI), and L1-norm-based feature selection have been shown to enhance performance in EEG emotion recognition tasks [11]. These techniques systematically select the most informative features, thereby reducing noise and computational complexity, which may lead to more robust model performance.

Another limitation is the ***lack of analysis of the influence of weights during testing***, such as using SHAP (SHapley Additive exPlanations) [53], or examining the correlation between high weight values and specific emotional states using statistical tests like the Mann-Whitney U test or Spearman’s rank correlation coefficient. This analysis is particularly applicable in the learning-based approach, where, without applying a softmax function to the weights, they can assume negative or positive values depending on their association with different classes, allowing for a more exhaustive analysis. However, this is not feasible with EEG-Glimpse since it only captures the magnitude of the emotional response and not its specific type.

Moreover, some ***limitations of EEG-Glimpse*** arise from its reliance on the nature of the extracted EEG features. Since EEG-Glimpse aims to capture temporal affective consistency directly from these features using a distance-based approach, inconsistencies can significantly affect its performance. Therefore, it would be advisable to thoroughly evaluate these features and perhaps select ones of the same nature. By focusing on homogeneous feature sets—such as exclusively time-domain or frequency-domain features, the model might better detect common emotional responses and enhance its capacity to capture temporal affective consistency. Moreover, although the objective of our consistency-based solutions is to quantify the common emotional response (temporal affective consistency), the learning-based approach depends on the protocol and might capture the emotional response of only a single subject in a subject-dependent model. In contrast, EEG-Glimpse focuses purely on consistency, requiring data from multiple subjects, which gives us less flexibility in contexts with only one subject.

Finally, we propose new methods to capture temporal affective consistency while improving prediction simultaneously, such as the use of transformers with self-attention mechanisms. Some approaches have been made in this direction, but they focus on spatial information [54]. Combining both temporal and spatial approaches could enhance emotional recognition, as well as allow us to quantify temporal affective consistency more effectively.

5.4. Implications and Possible Applications

If the results and findings are more thoroughly and precisely confirmed, they open the door to numerous applications and implications:

Human-Computer Interaction Temporal affective consistency scores can enhance emotional understanding in systems like personal assistants and virtual reality, enabling more emotionally adaptive interactions.

Mental Health and Neurofeedback: Consistent emotional monitoring offers possibilities for personalized neurofeedback, thus allowing physiological research in psychological therapies, enabling exploration of emotional responses before and after therapy, as well as observation of the effects of specific psychological medications on emotional responses.

Affective Video Annotation Refined consistency scores could automate affective tagging of videos by identifying key emotional moments based on EEG signals, making content categorization more efficient and personalized for viewers.

Education and Learning Emotion-aware systems can adjust teaching methods based on emotional engagement, making learning environments more responsive to students' needs.

5.5. Final Conclusion

This thesis successfully tackled the challenge of temporal affective consistency analysis in EEG-based emotion recognition, proposing, evaluating and comparing novel methods for quantifying and utilizing temporal emotional patterns. The research demonstrated that consistency-weighted approaches outperform traditional uniform weighting, particularly through learning-based methods. Despite limitations such as model performance and the need for refined feature selection, the study provides a foundation for future improvements in emotion recognition systems, particularly in handling intersubject variability. With further advancements, these contributions might open doors for more adaptive and personalized applications in areas like HCI, mental health, entertainment, and education.

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A. Appendix

This appendix provides supplementary information to support the main findings and methodology presented in this study. It includes additional data analysis, methodological details, and a more comprehensive set of results.

A.1. Methodology Details

In this section we will show additional details pertaining to the methodology.

As commented in the Methodology chapter, the optimizer of the EEG-CNN was applied with an split technique, where the weights of the contextual layer are updated following the standard SGD, while those of the dense layers are updated incorporating momentum and weight decay. The Code A.1 shows the Python implementation.

```python
optimizer = torch.optim.SGD([
                        {'params': network.w, 'lr': learning_rate},
                        {'params': [param for name, param in network.named_parameters() if name != 'w'],
                         'lr': rate, 'momentum': momentum, 'weight_decay': weight}
                        ])
```

Listing A.1: *Split technique in PyTorch SGD optimizer*

Algorithm 1 shows the simplified version of the Stochastic Gradient Descent algorithm with momentum, learning rate, and weight decay, implemented in the EEG-CNN with PyTorch and its torch.optim.SGD class:

Algorithm 1 Simplified Stochastic Gradient Descent (SGD)

Require: Initial parameters w_0, learning rate η, momentum coefficient μ, weight decay coefficient λ

Initialize momentum buffer $m_0 = 0$

for $t = 0$ to $T - 1 \ldots$ **do**

Sample a minibatch of data $\mathcal{B}_t$

Compute gradient of the loss:

$$g_t = \nabla_w L(w_t; \mathcal{B}_t)$$

if $\lambda \neq 0$ **then**

Apply weight decay:

$$g_t \leftarrow g_t + \lambda w_t$$

end if

if $\mu \neq 0$ **then**

Update momentum buffer:

$$m_t \leftarrow \mu m_{t-1} + g_t$$

Update parameters:

$$w_{t+1} \leftarrow w_t - \eta m_t$$

else

Update parameters without momentum:

$$w_{t+1} \leftarrow w_t - \eta g_t$$

end if

end for

A.2. Experimental Work

In this section we will show more detailed information related to the experimental work: more detailed descriptions (code) of the algorithms, why the hyperparameters were chosen, details of the procedures and extended results description.

Figure A.1 shows the proportion of High, Low, and Neutral Arousal and Valence categories across the DEAP and MAHNOB dataset.

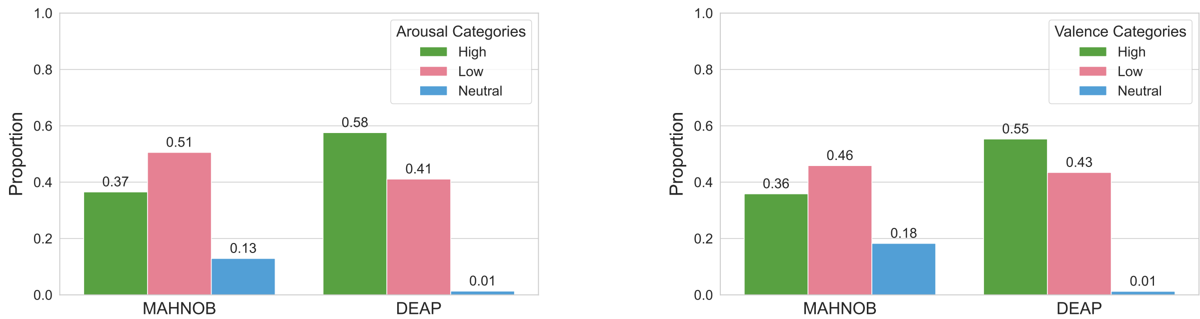

Figure A.1: *Proportion of High, Low, and Neutral Arousal and Valence categories across DEAP and MAHNOB datasets.*

As explained in the Results and Methodology chapters, the hyperparameters were chosen after applying Bayesian optimization using the K-fold cross Validation protocol explained above. In this way, to simplify the process for the other protocols (LOSO, LOVO and SDM), the same configuration was chosen for each one, based on the most repeated hyperparameter configuration throughout the k-Fold validation process.

For the Support Vector Machine model, Figure A.2 shows the distribution of the hyperparameters over all the models. However, the most common configurations obtained differ depending on whether the model classifies the arousal or valence class, see Table A.1.

Table A.1: *Most common SVM configurations provided by Bayesian Optimization with k-Fold Cross Validation.*

Class	Kernel	C	Gamma	Count
Valence	rbf	100	1	10
	rbf	100	0.01	9
	rbf	100	10	7
	rbf	1	100	3
	rbf	10	0.01	3
Arousal	rbf	10	100	7
	rbf	10	10	4
	rbf	100	100	4
	rbf	100	0.1	4
	rbf	1	100	3

In the case of the k-Nearest Neighbours, Table A.2 shows a summary of the different configurations obtained grouped according to valence or arousal class.

On the other hand, the EEG Context Neural Network is optimized following the same method as the SVM. In this case, we look at the most common number of neurons per each layer configuration and the most common parameters, after optimization across all models.

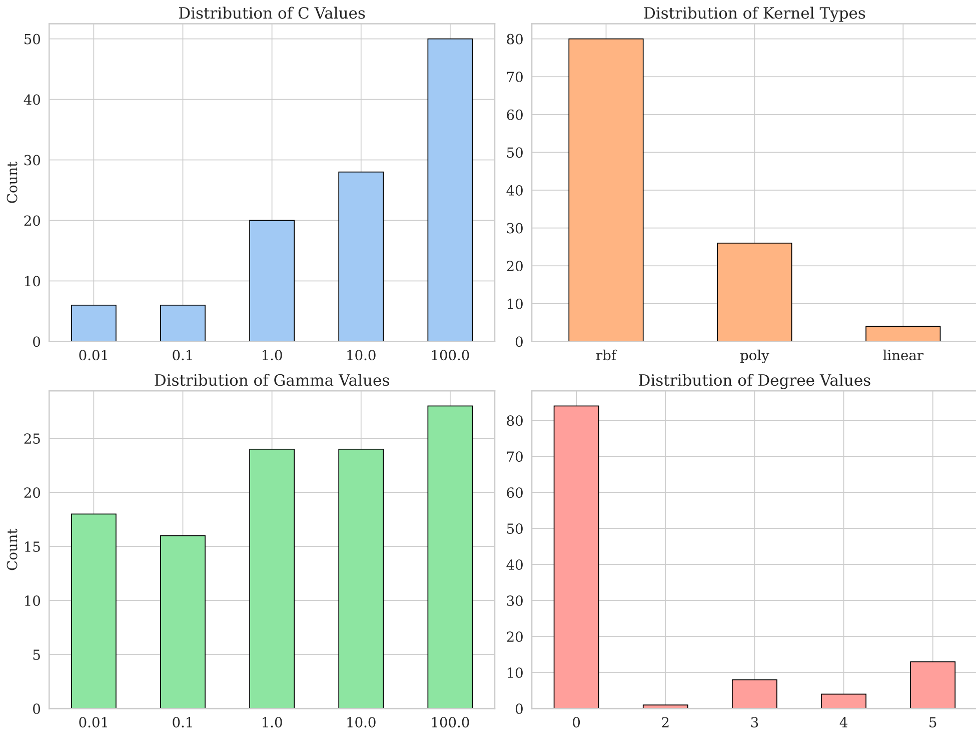

Figure A.2: *Hyperparameter distribution of the Support Vector Machine over the models, after applying Bayesian Optimization with k-Fold Cross Validation.*

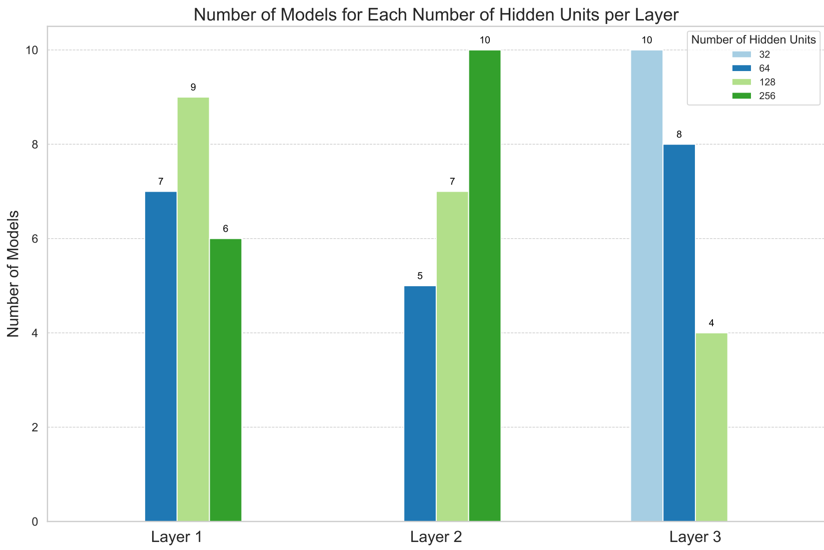

Figure A.3: *Distribution of the number of hidden units per layer over the models, after applying Bayesian Optimization with K-fold Cross Validation.*

However, although in Figure A.3 we can see that the most common number of neurons for the first layer is 128, the most common architecture has 256, 256 and 32 numbers of neurons, respectively. However, this combination is only repeated 4 times, which indicates the great variety of structures provided by the optimization, see Table A.3.

Regarding the rest of the hyperparameters, Figure A.4 shows their distribution over the models.

Table A.2: *Most common K-NN configurations for the number of neighbors, and distance metrics provided by Bayesian Optimization with k-Fold Cross Validation.*

Class	Num. Neighbours	Weight	Metric	Count
Valence	4	distance	chebyshev	6
	7	distance	euclidean	5
	6	distance	chebyshev	4
	1	uniform	euclidea	4
	7	distance	chebyshev	4
Arousal	7	distance	euclidean	9
	5	distance	chebyshev	6
	9	distance	chebyshev	6
	8	distance	chebyshev	6
	7	distance	chebyshev	4

Table A.3: *Most common architectures of number of hidden units per layer provided by the Bayesian Optimizacion with K-Fold Cross Validation.*

Configuration	Count
(256, 256, 32)	4
(128, 128, 128)	2
(128, 64, 64)	2
(128, 256, 32)	2
(256, 256, 64)	1

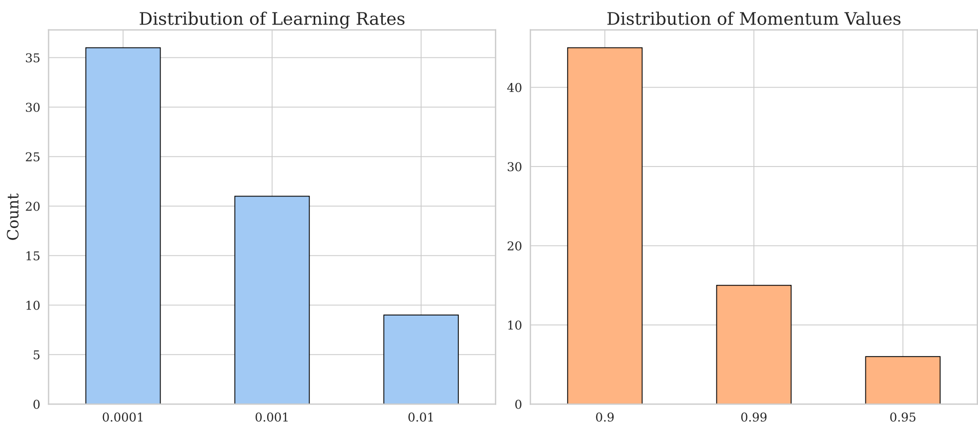

Figure A.4: *Hyperparameter distribution of the EEG Context Neural Network over the models, after applying Bayesian Optimization with K-Fold Cross Validation.*

A.3. Detailed Results

Finally, in this section we include a detailed description of all the results obtained, including supplementary tables that provide further information on the results of the study.

A.3.1. K-Fold Cross Validation

In this subsection are included the detailed results of the protocol K-Fold Cross Validation: Table A.8, Table A.6, Table A.7, Table A.4 and Table A.5. These tables compile the Balanced Accuracy, Accuracy and F1-Score results for the different models, datasets, window sizes and consistency solutions. In each table, the best results obtained for the Balanced Accuracy, both for Valence classification and Arousal Classification, are marked in bold and large type.

Table A.4: *Performance results of the K-fold Cross Validation protocol using different consistency solutions on DEAP dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - Support Vector Machine (SVM)							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.583	0.736	0.500	0.465	0.500	0.463
	2	0.583	0.736	0.500	0.559	0.714	0.502
	4	0.591	0.740	0.509	0.567	0.721	0.509
	6	0.583	0.736	0.500	0.559	0.714	0.502
	10	0.583	0.736	0.500	0.559	0.717	0.500
	12	0.583	0.736	0.500	0.583	0.624	0.578
	15	0.583	0.736	0.500	0.559	0.717	0.500
Uniform	1	0.488	0.564	0.472	0.543	0.678	0.501
	2	0.591	0.723	0.525	0.520	0.667	0.474
	4	0.583	0.736	0.500	0.543	0.691	0.493
	6	0.583	0.736	0.500	0.559	0.714	0.502
	10	0.583	0.736	0.500	0.559	0.717	0.500
	12	0.583	0.736	0.500	0.606	0.709	0.572
	15	0.583	0.736	0.500	0.559	0.717	0.500
First	1	0.583	0.736	0.500	0.520	0.555	0.518
	2	0.583	0.736	0.500	0.520	0.667	0.474
	4	0.591	0.740	0.509	0.543	0.691	0.493
	6	0.583	0.736	0.500	0.559	0.714	0.502
	10	0.583	0.736	0.500	0.559	0.717	0.500
	12	0.583	0.736	0.500	0.614	0.735	0.568
	15	0.583	0.736	0.500	0.559	0.717	0.500
Last	1	0.583	0.736	0.500	0.543	0.678	0.501
	2	0.583	0.736	0.500	0.520	0.667	0.474
	4	0.583	0.736	0.500	0.543	0.691	0.493
	6	0.583	0.736	0.500	0.559	0.714	0.502
	10	0.583	0.736	0.500	0.559	0.717	0.500
	12	0.583	0.736	0.500	0.559	0.717	0.500
	15	0.583	0.736	0.500	0.559	0.717	0.500
EEG-Glimpse	1	0.583	0.736	0.500	0.496	0.508	0.500
	2	0.591	0.740	0.509	0.535	0.563	0.535
	4	0.567	0.678	0.524	0.504	0.526	0.505
	6	0.583	0.736	0.500	0.528	0.583	0.519
	10	0.528	0.595	0.514	0.551	0.578	0.551
	12	0.614	0.699	**0.583**	0.528	0.583	0.519
	15	0.583	0.736	0.500	0.559	0.717	0.500
Learned	1	0.583	0.736	0.500	0.535	0.670	0.494
	2	0.575	0.730	0.493	0.543	0.691	0.493
	4	0.583	0.736	0.500	0.567	0.721	0.509
	6	0.583	0.736	0.500	0.559	0.717	0.500
	10	0.583	0.736	0.500	0.567	0.580	0.571
	12	0.583	0.736	0.500	0.583	0.629	0.576
	15	0.583	0.736	0.500	0.630	0.667	**0.626**

Table A.5: *Performance results of the K-fold Cross Validation protocol using different consistency solutions on MAHNOB dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - Support Vector Machine (SVM)							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.425	0.303	0.408	0.730	0.667	0.717
	4	0.500	0.091	0.442	0.595	0.400	0.561
	6	0.500	0.091	0.442	0.595	0.348	0.554
	10	0.500	0.333	0.473	0.541	0.514	0.543
Uniform	2	0.450	0.312	0.430	0.757	0.690	0.741
	4	0.500	0.091	0.442	0.568	0.385	0.537
	6	0.400	0.400	0.409	0.595	0.348	0.554
	10	0.500	0.333	0.473	0.595	0.571	0.598
First	2	0.450	0.312	0.430	0.676	0.600	0.662
	4	0.500	0.091	0.442	0.595	0.400	0.561
	6	0.500	0.091	0.442	0.595	0.348	0.554
	10	0.500	0.333	0.473	0.568	0.529	0.567
Last	2	0.400	0.333	0.394	0.541	0.452	0.528
	4	0.500	0.091	0.442	0.568	0.000	0.500
	6	0.500	0.333	0.473	0.649	0.435	0.609
	10	0.475	0.323	0.451	0.568	0.333	0.530
EEG-Glimpse	2	0.400	0.294	0.386	0.649	0.581	0.638
	4	0.475	0.400	0.467	0.568	0.000	0.500
	6	0.575	0.585	0.592	0.568	0.000	0.500
	10	0.525	0.387	0.503	0.541	0.000	0.476
Learned	2	0.475	0.432	0.474	0.568	0.200	0.515
	4	0.575	0.190	0.515	0.730	0.667	0.717
	6	0.625	0.595	**0.628**	0.649	0.480	0.616
	10	0.550	0.437	0.532	0.784	0.733	**0.772**

Table A.6: *Performance results of the K-fold Cross Validation protocol using different consistency solutions on DEAP dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - k-Nearest Neighbors (k-NN)							
		Arousal			**Valence**		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.543	0.646	0.509	0.504	0.540	0.502
	2	0.567	0.675	0.527	0.520	0.555	0.518
	4	0.528	0.630	0.496	0.480	0.507	0.481
	6	0.583	0.644	0.570	0.512	0.569	0.503
	10	0.661	0.726	0.640	0.575	0.625	0.567
	12	0.575	0.682	0.533	0.598	0.679	0.577
	15	0.622	0.704	0.593	0.488	0.558	0.476
Uniform	1	0.543	0.646	0.509	0.512	0.530	0.514
	2	0.591	0.679	0.560	0.528	0.577	0.521
	4	0.591	0.644	0.582	0.551	0.607	0.542
	6	0.575	0.671	0.541	0.512	0.557	0.507
	10	0.638	0.709	0.614	0.543	0.597	0.535
	12	0.598	0.698	0.559	0.488	0.519	0.488
	15	0.614	0.688	0.591	0.457	0.524	0.446
First	1	0.559	0.650	0.531	0.504	0.533	0.504
	2	0.583	0.686	0.543	0.551	0.584	0.550
	4	0.622	0.711	0.587	0.559	0.611	0.551
	6	0.567	0.663	0.535	0.551	0.601	0.544
	10	0.630	0.704	0.605	0.559	0.616	0.549
	12	0.575	0.686	0.531	0.449	0.533	0.433
	15	0.575	0.654	0.552	0.465	0.528	0.455
Last	1	0.583	0.667	0.556	0.496	0.568	0.483
	2	0.567	0.667	0.532	0.528	0.571	0.523
	4	0.598	0.653	0.588	0.465	0.493	0.465
	6	0.606	0.695	0.574	0.504	0.553	0.498
	10	0.638	0.709	0.614	0.488	0.552	0.478
	12	0.622	0.718	0.582	0.575	0.620	0.569
	15	0.614	0.684	0.594	0.535	0.609	0.520
EEG-Glimpse	1	0.591	0.687	0.555	0.480	0.515	0.479
	2	0.606	0.691	0.576	0.520	0.555	0.518
	4	0.630	0.715	0.597	0.504	0.553	0.498
	6	0.528	0.589	0.517	0.520	0.596	0.504
	10	0.598	0.683	0.570	0.551	0.601	0.544
	12	0.598	0.687	0.567	0.496	0.590	0.476
	15	0.606	0.688	0.579	0.465	0.553	0.448
Learned	1	0.575	0.649	0.555	0.496	0.562	0.485
	2	0.543	0.633	0.517	0.504	0.559	0.496
	4	0.606	0.658	0.598	0.583	0.639	0.572
	6	0.677	0.732	**0.661**	0.528	0.589	0.517
	10	0.591	0.687	0.555	0.472	0.504	0.472
	12	0.591	0.679	0.560	0.622	0.657	0.619
	15	0.591	0.690	0.552	0.661	0.691	**0.659**

Table A.7: *Performance results of the K-fold Cross Validation protocol using different consistency solutions on MAHNOB dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - k-Nearest Neighbors (k-NN)							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.475	0.323	0.451	0.568	0.385	0.537
	4	0.400	0.200	0.371	0.649	0.552	0.631
	6	0.475	0.400	0.467	0.568	0.500	0.560
	10	0.475	0.364	0.459	0.568	0.500	0.560
Uniform	2	0.450	0.312	0.430	0.568	0.385	0.537
	4	0.450	0.353	0.437	0.622	0.533	0.607
	6	0.475	0.400	0.467	0.622	0.588	0.622
	10	0.525	0.387	0.503	0.514	0.400	0.497
First	2	0.500	0.375	0.481	0.541	0.370	0.513
	4	0.400	0.294	0.386	0.649	0.519	0.624
	6	0.450	0.389	0.445	0.568	0.556	0.574
	10	0.500	0.412	0.488	0.649	0.581	0.638
Last	2	0.525	0.424	0.510	0.568	0.500	0.560
	4	0.400	0.250	0.379	0.649	0.552	0.631
	6	0.475	0.364	0.459	0.486	0.424	0.481
	10	0.500	0.375	0.481	0.649	0.581	0.638
EEG-Glimpse	2	0.425	0.410	0.431	0.595	0.444	0.568
	4	0.475	0.364	0.459	0.568	0.467	0.552
	6	0.525	0.457	0.518	0.595	0.483	0.576
	10	0.375	0.242	0.357	0.514	0.357	0.490
Learned	2	0.600	0.385	**0.560**	0.541	0.485	0.536
	4	0.550	0.500	0.547	0.595	0.516	0.583
	6	0.525	0.345	0.495	0.757	0.690	**0.741**
	10	0.525	0.424	0.510	0.703	0.621	0.686

Table A.8: *Performance results of the K-fold Cross Validation protocol, with the EEG Context Neural Network (EEG-CNN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

EEG Context Neural Network (EEG-CNN)							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
DEAP	1	0.575	0.682	0.533	0.567	0.721	0.509
	2	0.583	0.736	0.500	0.559	0.717	0.500
	4	0.583	0.736	0.500	0.441	0.000	0.500
	6	0.575	0.671	0.541	0.535	0.609	0.520
	10	0.598	0.687	0.567	0.543	0.628	0.524
	12	0.630	0.715	**0.597**	0.622	0.647	**0.622**
	15	0.598	0.698	0.559	0.567	0.621	0.558
MAHNOB	2	0.600	0.467	0.581	0.676	0.647	**0.677**
	4	0.600	0.500	0.586	0.622	0.632	0.637
	6	0.575	0.485	0.563	0.595	0.400	0.561
	10	0.650	0.563	**0.636**	0.595	0.400	0.561

A.3.2. Leave-one-Subject-out

In this subsection are included the detailed results of the protocol LOSO: Table A.11, Table A.12, Table A.10, Table A.9, Table A.13 and Table A.14. These tables compile the Balanced Accuracy, Accuracy and F1-Score results for the different models, datasets, window sizes and consistency solutions. In each table, the best results obtained for the Balanced Accuracy, both for Valence classification and Arousal Classification, are marked in bold and large type.

Table A.9: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on DEAP dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - Support Vector Machine (SVM) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.569	0.691	0.504	0.553	0.687	0.513
	2	0.569	0.699	0.502	0.544	0.679	0.505
	4	0.574	0.702	0.505	0.542	0.686	0.500
	6	0.570	0.686	0.501	0.541	0.682	0.506
	10	0.583	0.723	0.500	0.560	0.714	0.500
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
Uniform	1	0.568	0.689	0.504	0.559	0.692	0.518
	2	0.569	0.699	0.502	0.544	0.680	0.506
	4	0.574	0.702	0.505	0.541	0.685	0.499
	6	0.567	0.684	0.499	0.539	0.680	0.503
	10	0.583	0.723	0.500	0.560	0.714	0.500
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
First	1	0.569	0.692	0.510	0.553	0.687	0.512
	2	0.565	0.697	0.498	0.542	0.678	0.504
	4	0.575	0.703	0.506	0.543	0.686	0.501
	6	0.571	0.687	0.502	0.541	0.684	0.504
	10	0.583	0.723	0.500	0.560	0.714	0.500
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
Last	1	0.575	0.699	0.509	0.560	0.694	**0.520**
	2	0.569	0.699	0.504	0.546	0.683	0.506
	4	0.573	0.701	0.504	0.541	0.684	0.499
	6	0.569	0.687	0.500	0.541	0.679	0.508
	10	0.583	0.723	0.500	0.560	0.714	0.500
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
EEG-Glimpse	1	0.570	0.692	0.508	0.553	0.687	0.511
	2	0.569	0.699	0.503	0.546	0.683	0.507
	4	0.574	0.703	0.505	0.542	0.685	0.501
	6	0.574	0.700	0.502	0.541	0.683	0.503
	10	0.583	0.723	0.500	0.560	0.714	0.500
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
Learned	1	0.576	0.700	0.508	0.557	0.690	0.514
	2	0.577	0.701	**0.513**	0.542	0.681	0.502
	4	0.568	0.701	0.500	0.547	0.687	0.508
	6	0.571	0.701	0.500	0.543	0.689	0.504
	10	0.576	0.716	0.497	0.559	0.708	0.505
	12	0.583	0.723	0.500	0.560	0.714	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500

Table A.10: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on MAHNOB dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - Support Vector Machine (SVM) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
Uniform	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
First	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
Last	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
EEG-Glimpse	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
Learned	2	0.580	0.000	0.500	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500

Table A.11: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on DEAP dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - k-Nearest Neighbors (k-NN) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.537	0.571	0.495	0.518	0.590	0.508
	2	0.529	0.561	0.500	0.518	0.577	0.510
	4	0.521	0.547	0.497	0.507	0.564	0.506
	6	0.522	0.523	0.497	0.478	0.515	0.474
	10	0.515	0.560	0.499	0.525	0.567	0.507
	12	0.528	0.642	0.494	0.488	0.532	0.480
	15	0.553	0.635	0.530	0.502	0.529	0.504
Uniform	1	0.543	0.577	0.502	0.515	0.583	0.506
	2	0.532	0.565	0.501	0.519	0.581	0.510
	4	0.531	0.556	0.506	0.505	0.554	0.506
	6	0.511	0.517	0.486	0.489	0.526	0.483
	10	0.507	0.544	0.491	0.509	0.556	0.495
	12	0.535	0.647	0.505	0.495	0.546	0.488
	15	0.545	0.627	0.529	0.477	0.519	0.474
First	1	0.548	0.583	0.503	0.498	0.564	0.495
	2	0.548	0.583	0.517	0.518	0.581	0.512
	4	0.525	0.544	0.502	0.504	0.559	0.504
	6	0.519	0.522	0.496	0.487	0.524	0.481
	10	0.509	0.556	0.492	0.510	0.562	0.497
	12	0.555	0.669	0.516	0.510	0.553	0.510
	15	0.553	0.641	0.529	0.500	0.548	0.496
Last	1	0.544	0.575	0.502	0.500	0.575	0.495
	2	0.531	0.564	0.502	0.509	0.570	0.501
	4	0.528	0.552	0.500	0.507	0.560	0.507
	6	0.519	0.532	0.496	0.497	0.543	0.488
	10	0.503	0.547	0.490	0.514	0.557	0.495
	12	0.522	0.627	0.490	0.523	0.571	0.512
	15	0.533	0.599	0.523	0.507	0.539	0.508
EEG-Glimpse	1	0.544	0.579	0.507	0.522	0.598	0.513
	2	0.526	0.556	0.497	0.510	0.571	0.503
	4	0.536	0.559	0.506	0.510	0.563	0.509
	6	0.514	0.519	0.488	0.490	0.524	0.486
	10	0.523	0.568	0.511	0.507	0.553	0.492
	12	0.540	0.642	0.505	0.523	0.571	0.518
	15	0.540	0.629	0.515	0.499	0.549	0.493
Learned	1	0.540	0.566	0.502	0.521	0.593	0.515
	2	0.553	0.590	0.529	0.495	0.563	0.486
	4	0.517	0.538	0.496	0.532	0.591	**0.532**
	6	0.517	0.561	0.480	0.518	0.563	0.512
	10	0.570	0.633	0.545	0.529	0.571	0.525
	12	0.571	0.660	0.553	0.535	0.580	0.526
	15	0.577	0.658	**0.562**	0.529	0.591	0.523

Table A.12: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on MAHNOB dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - k-Nearest Neighbors (k-NN) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.484	0.199	0.480	0.570	0.427	0.561
	4	0.510	0.289	0.506	0.539	0.369	0.521
	6	0.502	0.269	0.504	0.537	0.355	0.519
	10	0.499	0.263	0.487	0.540	0.397	0.533
Uniform	2	0.491	0.198	0.483	0.579	0.429	0.569
	4	0.498	0.271	0.495	0.529	0.363	0.518
	6	0.506	0.282	0.509	0.541	0.363	0.528
	10	0.504	0.274	0.496	0.526	0.354	0.518
First	2	0.489	0.201	0.496	0.588	0.449	**0.579**
	4	0.492	0.263	0.490	0.514	0.358	0.505
	6	0.495	0.260	0.498	0.529	0.390	0.520
	10	0.500	0.272	0.488	0.539	0.366	0.524
Last	2	0.497	0.192	0.484	0.577	0.396	0.563
	4	0.499	0.256	0.494	0.530	0.360	0.513
	6	0.518	0.309	0.523	0.536	0.348	0.521
	10	0.498	0.267	0.490	0.518	0.380	0.514
EEG-Glimpse	2	0.487	0.215	0.487	0.569	0.416	0.557
	4	0.509	0.289	0.505	0.523	0.361	0.506
	6	0.473	0.271	0.492	0.562	0.398	0.558
	10	0.497	0.246	0.483	0.529	0.374	0.523
Learned	2	0.491	0.175	0.487	0.541	0.417	0.529
	4	0.498	0.237	0.494	0.500	0.340	0.491
	6	0.521	0.282	0.514	0.556	0.382	0.549
	10	0.543	0.330	**0.549**	0.580	0.416	0.570

Table A.13: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on DEAP dataset, with the EEG Context Neural Network (EEG-CNN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - EEG Context Neural Network (EEG-CNN) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.577	0.711	0.500	0.561	0.714	0.501
	2	0.583	0.723	0.500	0.553	0.702	0.496
	4	0.583	0.723	0.500	0.561	0.714	0.501
	6	0.583	0.723	0.500	0.552	0.695	0.497
	10	0.583	0.723	0.500	0.562	0.712	0.503
	12	0.583	0.723	0.500	0.554	0.693	0.500
	15	0.583	0.723	0.500	0.560	0.714	0.500
Uniform	1	0.583	0.723	0.500	0.558	0.711	0.498
	2	0.583	0.723	0.500	0.555	0.707	0.497
	4	0.579	0.718	0.499	0.553	0.698	0.498
	6	0.583	0.723	0.500	0.565	0.709	0.504
	10	0.583	0.723	0.500	0.557	0.710	0.498
	12	0.583	0.723	0.500	0.560	0.692	0.504
	15	0.583	0.723	0.500	0.555	0.691	0.501
First	1	0.583	0.723	0.500	0.560	0.714	0.500
	2	0.583	0.723	0.500	0.559	0.713	0.500
	4	0.583	0.723	0.500	0.553	0.699	0.497
	6	0.583	0.723	0.500	0.556	0.676	0.499
	10	0.583	0.723	0.500	0.555	0.697	0.500
	12	0.581	0.722	0.499	0.561	0.714	0.501
	15	0.583	0.723	0.500	0.554	0.691	0.500
Last	1	0.583	0.723	0.500	0.559	0.711	0.500
	2	0.583	0.721	0.503	0.555	0.707	0.498
	4	0.583	0.723	0.500	0.560	0.707	0.501
	6	0.579	0.711	0.502	0.554	0.693	0.499
	10	0.583	0.723	0.500	0.556	0.704	0.498
	12	0.583	0.723	0.500	0.561	0.692	0.500
	15	0.581	0.722	0.499	0.560	0.714	0.500
EEG-Glimpse	1	0.528	0.602	0.491	0.516	0.503	0.520
	2	0.532	0.582	0.504	0.526	0.584	0.502
	4	0.520	0.578	0.508	0.540	0.599	0.518
	6	0.540	0.623	0.523	0.540	0.578	0.512
	10	0.547	0.632	0.518	0.554	0.656	0.508
	12	0.548	0.642	0.504	0.537	0.652	0.497
	15	0.543	0.675	0.483	0.544	0.675	0.495
Learned	1	0.507	0.560	0.481	0.526	0.581	0.514
	2	0.518	0.570	0.502	0.527	0.575	0.513
	4	0.530	0.588	0.514	0.533	0.581	0.523
	6	0.520	0.553	0.506	0.505	0.540	0.494
	10	0.553	0.600	**0.525**	0.531	0.578	0.536
	12	0.526	0.562	0.508	0.547	0.585	0.534
	15	0.557	0.581	0.519	0.563	0.596	**0.554**

Table A.14: *Performance results of the Leave-One-Subject-Out (LOSO) protocol using different consistency solutions on MAHNOB dataset, with the EEG Context Neural Network (EEG-CNN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - EEG Context Neural Network (EEG-CNN) - LOSO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.580	0.000	0.500	0.557	0.000	0.496
	4	0.580	0.000	0.500	0.564	0.011	0.503
	6	0.580	0.000	0.500	0.560	0.009	0.499
	10	0.580	0.000	0.500	0.560	0.000	0.500
Uniform	2	0.577	0.012	0.499	0.557	0.009	0.496
	4	0.577	0.000	0.498	0.560	0.000	0.500
	6	0.577	0.000	0.498	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.019	0.499
First	2	0.583	0.015	0.505	0.560	0.000	0.500
	4	0.580	0.000	0.500	0.563	0.008	0.502
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
Last	2	0.577	0.000	0.498	0.557	0.000	0.496
	4	0.580	0.000	0.500	0.560	0.000	0.500
	6	0.580	0.000	0.500	0.560	0.000	0.500
	10	0.580	0.000	0.500	0.560	0.000	0.500
EEG-Glimpse	2	0.524	0.314	0.506	0.537	0.357	0.538
	4	0.535	0.149	0.500	0.534	0.161	0.514
	6	0.573	0.113	0.529	0.559	0.141	0.523
	10	0.571	0.027	0.502	0.540	0.017	0.494
Learned	2	0.526	0.309	0.500	0.571	0.423	**0.575**
	4	0.507	0.309	0.508	0.571	0.410	0.565
	6	0.500	0.314	0.518	0.530	0.328	0.535
	10	0.553	0.313	**0.539**	0.520	0.285	0.510

A.3.3. Leave-one-Video-out

In this subsection are included the detailed results of the protocol LOVO: Table A.15, Table A.16, Table A.17, Table A.18 and Table A.19. These tables compile the Balanced Accuracy, Accuracy and F1-Score results for the different models, datasets, window sizes and consistency solutions. In each table, the best results obtained for the Balanced Accuracy, both for Valence classification and Arousal Classification, are marked in bold and large type.

Table A.15: *Performance results of the Leave-One-Video-Out (LOVO) protocol on DEAP and MAHNOB dataset, with the EEG Context Neural Network (EEG-CNN) model.*

DEAP - EEG Context Neural Network (EEG-CNN) - LOVO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
DEAP	1	0.541	0.598	0.532	0.538	0.543	0.548
	2	0.566	0.620	0.558	0.547	0.552	0.539
	4	0.571	0.625	0.562	0.528	0.522	0.569
	6	0.551	0.603	0.538	0.531	0.530	0.546
	10	0.568	0.619	0.560	0.546	0.541	0.540
	12	0.580	0.627	0.556	0.568	0.563	**0.589**
	15	0.581	0.624	**0.572**	0.561	0.554	0.546
MAHNOB	2	0.561	0.407	0.573	0.538	0.328	0.537
	4	0.571	0.423	0.590	0.538	0.294	0.520
	6	0.563	0.361	0.543	0.545	0.324	**0.593**
	10	0.610	0.413	**0.615**	0.509	0.310	0.563

Table A.16: *Performance results of the Leave-One-Video-Out (LOVO) protocol using different consistency solutions on DEAP dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - Support Vector Machine (SVM) - LOVO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.586	0.718	0.502	0.580	0.633	0.544
	2	0.584	0.717	0.500	0.567	0.629	0.516
	4	0.582	0.715	0.497	0.553	0.629	0.503
	6	0.582	0.715	0.497	0.565	0.634	0.519
	10	0.584	0.717	0.500	0.560	0.660	0.500
	12	0.584	0.717	0.500	0.560	0.660	0.500
	15	0.584	0.717	0.500	0.560	0.660	0.500
Uniform	1	0.586	0.718	0.502	0.583	0.636	0.550
	2	0.584	0.717	0.500	0.567	0.629	0.516
	4	0.582	0.715	0.497	0.554	0.631	0.504
	6	0.582	0.715	0.497	0.563	0.634	0.519
	10	0.584	0.717	0.500	0.560	0.660	0.500
	12	0.584	0.717	0.500	0.560	0.660	0.500
	15	0.584	0.717	0.500	0.560	0.660	0.500
First	1	0.586	0.718	0.502	0.570	0.629	0.535
	2	0.584	0.717	0.500	0.570	0.631	0.518
	4	0.581	0.714	0.496	0.554	0.630	0.504
	6	0.580	0.713	0.495	0.563	0.633	0.514
	10	0.584	0.717	0.500	0.560	0.660	0.500
	12	0.584	0.717	0.500	0.560	0.660	0.500
	15	0.584	0.717	0.500	0.560	0.660	0.500
Last	1	0.584	0.717	0.500	0.579	0.634	0.543
	2	0.584	0.717	0.500	0.569	0.630	0.515
	4	0.582	0.715	0.497	0.556	0.631	0.511
	6	0.583	0.716	0.499	0.566	0.634	0.521
	10	0.584	0.717	0.500	0.557	0.658	0.498
	12	0.584	0.717	0.500	0.560	0.660	0.500
	15	0.584	0.717	0.500	0.560	0.660	0.500
EEG-Glimpse	1	0.586	0.718	0.502	0.584	0.636	**0.550**
	2	0.584	0.717	0.500	0.570	0.631	0.519
	4	0.581	0.714	0.496	0.553	0.631	0.503
	6	0.583	0.716	0.499	0.566	0.636	0.524
	10	0.584	0.717	0.500	0.559	0.660	0.500
	12	0.584	0.717	0.500	0.560	0.660	0.500
	15	0.584	0.717	0.500	0.560	0.660	0.500

Table A.17: *Performance results of the Leave-One-Video-Out (LOVO) protocol using different consistency solutions on MAHNOB dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - Support Vector Machine (SVM) - LOVO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.575	0.000	0.529	0.569	0.025	0.593
	4	0.575	0.000	0.529	0.564	0.000	0.588
	6	0.575	0.000	0.529	0.564	0.000	0.588
	10	0.575	0.000	0.529	0.564	0.000	0.588
Uniform	2	0.575	0.000	0.529	0.569	0.025	0.593
	4	0.575	0.000	0.529	0.564	0.000	0.588
	6	0.575	0.000	0.529	0.564	0.000	0.588
	10	0.575	0.000	0.529	0.564	0.000	0.588
First	2	0.575	0.000	0.529	0.561	0.000	0.586
	4	0.575	0.000	0.529	0.564	0.000	0.588
	6	0.575	0.000	0.529	0.564	0.000	0.588
	10	0.575	0.000	0.529	0.564	0.000	0.588
Last	2	0.575	0.000	0.529	0.571	0.030	**0.594**
	4	0.575	0.000	0.529	0.564	0.000	0.588
	6	0.575	0.000	0.529	0.564	0.000	0.588
	10	0.575	0.000	0.529	0.564	0.000	0.588
EEG-Glimpse	2	0.575	0.000	0.529	0.569	0.025	0.593
	4	0.575	0.000	0.529	0.564	0.000	0.588
	6	0.575	0.000	0.529	0.564	0.000	0.588
	10	0.575	0.000	0.529	0.564	0.000	0.588

Table A.18: *Performance results of the Leave-One-Video-Out (LOVO) protocol using different consistency solutions on DEAP dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - k-Nearest Neighbors (k-NN) - LOVO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.599	0.655	0.582	0.562	0.547	0.531
	2	0.611	0.662	0.602	0.566	0.542	**0.548**
	4	0.612	0.671	0.591	0.537	0.526	0.530
	6	0.610	0.673	0.588	0.533	0.517	0.508
	10	0.613	0.665	0.600	0.539	0.545	0.538
	12	0.610	0.678	0.591	0.508	0.509	0.506
	15	0.612	0.673	0.592	0.508	0.515	0.479
Uniform	1	0.600	0.656	0.585	0.556	0.549	0.525
	2	0.613	0.664	0.603	0.567	0.541	0.547
	4	0.610	0.670	0.588	0.538	0.525	0.512
	6	0.601	0.665	0.578	0.538	0.522	0.513
	10	0.612	0.665	0.595	0.545	0.543	0.514
	12	0.609	0.678	0.592	0.501	0.501	0.483
	15	0.609	0.673	0.593	0.501	0.520	0.503
First	1	0.598	0.656	0.586	0.544	0.541	0.512
	2	0.614	0.664	0.604	0.566	0.538	0.546
	4	0.616	0.675	0.591	0.538	0.523	0.528
	6	0.614	0.675	0.590	0.537	0.523	0.517
	10	0.601	0.651	0.587	0.534	0.528	0.511
	12	0.598	0.675	0.563	0.517	0.512	0.502
	15	0.599	0.662	0.580	0.506	0.514	0.516
Last	1	0.592	0.644	0.579	0.559	0.552	0.542
	2	0.619	0.667	0.610	0.555	0.532	0.536
	4	0.613	0.673	0.592	0.540	0.528	0.514
	6	0.610	0.673	0.589	0.531	0.510	0.511
	10	0.614	0.668	0.599	0.544	0.549	0.528
	12	0.606	0.680	0.587	0.531	0.522	0.532
	15	0.624	0.683	**0.618**	0.523	0.530	0.501
EEG-Glimpse	1	0.597	0.650	0.582	0.568	0.558	0.537
	2	0.617	0.663	0.601	0.563	0.545	0.556
	4	0.621	0.679	0.603	0.538	0.531	0.535
	6	0.611	0.672	0.588	0.527	0.511	0.509
	10	0.612	0.661	0.607	0.527	0.532	0.532
	12	0.601	0.677	0.581	0.501	0.501	0.497
	15	0.588	0.654	0.563	0.532	0.541	0.524

Table A.19: *Performance results of the Leave-One-Video-Out (LOVO) protocol using different consistency solutions on MAHNOB dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - k-Nearest Neighbors (k-NN) - LOVO							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.528	0.310	0.557	0.548	0.307	0.590
	4	0.577	0.392	0.595	0.496	0.251	0.516
	6	0.566	0.373	0.588	0.499	0.297	0.552
	10	0.580	0.376	0.577	0.491	0.291	0.527
Uniform	2	0.523	0.306	0.554	0.543	0.311	0.590
	4	0.576	0.381	0.592	0.509	0.252	0.539
	6	0.570	0.374	0.593	0.537	0.320	0.582
	10	0.564	0.365	0.564	0.478	0.282	0.502
First	2	0.521	0.286	0.552	0.563	0.312	**0.608**
	4	0.570	0.365	0.563	0.503	0.260	0.535
	6	0.578	0.391	0.593	0.522	0.314	0.580
	10	0.570	0.386	0.584	0.509	0.299	0.542
Last	2	0.525	0.320	0.550	0.536	0.296	0.578
	4	0.578	0.392	**0.596**	0.511	0.256	0.527
	6	0.575	0.382	0.595	0.508	0.302	0.565
	10	0.567	0.379	0.564	0.501	0.322	0.524
EEG-Glimpse	2	0.523	0.299	0.553	0.536	0.292	0.581
	4	0.569	0.381	0.586	0.487	0.244	0.521
	6	0.549	0.350	0.537	0.539	0.327	0.592
	10	0.569	0.366	0.577	0.479	0.287	0.503

A.3.4. Subject-Dependent Model

In this subsection are included the detailed results of the protocol SDM: Table A.20, Table A.21, Table A.22, Table A.23, Table A.24 and Table A.25. These tables compile the Balanced Accuracy, Accuracy and F1-Score results for the different models, datasets, window sizes and consistency solutions. In each table, the best results obtained for the Balanced Accuracy, both for Valence classification and Arousal Classification, are marked in bold and large type.

Table A.20: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on DEAP dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - Support Vector Machine (SVM) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.729	0.341	0.729	0.626	0.626	0.626
	2	0.729	0.341	0.729	0.626	0.626	0.626
	4	0.733	0.341	0.733	0.626	0.626	0.626
	6	0.748	0.341	0.748	0.626	0.626	0.626
	10	0.733	0.341	0.733	0.626	0.626	0.626
	12	0.729	0.341	0.729	0.626	0.626	0.626
	15	0.738	0.341	0.738	0.626	0.626	0.626
Uniform	1	0.729	0.341	0.729	0.626	0.626	0.626
	2	0.729	0.341	0.729	0.626	0.626	0.626
	4	0.733	0.341	0.733	0.626	0.626	0.626
	6	0.748	0.341	0.748	0.626	0.626	0.626
	10	0.738	0.341	0.738	0.626	0.626	0.626
	12	0.733	0.341	0.733	0.626	0.626	0.626
	15	0.738	0.341	0.738	0.626	0.626	0.626
First	1	0.729	0.341	0.729	0.626	0.626	0.626
	2	0.729	0.341	0.729	0.626	0.626	0.626
	4	0.733	0.341	0.733	0.626	0.626	0.626
	6	0.748	0.341	0.748	0.626	0.626	0.626
	10	0.743	0.341	0.743	0.626	0.626	0.626
	12	0.724	0.341	0.724	0.626	0.626	0.626
	15	0.738	0.341	0.738	0.626	0.626	0.626
Last	1	0.729	0.341	0.729	0.626	0.626	0.626
	2	0.729	0.341	0.729	0.626	0.626	0.626
	4	0.733	0.341	0.733	0.626	0.626	0.626
	6	0.753	0.341	**0.753**	0.626	0.626	**0.626**
	10	0.733	0.341	0.733	0.626	0.626	0.626
	12	0.724	0.341	0.724	0.626	0.626	0.626
	15	0.738	0.341	0.738	0.626	0.626	0.626
EEG-Glimpse	1	0.729	0.341	0.729	0.626	0.626	0.626
	2	0.729	0.341	0.729	0.626	0.626	0.626
	4	0.733	0.341	0.733	0.626	0.626	0.626
	6	0.748	0.341	0.748	0.626	0.626	0.626
	10	0.743	0.341	0.743	0.626	0.626	0.626
	12	0.743	0.341	0.743	0.626	0.626	0.626
	15	0.729	0.341	0.729	0.626	0.626	0.626
Learned	1	0.730	0.341	0.730	0.626	0.626	0.626
	2	0.730	0.341	0.730	0.626	0.626	0.626
	4	0.730	0.341	0.730	0.626	0.626	0.626
	6	0.733	0.341	0.733	0.626	0.626	0.626
	10	0.744	0.341	0.744	0.626	0.626	0.626
	12	0.740	0.341	0.740	0.626	0.626	0.626
	15	0.741	0.341	0.741	0.626	0.626	0.626

Table A.21: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on MAHNOB dataset, with the Support Vector Machine (SVM) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - Support Vector Machine (SVM) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.464	0.259	0.464	0.629	0.000	0.629
	4	0.464	0.259	0.464	0.629	0.000	0.629
	6	0.464	0.259	0.464	0.629	0.000	0.629
	10	0.464	0.259	0.464	0.629	0.000	0.629
Uniform	2	0.464	0.259	0.464	0.629	0.000	0.629
	4	0.460	0.259	0.460	0.629	0.000	0.629
	6	0.464	0.259	0.464	0.629	0.000	0.629
	10	0.464	0.259	0.464	0.629	0.000	0.629
First	2	0.464	0.259	0.464	0.629	0.000	0.629
	4	0.460	0.259	0.460	0.629	0.000	0.629
	6	0.464	0.259	0.464	0.629	0.000	0.629
	10	0.464	0.259	0.464	0.629	0.000	0.629
Last	2	0.460	0.259	0.460	0.629	0.000	0.629
	4	0.464	0.259	0.464	0.629	0.000	0.629
	6	0.454	0.259	0.454	0.629	0.000	0.629
	10	0.464	0.259	0.464	0.629	0.000	0.629
EEG-Glimpse	2	0.464	0.259	0.464	0.629	0.000	0.629
	4	0.460	0.259	0.460	0.629	0.000	0.629
	6	0.464	0.259	0.464	0.629	0.000	0.629
	10	0.464	0.259	0.464	0.629	0.000	0.629
Learned	2	0.464	0.259	0.464	0.629	0.000	0.629
	4	0.463	0.259	0.463	0.629	0.000	0.629
	6	0.469	0.265	**0.469**	0.629	0.000	**0.629**
	10	0.444	0.259	0.444	0.629	0.000	0.629

Table A.22: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on DEAP dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - k-Nearest Neighbors (k-NN) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.559	0.294	0.559	0.681	0.414	0.681
	2	0.582	0.276	0.582	0.589	0.326	0.589
	4	0.525	0.211	0.525	0.610	0.450	0.610
	6	0.649	0.355	0.649	0.548	0.374	0.548
	10	0.464	0.191	0.464	0.616	0.376	0.616
	12	0.420	0.151	0.420	0.504	0.339	0.504
	15	0.453	0.231	0.453	0.565	0.360	0.565
Uniform	1	0.559	0.294	0.559	0.673	0.405	0.673
	2	0.577	0.276	0.577	0.609	0.347	0.609
	4	0.525	0.211	0.525	0.613	0.446	0.613
	6	0.657	0.355	**0.657**	0.543	0.365	0.543
	10	0.447	0.173	0.447	0.579	0.365	0.579
	12	0.414	0.151	0.414	0.503	0.309	0.503
	15	0.462	0.231	0.462	0.539	0.356	0.539
First	1	0.565	0.297	0.565	0.678	0.431	0.678
	2	0.583	0.289	0.583	0.638	0.373	0.638
	4	0.516	0.216	0.516	0.612	0.445	0.612
	6	0.652	0.351	0.652	0.517	0.351	0.517
	10	0.457	0.183	0.457	0.551	0.386	0.551
	12	0.437	0.134	0.437	0.482	0.342	0.482
	15	0.451	0.220	0.451	0.575	0.347	0.575
Last	1	0.558	0.293	0.558	0.703	0.450	**0.703**
	2	0.560	0.271	0.560	0.589	0.374	0.589
	4	0.530	0.216	0.530	0.615	0.452	0.615
	6	0.645	0.343	0.645	0.535	0.340	0.535
	10	0.462	0.197	0.462	0.506	0.342	0.506
	12	0.442	0.173	0.442	0.513	0.365	0.513
	15	0.462	0.240	0.462	0.591	0.432	0.591
EEG-Glimpse	1	0.532	0.297	0.532	0.656	0.414	0.656
	2	0.560	0.280	0.560	0.552	0.333	0.552
	4	0.530	0.213	0.530	0.582	0.428	0.582
	6	0.606	0.282	0.606	0.612	0.414	0.612
	10	0.484	0.229	0.484	0.553	0.389	0.553
	12	0.540	0.220	0.540	0.550	0.384	0.550
	15	0.518	0.262	0.518	0.543	0.446	0.543
Learned	1	0.573	0.298	0.573	0.671	0.434	0.671
	2	0.543	0.263	0.543	0.652	0.433	0.652
	4	0.506	0.219	0.506	0.622	0.465	0.622
	6	0.567	0.294	0.567	0.590	0.421	0.590
	10	0.506	0.245	0.506	0.592	0.404	0.592
	12	0.499	0.233	0.499	0.615	0.439	0.615
	15	0.488	0.270	0.488	0.604	0.432	0.604

Table A.23: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on MAHNOB dataset, with the k-Nearest Neighbors (k-NN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - k-Nearest Neighbors (k-NN) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.747	0.355	**0.747**	0.661	0.089	0.661
	4	0.655	0.338	0.655	0.653	0.179	0.653
	6	0.632	0.349	0.632	0.575	0.098	0.575
	10	0.636	0.310	0.636	0.644	0.103	0.644
Uniform	2	0.747	0.355	0.747	0.651	0.089	0.651
	4	0.646	0.338	0.646	0.662	0.179	0.662
	6	0.690	0.398	0.690	0.614	0.095	0.614
	10	0.636	0.310	0.636	0.629	0.103	0.629
First	2	0.737	0.345	0.737	0.639	0.099	0.639
	4	0.641	0.333	0.641	0.653	0.169	0.653
	6	0.598	0.335	0.598	0.612	0.095	0.612
	10	0.649	0.324	0.649	0.658	0.114	0.658
Last	2	0.747	0.355	**0.747**	0.663	0.091	**0.663**
	4	0.655	0.338	0.655	0.620	0.179	0.620
	6	0.686	0.373	0.686	0.622	0.106	0.622
	10	0.648	0.323	0.648	0.586	0.064	0.586
EEG-Glimpse	2	0.747	0.355	**0.747**	0.631	0.091	0.631
	4	0.646	0.338	0.646	0.626	0.169	0.626
	6	0.683	0.391	0.683	0.625	0.085	0.625
	10	0.636	0.310	0.636	0.643	0.118	0.643
Learned	2	0.717	0.323	0.717	0.610	0.067	0.610
	4	0.628	0.307	0.628	0.632	0.111	0.632
	6	0.571	0.278	0.571	0.591	0.069	0.591
	10	0.500	0.235	0.500	0.553	0.064	0.553

Table A.24: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on DEAP dataset, with the EEG Context Neural Network (EEG-CNN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

DEAP - EEG Context Neural Network (EEG-CNN) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	1	0.564	0.380	0.564	0.512	0.335	0.512
	2	0.555	0.366	0.555	0.523	0.341	0.523
	4	0.576	0.392	0.576	0.525	0.338	0.525
	6	0.592	0.391	0.592	0.502	0.328	0.502
	10	0.570	0.394	0.570	0.528	0.337	0.528
	12	0.567	0.388	0.567	0.514	0.343	0.514
	15	0.559	0.392	0.559	0.523	0.356	0.523
Uniform	1	0.565	0.382	0.565	0.513	0.333	0.513
	2	0.587	0.397	0.587	0.524	0.328	0.524
	4	0.571	0.382	0.571	0.525	0.335	0.525
	6	0.586	0.394	0.586	0.532	0.341	0.532
	10	0.572	0.401	0.572	0.530	0.344	0.530
	12	0.566	0.390	0.566	0.496	0.334	0.496
	15	0.574	0.396	0.574	0.514	0.335	0.514
First	1	0.585	0.402	0.585	0.545	0.347	0.545
	2	0.589	0.399	0.589	0.513	0.331	0.513
	4	0.586	0.392	0.586	0.524	0.331	0.524
	6	0.582	0.400	0.582	0.520	0.325	0.520
	10	0.560	0.386	0.560	0.551	0.363	0.551
	12	0.606	0.415	0.606	0.501	0.335	0.501
	15	0.576	0.400	0.576	0.519	0.344	0.519
Last	1	0.562	0.378	0.562	0.506	0.325	0.506
	2	0.586	0.394	0.586	0.507	0.330	0.507
	4	0.577	0.400	0.577	0.555	0.362	0.555
	6	0.576	0.391	0.576	0.532	0.336	0.532
	10	0.575	0.395	0.575	0.523	0.353	0.523
	12	0.574	0.396	0.574	0.534	0.352	0.534
	15	0.588	0.393	0.588	0.514	0.333	0.514
EEG-Glimpse	1	0.609	0.431	0.609	0.584	0.382	0.584
	2	0.614	0.441	0.614	0.587	0.380	0.587
	4	0.611	0.446	0.611	0.578	0.393	0.578
	6	0.618	0.443	0.618	0.578	0.386	0.578
	10	0.603	0.428	0.603	0.575	0.392	0.575
	12	0.607	0.442	0.607	0.545	0.386	0.545
	15	0.591	0.428	0.591	0.541	0.373	0.541
Learned	1	0.608	0.417	0.608	0.578	0.364	0.578
	2	0.615	0.424	0.615	0.594	0.376	0.594
	4	0.613	0.440	0.613	0.580	0.381	0.580
	6	0.603	0.442	0.603	0.604	0.395	**0.604**
	10	0.625	0.459	0.625	0.587	0.396	0.587
	12	0.618	0.456	0.618	0.581	0.403	0.581
	15	0.629	0.460	**0.629**	0.578	0.407	0.578

Table A.25: *Performance results of the Subject-Dependent-Model (SDM) protocol using different consistency solutions on MAHNOB dataset, with the EEG Context Neural Network (EEG-CNN) model. The table includes evaluation metrics such as accuracy, F1-Score, and balanced accuracy, assessed across different time window sizes.*

MAHNOB - EEG Context Neural Network (EEG-CNN) - SMD							
		Arousal			Valence		
Consistency	**Window Size**	**Accuracy**	**F1-Score**	**Bal. Accuracy**	**Accuracy**	**F1-Score**	**Bal. Accuracy**
Centered	2	0.512	0.182	0.512	0.471	0.185	0.471
	4	0.537	0.207	0.537	0.504	0.210	0.504
	6	0.575	0.230	0.575	0.474	0.177	0.474
	10	0.532	0.215	0.532	0.482	0.188	0.482
Uniform	2	0.522	0.207	0.522	0.460	0.193	0.460
	4	0.519	0.197	0.519	0.529	0.215	0.529
	6	0.583	0.233	0.583	0.458	0.183	0.458
	10	0.578	0.233	0.578	0.520	0.204	0.520
First	2	0.547	0.230	0.547	0.537	0.232	0.537
	4	0.512	0.207	0.512	0.537	0.234	0.537
	6	0.558	0.228	0.558	0.488	0.188	0.488
	10	0.524	0.194	0.524	0.515	0.202	0.515
Last	2	0.535	0.194	0.535	0.575	0.234	**0.575**
	4	0.558	0.235	0.558	0.471	0.174	0.471
	6	0.558	0.220	0.558	0.458	0.155	0.458
	10	0.560	0.205	0.560	0.499	0.202	0.499
EEG-Glimpse	2	0.565	0.187	0.565	0.490	0.139	0.490
	4	0.611	0.230	0.611	0.496	0.183	0.496
	6	0.611	0.194	**0.611**	0.507	0.166	0.507
	10	0.517	0.194	0.517	0.504	0.183	0.504
Learned	2	0.565	0.189	0.565	0.529	0.166	0.529
	4	0.578	0.199	0.578	0.431	0.104	0.431
	6	0.604	0.210	0.604	0.477	0.153	0.477
	10	0.588	0.202	0.588	0.493	0.158	0.493